Theory of Causal Arisings

Denis Britov

April 19, 2026

Publication DOI: 10.5281/zenodo.19654196

Contents

1 Introduction **4**

2 The problem of arising and emergence in contemporary science **4**

3 Historical approaches to explaining multi-level organization **5**

3.1 Limitations of existing system theories and ontologies 7

4 Ontological units of causal arisings and their combinations **8**

4.1 Level phases as the sequential architecture of organizational development 9

4.2 Interaction modes as fundamental operators of causal configuration 10

4.3 Tabulated ontology of interaction modes 11

4.4 Causal directions as structural orientations of influence 15

4.5 Tabulated ontology of causal directions 15

4.6 Type roles as structural categories of emergent phenomena 18

4.7 Tabulated ontology of type roles 19

4.8 Carrier groups as ontological classes of quantitative causality 22

4.9 Tabulated ontology of carrier groups 24

4.10 Carrier types as quantitative operators of causality 28

4.11 Tabulated ontology of carrier types 29

4.12 Complexity levels as the hierarchical architecture of multi-scale organization 33

4.13 Tabulated architecture of complexity levels 33

5 Arising and causal configurations 36

5.1 Configuration as a condition of arising 36

5.2 Formal definition of a configuration in code form 37

5.3 Configurational codes as compositions of mechanisms 39

5.4 Example 1: DNA replication as a configurational code 39

5.5 Example 2: Virality of digital content as a configurational code 40

5.6 Example 3: Simple protein-based causal map in code form 41

5.7 How to compute configurational codes from data 42

5.8 Configuration $\neq$ result 43

5.9 Role of configurational codes in the overall theory 43

6 Principle of structural potency and ontology of possibility 44

6.1 The problem of possibility as a fundamental lacuna in science 44

6.2 What is structural potency 45

6.3 Emergence as the realization of structural potency 48

6.4 How is emergence different from a random fluctuation? 50

6.5 Potency and multilevel causality 51

6.6 From theory of possibility to engineering of arising 53

7 Causal maps of arising: structure, tensions, and prediction 53

7.1 Why mapping is necessary 53

7.2 The node as an ontological unit 55

7.3 The edge as an operator of arising 56

7.4 Ontological stratification of parent relations 58

7.5 Tension nodes and points of potential 61

7.6 Example: a simple causal map 63

7.7 How the map becomes predictive 65

7.8 Connection to a computable model 66

8 Causal completeness and the closure of arising conditions in ontological graphs 69

9 Lagrangian admissibility and the calculation of transitions 79

9.1 Why a Lagrangian description is needed 79

9.2 General form of the transition Lagrangian 80

9.3 Mapping causal configurations into Lagrangian terms 82

9.4 Domain-specific interpretation of subtractive structure 85

9.5 Transition criteria, thresholds, and regime change 88

9.6 Relation between Lagrangian selection and structural potency 92

9.7 Worked cross-level examples 94

9.8 Methodological limits of the generalized Lagrangian 97

10 Conclusion and further development of the theory 100

1 Introduction

The starting point of the "Theory of Causal Arisings" was a simple but consequential assumption: the world is not composed of fundamental entities, but of interactions, and those interactions are themselves the result of prior interactions. There is no final layer of irreducible elements. Every object, process, pattern, property, or effect exists only as a product of specific configurations of causal relations that have already formed before it.

From this perspective, describing the world as a collection of finished things or universal laws is methodologically insufficient. What actually exists are chains and networks of arisings, where each arising is conditioned by other arisings. For this reason, the project does not attempt to describe the full complexity of reality in a single unified model or formula. Instead, it treats each arising as a distinct ontological node and records the concrete causal relations through which it becomes admissible.

Each arising is described through its parent arisings, the directions of causation connecting them, and the interaction modes by which their relations operate. Taken together, these local descriptions form a multi-level causal map in which the structure of the real world can be decomposed without assuming any privileged foundational layer. Complexity becomes readable not by reduction to a single principle, but by explicit reconstruction of how configurations of arisings give rise to further arisings.

Within this framework, the absence of a single universal formula is not a limitation but a derived result. Formulas, laws, and models appear, transform, and disappear as local expressions of structural admissibility tied to specific configurations. The real world is thus understood as a genuinely complex system whose organization is divisible without limit, not into elemental substances, but into conditions of admissibility that recursively generate new levels of organization.

2 The problem of arising and emergence in contemporary science

Contemporary scientific disciplines rely on concepts that describe how new properties, structures, and organizational levels appear within complex systems. The general term *arising* denotes any transition resulting in the formation of new objects, states, or dependencies inside a broader system. Arising captures the fact of transition without specifying the degree of novelty, reducibility, or mechanistic tractability.

A specific class of arising is *emergence*, defined as the appearance of properties or patterns that cannot be reduced to the characteristics of constituent elements or to local interactions alone. Emergence expresses a qualitative transformation that cannot be accounted for by the sum or combination of lower-level properties. Typical examples include the appearance of consciousness from neural dynamics, the arising of living systems from biochemical processes, and the formation of collective behavior from individual actions.

In the English-language literature, the term *emergence* is applied both to simple arising and to

emergence in the strict sense. This terminological overlap complicates methodological analysis. In one case, the researcher refers only to the formation of a new state such as a molecule formed from atoms. In another case, the same word denotes irreducible novel properties, such as subjective experience. The lack of distinction merges separate explanatory strategies, reducing analytic precision. This project employs the term *arising* in a broad sense, encompassing simple transitions as well as qualitative transformations, with a central focus on configurations of causality rather than on reducibility or irreducibility.

The central problem arises from the fact that contemporary science contains many descriptions of specific arising processes but lacks a unified ontology capable of systematizing them as one class of causal transitions. Individual disciplines—physics, chemistry, biology, cognitive science, and others—use their own terminologies, methodological assumptions, and representational schemes. The heterogeneity of conceptual frameworks prevents the formulation of a universal map of causal arisings. As a result, no formal structure exists that would make it possible to document, compare, and analyze transitions between organizational levels, identify recurring invariants, and define conditions under which transitions to new qualities become possible.

3 Historical approaches to explaining multi-level organization

The idea of multi-level organization has a long intellectual history. Different scientific domains developed independent explanatory frameworks intended to account for how complex structures arise from simpler constituents.

Physics

Early natural philosophy introduced the first conceptions of stratified reality. Atomist thinkers such as Leucippus and Democritus proposed that indivisible elementary units generate observable diversity through their combinations. In the modern period, classical mechanics established a universal mathematical framework in which macroscopic motion followed deterministic laws. Within this paradigm, complexity was interpreted primarily as the aggregate result of simple mechanical interactions rather than as the appearance of qualitatively novel organizational properties.

The twentieth century revealed the limitations of classical reductionism. Quantum theory introduced superposition, entanglement, and uncertainty, thereby demonstrating the existence of properties not reducible to classical particle descriptions. Later developments in nonlinear dynamics and complexity theory identified mechanisms of self-organization and phase transition, making it possible to describe physical structures whose formation cannot be fully captured by classical reduction alone.

Chemistry

Chemical systems exhibit multi-level organization through bonding, molecular structure, and context-dependent interaction. Chemical bonds between atoms generate new properties at molecular and macroscopic scales. Catalysis provides a clear example: catalytic efficacy depends not only on chemical composition but also on spatial structure and environmental conditions. Such cases demonstrate that new properties arise through relational and configurational factors rather than through composition alone.

Across the history of chemistry, explanatory frameworks evolved from early conceptions of elemental building blocks to modern theories emphasizing bonding patterns, geometry, and contextual interaction. This development corresponds to a broader shift from strict reductionism toward systemic interpretation.

Biology

Biological theory historically oscillated between vitalistic explanation and mechanistic reductionism. Vitalism posited a special non-reducible force specific to living systems. Mechanistic approaches attempted to derive biological phenomena entirely from physical and chemical laws. Systems biology and general systems theory introduced an alternative perspective in which living systems were described as hierarchical, open, and dynamically regulated organizations. Evolutionary theory added a historical dimension by showing how new forms of stability, cooperation, and functional organization arise through selection processes.

Social sciences

Social research developed multiple approaches to multi-level organization. Marxist theory described society as a structured system in which economic relations shape large-scale ideological and institutional patterns. Structuralist approaches interpreted culture and social systems as networks of structural relations. Actor-network theory emphasized heterogeneous networks of human and non-human actants, demonstrating how social properties arise through distributed interaction rather than through isolated individual agency alone.

Cognitive sciences

The study of cognition likewise exhibits a shift in explanatory strategy. Behaviorism restricted analysis to observable behavior and external stimuli. Cognitive psychology and neuroscience redirected attention toward internal representation, information processing, and neural substrates. Research in artificial intelligence further demonstrated that cognitive capacities may arise from coordinated interactions within large networks of relatively simple computational units. This development renewed the question of how arising should be characterized in cognitive systems.

Conclusion

Historical analysis across scientific domains reveals a broad movement from reductionist explanation toward multi-level and systemic frameworks. Despite this convergence, each discipline developed its own conceptual vocabulary and explanatory structure independently. No universal ontology exists that can represent physical, biological, cognitive, and social arisings within a single coherent framework. This absence indicates the need for a unified theoretical model capable of analyzing causal arisings across domains.

3.1 Limitations of existing system theories and ontologies

Modern sciences of complexity contain several major directions that have significantly advanced the understanding of multi-level organization. Each of them, however, retains methodological limits that prevent the construction of a universal framework for analyzing causal transitions.

Systems theory

General systems theory, developed by Ludwig von Bertalanffy, introduced principles of integrity, hierarchy, openness, homeostasis, and feedback. These concepts made it possible to describe biological, ecological, and social systems as coherent wholes. Despite this contribution, systems theory does not provide a formal representation of multi-level transitions. It identifies levels of organization, but it does not specify the conditions under which a transition from one level to the next becomes possible. Nor does it formalize the mechanisms through which new organizational qualities arise.

Ontologies

Philosophical and applied ontologies such as OntoUML and Basic Formal Ontology focus primarily on the classification of entities and their relations. Their structural clarity is valuable for organizing domain knowledge. However, they are oriented mainly toward static entities, whereas processes, transitions, and causal relations remain secondary, indirect, or underrepresented. This limitation restricts their capacity to express dynamic organization and transformational structure.

Complexity theory

Complexity theory and nonlinear dynamics describe critical states, attractors, chaos, and phase transitions. These frameworks illuminate how systems behave near instability thresholds and provide important tools for modeling self-organization. Their primary focus, however, lies on dynamic regimes rather than on the ontological classification of causal transitions. They describe behavioral modes, but they do not supply a general ontology capable of representing the mechanisms through which new levels of organization arise.

Conclusion

Systems theory, applied ontology, and complexity theory each capture important aspects of organized systems. None of them, however, provides a universal treatment of arising. Systems theory describes integrity without formally reconstructing transitions. Ontological frameworks classify entities but insufficiently represent processes and causal transformation. Complexity theory captures dynamic regimes but does not provide a general classification of causal transitions. These limitations indicate the need for an ontological model explicitly centered on arising and causal transition as foundational components of the multi-level structure of the world.

4 Ontological units of causal arisings and their combinations

> The ontological typology of causal arisings is a formalized system of distinctions that defines the functional positions of elements, the admissible directions of their causal interactions, and the structural schemes through which these interactions combine within a level of organization. This typology establishes the basic vocabulary for describing how configurations form, interact, and produce new organizational outcomes.

To construct a universal framework for the analysis of causal transitions, it is necessary to introduce a set of ontological units that capture the essential dimensions of arising. These units specify not only what kinds of elements participate in a transition, but also how they interact, within which causal regime they operate, and at which phase of organizational development their contribution becomes decisive.

The classification employed in this work consists of several coordinated components. Interaction modes describe the fundamental patterns of influence between elements, specifying whether they act cooperatively, restrictively, through feedback, by initiation, or through the realization of potential. Causal directions define the orientation of influence across organizational levels, distinguishing upward, downward, horizontal, diagonal, and decoupled forms of causation.

Type roles characterize the structural nature of emergent phenomena, differentiating processes, objects, patterns, properties, and effects according to their functional contribution to organization. Level phases provide a temporal and structural sequence that describes how each level progresses from differentiation to integration, self-organization, and threshold transition. Finally, levels themselves form a hierarchical architecture ranging from the quantum domain to social and speculative domains, each representing a distinct mode of organization with its own constraints, potentials, and transition rules.

Taken together, these units form a coherent ontological system for the detailed analysis of causal configurations. Their explicit coordination makes it possible to trace multi-level transitions, compare heterogeneous phenomena, and formalize the conditions under which new organizational qualities arise. The system therefore serves both descriptive and constructive purposes, supporting empirical mapping, theoretical integration, and the engineering of complex emergent processes.

4.1 Level phases as the sequential architecture of organizational development

Level phases describe the internal progression through which each level of organization forms, stabilizes, restructures, and ultimately transitions into a higher-order configuration. They provide a temporal and structural decomposition of arising by identifying which mechanisms dominate at each stage and how causal contributions are reorganized as complexity increases. Whereas interaction modes define the operators of influence and causal directions define its orientation, level phases define the developmental logic within which these influences become active.

Within the ontology of causal arisings, level phases formalize a universal four-step progression observable across physical, biological, cognitive, and social systems. This progression begins with the differentiation of basic components, continues with their cooperative integration, advances toward regulated self-organization, and culminates in a threshold transition that generates a new organizational level. By capturing this progression explicitly, level phases make it possible to compare heterogeneous systems through a shared structural sequence.

The classification consists of four phases. The differentiation phase establishes the minimal set of components required for subsequent organization and is characterized by distinct identities and predominantly local interactions. The integration phase produces stable composite units and reproducible interaction patterns. The self-organization phase introduces regulatory structures, feedback loops, and adaptive coordination, thereby giving the system partial autonomy. The threshold transition phase marks the arising of a new organizational logic, as internal structures reach a qualitative limit and enable the formation of the next level.

Each phase performs a specific structural function. Differentiation provides the elements; integration produces cohesive units; self-organization stabilizes internal dynamics; and threshold transition opens access to a qualitatively new causal regime. These phases are not merely chronological stages but structural conditions, each expressing a distinct configuration of constraints, potentials, and permissible interactions.

The following table presents the four level phases in concise form. It outlines their definitions, characteristic signs, dominant interaction modes, and roles within the architecture of multi-level organization. This reference supports the subsequent analysis, in which identification of a phenomenon's phase becomes essential for tracing its causal role within broader transitions.

Phase	**Definition**	**Structural role**	**Characteristic signs**	**Predominant interaction modes**
LP_A — Differentiation	Formation of the minimal set of distinguishable elements with the properties required for subsequent organization.	Establishes the initial ontological basis of a level by producing the elementary units from which later organization can be constructed.	Emergence of distinguishable components, stabilization of basic identities, predominance of local interactions, and absence of system-wide coordination.	Dispositional causality, local triggering, intrinsic parameter formation, and emergence within minimal interaction contexts.

Phase	Definition	Structural role	Characteristic signs	Predominant interaction modes
LP_B — Integration	Cooperative combination of differentiated elements into stable patterns, assemblies, or structural units.	Transforms isolated elements into reproducible composite organizations and creates the first level-specific forms of coherence.	Formation of composite units, emergence of stable interaction patterns, mutual reinforcement or constraint, and partial systemic coherence.	Cooperative dynamics, synergy, structural triggering, semantic alignment, and mutual stabilization.
LP_C — Self-organization	Emergence of regulatory structures that stabilize, coordinate, and adaptively modulate the internal dynamics of the system.	Introduces endogenous regulation, feedback architecture, and partial autonomy, thereby enabling robust persistence under perturbation.	Appearance of feedback loops, dynamic equilibrium, adaptive coordination, resilience to perturbation, and internal differentiation of roles.	Systemic dynamics, feedback control, lateral loops, structural constraints, and regulated flows.
LP_D — Threshold transition	Critical transformation in which the system acquires qualitatively new properties irreducible to the previous configuration, thereby initiating a new organizational level.	Completes the developmental cycle of the current level and opens access to a distinct causal regime requiring higher-order description.	Appearance of new causality, systemic discontinuity, loss of reducibility, emergence of a new coherence domain, and triggering of cross-level effects.	Cross-scale influence, structural triggering, emergent constraints, semantic phase shift, and transition to a new organizational logic.

Table 1: Level phases as the sequential architecture of organizational development.

4.2 Interaction modes as fundamental operators of causal configuration

Interaction modes represent the primary operators through which elements of a system influence one another and participate in the formation of higher-order structures. Each mode defines a distinct pattern of causal contribution, specifying how individual components combine, modify, constrain, or activate collective behavior. These modes do not describe the elements themselves; rather, they describe the principles by which relations between elements generate admissible configurations.

Within the ontology of causal arisings, interaction modes function as the minimal building blocks of causal logic. They determine how local interactions accumulate into macro-level effects, how permissive conditions are converted into active transitions, and how dynamic closures produce stable structures. Their classification makes it possible to represent the internal grammar of arising, in which each mode operates as a formal operator shaping both the trajectory and the admissibility of system-wide transformation.

The classification employed in this framework distinguishes five principal interaction modes: cooperative enhancement, structural constraint, feedback organization, initiation triggering, and realization. Each mode is associated with its own activation conditions, characteristic failure modes, regulated flows of energy, matter, or information, and formal mathematical expressions. Interaction modes therefore provide a rigorous basis for the analysis of causal transitions at any level of organization.

Within multi-level organization, the role of interaction modes is twofold. First, they define which configurations are structurally possible at a given level by specifying the compatibility, stability, and directional organization of influence. Second, they function as a universal repertoire of transformation operators whose combinations generate new organizational forms. Analysis of the composition and sequencing of interaction modes therefore makes it possible to reconstruct how a configuration moves from latent structural potency to stable emergent form.

The following table presents the five interaction modes in a structured format. It summarizes their definitions, characteristic roles across organizational phases, activation conditions, typical failure modes, regulated flows, and corresponding mathematical expressions. This tabular reference will support the subsequent chapters, in which interaction modes are used to describe causal mechanisms across physical, biochemical, cognitive, and social domains.

4.3 Tabulated ontology of interaction modes

Because the ontology of interaction modes contains several distinct dimensions of description, its presentation is divided into three complementary tables. The first table establishes the conceptual identity of each mode and its structural role across developmental phases. The second table specifies the conditions under which each mode becomes operative, the characteristic ways in which it fails, and the principal forms of flow it regulates. The third table summarizes the formal mathematical expressions associated with each mode together with their indicative validation logic. Taken together, these tables provide a compact but analytically complete reference for the use of interaction modes throughout the theory.

Conceptual identity and phase-specific role of interaction modes

The following table presents the conceptual core of each interaction mode. It specifies the formal code and name, the general definition, and the role that the mode tends to assume within each of the four level phases. This table is intended to function as the primary conceptual reference for identifying the structural contribution of a mode within a causal configuration.

Interaction mode	**Definition**	LP_A	LP_B	LP_C	LP_D
IM_C — Cooperation	Local mutual enhancement between elements, leading to stable patterns and modules through synergistic interaction.	Rudimentary compatibility between differentiated elements; preparation of later pattern formation.	Dominant integration mechanism through which stable patterns and modules are assembled.	Supports alignment of modes in the presence of feedback and distributed coordination.	Strengthens threshold transition under critical saturation, although it does not define the transition by itself.
IM_K — Constraint	Restriction of the state space through form, context, or rule, thereby channeling trajectories and fixing admissibility.	Performs early filtering of incompatible modes at the minimal level.	Tunes and delimits the structure of assembled patterns.	Stabilizes and regulates existing modes, typically with reduced dominance relative to integration.	Filters new organizational logic and reorganizes admissibility conditions at the transition threshold.

Interaction mode	**Definition**	LP_A	LP_B	LP_C	LP_D
IM_F — Feedback	Closed cycles of influence, including lateral and diagonal forms, that stabilize, amplify, or redirect the system's mode.	Appears only weakly, typically in local preparatory mini-loops.	Introduces weak stabilization contours within collected patterns.	Becomes dominant in regulation and mode retention.	Generates cross-scale loops capable of initiating or shaping transition.
IM_I — Initiation	Local triggering act or condition that converts latent permissiveness into active unfolding of a configuration.	Triggers differentiation and local process onset.	Initiates cooperative assembly once sufficient density or compatibility has been reached.	Activates regulatory loops that remain latent until triggering conditions are satisfied.	Triggers threshold transition through minimal internal or external impulse.
IM_R — Realization	Transformation of an admissible configuration of potential into an observable effect, including fixation of a new function or quality.	Appears only weakly as local micro-realizations indicating permissiveness.	Produces partial manifestations within assembling patterns.	Realizes functional effects at the level of contour or mode.	Fixes the effect and marks the birth of a new contour or organizational level.

Table 2: Conceptual identity and phase-specific structural role of interaction modes.

Activation conditions, failure modes, and regulated flows

The next table presents the operational profile of each interaction mode. It identifies the conditions under which a mode becomes active, the characteristic ways in which its operation may break down, and the principal flows of energy, information, and matter that it regulates. This table is intended to support causal diagnosis and comparison across heterogeneous domains.

Interaction mode	**Activation conditions**	**Failure modes**	**Energy flow**	**Information flow**	**Matter flow**
IM_C — Cooperation	Functional complementarity or compatible interfaces; sufficient local density of connections; common contextual or resource support excluding competing modes.	False synergy caused by hidden dependence on an external node; trapping in a local optimum through excessive cooperation and loss of variability; creeping cooperation with progressive blurring of node roles.	Reduction of dissipation through alignment of action.	Increase of redundancy and reliability of encoding.	Local condensation and assembly of structures.

Interaction mode	Activation conditions	Failure modes	Energy flow	Information flow	Matter flow
IM_K — Constraint	Presence of a stable form, rule, or boundary; alignment between the scale of constraint and the scale of dynamics; resource ceilings or norms excluding unsuitable modes.	Overconstraint leading to fragility or failure under perturbation; mismatch between the scale of constraints and microdynamics; illegitimate constraint producing meta-conflict at the rule level.	Channeling of flows and reduction of excessive degrees of freedom.	Increase in predictability through reduction of conditional entropy.	Formation of admissible geometry, including niches, orbitals, and packing templates.
IM_F — Feedback	Presence of a closed loop in the interaction graph; effective loop gain of appropriate magnitude; coordination of timescales in the presence of delays.	Destabilization through unbounded positive feedback; autonomous oscillation or chaos produced by delay-amplification interaction; overdamping with consequent loss of adaptability.	Loop-power redistribution with retention of operating mode.	Amplification or suppression of signals and reduction of regulation error.	Recycling of material flows within closed loops.
IM_I — Initiation	Presence of structural potential allowing transition; exceedance of a local threshold θ in amplitude, gradient, or concentration; access to a resource or channel for propagation.	Insufficient amplitude with threshold not reached; false triggering through isolated surges without supporting configuration; premature activation in the absence of prepared contours.	Overcoming the activation or energetic barrier.	Trigger signal in the form of an impulse, event, or timestamp.	Seed or nucleus of formation, or a local reagent source.
IM_R — Realization	Fulfillment of permissive conditions through potential plus activating connections; availability of a fixation or observation channel; closure of the contour under small perturbation.	Latent potency without fixation; disruption of realization by conflicting constraints; replacement of effect by measurement artefact.	Transition to a mode with a new power distribution.	Explicit observable change in signal, state, or code.	Material manifestation, for example in morphogenesis or product release.

Table 3: Activation conditions, failure modes, and regulated flows of interaction modes.

Mathematical form and validation logic

The final table summarizes the formal mathematical expressions associated with each interaction mode and the corresponding logic of validation. The equations should not be interpreted as a single universal formalism, but as representative operator-level expressions that capture the characteristic structure of each mode. The validation column indicates the kinds of empirical or formal signatures that would support or challenge the attribution of the mode within a given configuration.

Interaction mode	**Mathematical form**	**Indicative predictions**	**Falsification criteria**
IM_C — Cooperation	Positive contribution in effective equations; in the CRT setting, cooperation is represented by a cubic coupling term: $$L \supset \lambda(\psi_1\psi_2\psi_3 + c.c.), \qquad \lambda > 0. \quad (1)$$	Growth of clustering and k-core structure should precede the formation of stable patterns.	Rupture of a small number of key cooperative edges should destroy the pattern while leaving the intrinsic parameters of the elements unchanged.
IM_K — Constraint	Constraints are represented by admissibility conditions and Lagrange multipliers, for example: $$\min S[\psi] \quad \text{subject to} \quad \phi(\psi) \leq 0, \quad (2)$$ $$L_{\text{eff}} = L + \sum_i \lambda_i \phi_i(\psi). \quad (3)$$	Tightening constraint should reduce the set of achievable trajectories and increase the repeatability of the effect.	Loosening or removing constraint should expand the behavioral repertoire and reduce repeatability; failure of this pattern would count against the attribution.
IM_F — Feedback	Closed-loop update structure with effective gain: $$x_{t+1} = Ax_t + By_t \quad (4)$$ $$y_t = Cx_t \Rightarrow x_{t+1} = (A + BC)x_t, \quad (5)$$ with criticality estimated by $$\rho(A + BC) \gtrless 1. \quad (6)$$	Increase of spectral radius in the circuit should precede a mode change; introduction of negative feedback should reduce output variance.	Closure of explicit cycles should alter system mode under otherwise equal conditions; if it does not, and hidden loops are excluded, the attribution is weakened.
IM_I — Initiation	Threshold or escape dynamics, for example: $$u' = f(u) + H(g(u) - \theta), \quad (7)$$ $$\lambda_{\text{escape}} \propto \exp(-\Delta E/D). \quad (8)$$	Where structural potential exists, a small impulse should initiate sustained unfolding; where it does not, even large impulses should produce only damped responses.	In the absence of structural potential, small impulses should not systematically generate stable novelty.
IM_R — Realization	Mapping from potential and conditions to observable effect: $$R : (P, C) \to E, \quad (9)$$ $$E = \mathcal{O}[\psi], \quad (10)$$ and in the CRT-node context: $$\Gamma(t) = \int \lvert\Psi\rvert^2 \, dx. \quad (11)$$	Given potency and triggering conditions, a stable effect-node should arise; fixation of the effect should be accompanied by emergence of a new closed circuit including that node.	A stably fixed effect without closed loops, under control for hidden loops, would count against the realization attribution.

Table 4: Mathematical form and validation logic of interaction modes.

4.4 Causal directions as structural orientations of influence

Causal directions define the orientation and scope of influence between components of a system across its organizational hierarchy. They specify how interactions propagate, how constraints or potentials are transmitted between levels, and how distributed dynamics combine into observable effects. Whereas interaction modes describe the internal logic of influence, causal directions describe its pathway and structural position within the larger organization of the system.

Within the ontology of causal arisings, causal directions function as fundamental descriptors of the relation between micro-level and macro-level processes. They determine whether an effect originates in local interaction, in higher-level constraint, in lateral coupling, or in cross-level circuits that simultaneously involve multiple scales. This dimension is indispensable for tracing the formation of new organizational qualities, because the orientation of causality determines which mechanisms are operative and which transformations are structurally admissible.

The classification includes upward, downward, horizontal, diagonal, decoupled, and meta-level forms of causation. Upward causation describes how lower-level components generate higher-level structures. Downward causation captures how organized wholes impose constraints and admissibility conditions on their parts. Horizontal causation denotes interactions within the same level, including cooperative, competitive, and synergistic forms. Diagonal causation accounts for cross-level circuits in which influences propagate simultaneously along multiple causal axes. Decoupled causation describes interactions governed by effective macro-parameters that bypass detailed micro-description. Metacausation addresses the logical and semantic frameworks within which causality itself becomes definable.

Each causal direction performs a distinct organizational function. Upward causation supports the buildup of complexity through local interaction. Downward causation stabilizes or reconfigures microdynamics through higher-level structure. Horizontal causation produces distributed patterning and lateral coordination. Diagonal causation links scales into integrated circuits of mutual dependence. Decoupled causation enables effective theory and large-scale description. Metacausation defines the logical preconditions under which causal relation can be formulated at all. Taken together, these directions map the full spectrum of structurally possible orientations of influence.

The following table presents these causal directions in formalized form. It summarizes their definitions, functional roles, characteristic regimes of operation, and principal domains of application. This reference will support the subsequent analysis of how specific emergent phenomena combine multiple causal orientations in the generation of new organizational levels.

4.5 Tabulated ontology of causal directions

Because causal directions combine definitional, diagnostic, and formal dimensions, their presentation is divided into three complementary tables. The first table establishes the conceptual identity of each causal direction. The second specifies the formal criteria and detection tests by which each direction may be identified in analysis. The third summarizes the representative mathematical forms and the principal failure modes associated with misclassification or misuse. Taken together, these tables provide a compact analytical reference for tracing how influence is structurally oriented within and across levels of organization.

Conceptual identity of causal directions

The following table presents the six causal directions as the primary structural orientations of influence recognized in the ontology of causal arisings. It specifies their formal codes, names, and general definitions. This table functions as the conceptual point of reference for distinguishing how influence is oriented across hierarchical organization.

Causal direction	**Definition**
CD_U — Upward causation	Causal direction in which interactions and configurations at a lower descriptive level generate structures, properties, or dynamics at a higher level. Influence is expressed through aggregation, coarse-graining, or composition.
CD_D — Downward causation	Causal direction in which higher-level structures or states constrain, modulate, or parameterize the behavior of lower-level components. The macro-level shapes the admissible space of micro-dynamics.
CD_H — Horizontal causation	Causal interaction occurring between components or processes at the same descriptive level, forming local dynamics without hierarchical asymmetry.
CD_{DI} — Diagonal causation	Causal direction characterized by closed or intertwined paths of influence spanning multiple levels simultaneously, integrating upward, downward, and horizontal components into a single circuit.
CD_{DC} — Decoupled causation	Causal direction in which interactions between higher-level entities are governed by effective parameters that abstract away micro-level detail, such that microstates are statistically or structurally marginalized.
CD_M — Metacausation	Causal direction operating at the level of causal admissibility itself, defining when and how causal relations can be formulated, compared, or validated.

Table 5: Conceptual identity of causal directions.

Formal criteria and detection tests

The next table specifies the formal criteria that characterize each causal direction together with the principal detection tests by which it may be identified in empirical, theoretical, or model-based analysis. These criteria are not intended as strict necessary-and-sufficient conditions in every case, but as a structured diagnostic framework for distinguishing different orientations of causal influence.

Causal direction	**Formal criteria**	**Detection tests**
CD_U — Upward causation	Macrovariables are functions of microvariables; higher-level description is undefined without lower-level interactions; removal of micro-interactions eliminates the macro-phenomenon.	Altering micro-interactions changes macro-behavior; macro-structure cannot be postulated independently; multiple microstates map onto the same macrostate.
CD_D — Downward causation	Micro-dynamics include parameters or constraints derived from macrostate; macrostate acts as a boundary condition or control variable; micro-equations change form depending on macro-context.	Changing macrostate alters micro-level trajectories; micro-components behave differently under identical local conditions but different macro-contexts; macro-removal restores a broader microstate space.

Causal direction	Formal criteria	Detection tests
CD_H — Horizontal causation	Interacting variables belong to the same level of description; no coarse-graining or level transition is required; causality is symmetric or peer-based.	Removing one peer directly affects another at the same level; causal influence does not propagate through a higher-level variable; the interaction graph remains intra-level.
CD_{DI} — Diagonal causation	Presence of feedback loops crossing levels; mutual dependence between micro- and macrovariables; no single causal direction is sufficient to explain the dynamics.	Breaking either upward or downward linkage destroys the phenomenon; stability depends on cross-level loop gain; emergent behavior appears only in the closed multi-level system.
CD_{DC} — Decoupled causation	Effective equations omit explicit microvariables; macro-dynamics remain valid across micro-variation; renormalized or averaged parameters exist.	Changing micro-details does not affect macro-behavior; macro-predictions remain accurate without micro-resolution; explicit micro-reconstruction is unnecessary.
CD_M — Metacausation	Concerns rules of causation rather than causal flows; operates on models, logics, or representational frameworks; defines constraints on what counts as a cause.	Altering meta-rules changes admissible causal explanations; causality fails without these conditions; no direct empirical signal exists without lower-level instantiation.

Table 6: Formal criteria and detection tests for causal directions.

Mathematical form and principal failure modes

The final table summarizes representative mathematical forms associated with each causal direction together with the principal failure modes that arise when a direction is misidentified or overextended beyond its proper domain. The equations should be read as schematic formalizations of structural orientation rather than as exhaustive descriptions of all possible implementations.

Causal direction	Mathematical form	Principal failure modes
CD_U — Upward causation	Constructive mappings from microstates to macrostates: $$X = \Phi(x_1, \ldots, x_n), \quad (12)$$ $$\dot{X} = F(\{x_i\}). \quad (13)$$	Confusing correlation with generation; ignoring the aggregation scale required for macro-level formation.
CD_D — Downward causation	Constraints or control parameters imposed on micro-dynamics: $$\dot{x}_i = f(x_i; X), \quad (14)$$ $$\phi(x_i, X) \leq 0. \quad (15)$$	Misinterpreting feedback as pure downward control; attributing global influence without explicit constraints or control variables.

Causal direction	Mathematical form	Principal failure modes
CD_H — Horizontal causation	Direct coupling between peer variables: $$\dot{x}_i = f(x_i, x_j), \quad (16)$$ $$A_{ij} \neq 0. \quad (17)$$	Assuming horizontal coupling where hidden macro-mediation is operative; collapsing genuinely collective effects into intra-level linkage alone.
CD_{DI} — Diagonal causation	Coupled multi-level dynamics with feedback: $$\dot{x} = f(x, X), \quad (18)$$ $$\dot{X} = F(X, x), \quad (19)$$ $$\rho(A + BC) \geq 1. \quad (20)$$	Reducing diagonal causation to simple feedback; ignoring coordination of time scales across levels.
CD_{DC} — Decoupled causation	Effective theory representation: $$\dot{X} = F_{\text{eff}}(X), \quad (21)$$ $$\int \mathcal{D}x\, e^{-S[x]} \to S_{\text{eff}}[X]. \quad (22)$$	Applying decoupling outside its regime of validity; confusing epistemic ignorance with genuine effective independence.
CD_M — Metacausation	Meta-rules and logical admissibility structures: $$C : (M, \mathcal{L}) \to \{\text{valid causal relations}\}, \quad (23)$$ $$\models_{\mathcal{L}} (A \to B). \quad (24)$$	Treating metacausation as physical interaction; embedding meta-rules directly into object-level dynamics.

Table 7: Mathematical form and principal failure modes of causal directions.

4.6 Type roles as structural categories of emergent phenomena

Type roles classify emergent phenomena according to their functional position within an organizational configuration. Whereas interaction modes specify how elements influence one another, and causal directions specify how that influence is oriented across levels, type roles specify the kind of structural contribution made by the resulting phenomenon. They indicate whether a phenomenon functions primarily as a transformation, a stabilized configuration, a recurrent regularity, a system-level attribute, or a large-scale effect.

Within the ontology of causal arisings, type roles constitute a minimal structural grammar capturing the principal ways in which emergent units participate in multi-level organization. Each role reflects a distinct manner in which a phenomenon integrates causal contributions, relates to other units, and shapes the admissible space of configurations. By distinguishing these roles, it becomes possible to compare heterogeneous emergent processes across physics, chemistry, biology, cognition, and social systems within a unified structural vocabulary.

The classification includes five primary roles. The process role denotes temporally extended transformations driven by coordinated interaction and feedback-conditioned progression. The object role identifies persistent configurations stabilized by compatible constraints and interaction patterns. The pattern role denotes

reproducible regularities that structure local or global organization. The property role captures system-level attributes irreducible to individual components or simple aggregates. The effect role designates large-scale outcomes produced by integrated microdynamics and cross-level closure.

Each type role corresponds to a distinct mode of participation in organization. Processes transform system states; objects maintain stable structures; patterns guide and constrain organization through recurrence; properties express collective capacities; and effects propagate consolidated outcomes across scales. These distinctions do not describe intrinsic substance but functional contribution, thereby allowing each emergent phenomenon to be positioned according to the role it performs within a causal configuration.

The following table presents these type roles in structured form. It summarizes their definitions, characteristic causal profiles, typical domains of appearance, and the kinds of transitions they support. This reference will underpin the subsequent analysis in which emergent phenomena are classified according to their structural role within the broader architecture of causal arising.

4.7 Tabulated ontology of type roles

Because type roles combine conceptual, diagnostic, and formal dimensions, their presentation is divided into three complementary tables. The first table introduces the conceptual identity of each role. The second specifies the formal criteria and detection tests by which a phenomenon may be assigned to a given role. The third summarizes representative mathematical forms together with the principal failure modes associated with misclassification. Taken together, these tables provide a compact analytical reference for classifying emergent phenomena according to their structural role within causal configurations.

Conceptual identity of type roles

The following table presents the five type roles as the principal structural categories through which emergent phenomena participate in organization. It specifies their formal codes, names, and general definitions. This table serves as the primary conceptual reference for distinguishing the functional contribution made by a phenomenon within a causal configuration.

Type role	**Definition**
TR_{PR} — Process	Emergent phenomenon understood as a time-extended transformation that exists only through ordered change of system state. A process is present when the phenomenon cannot be defined without an explicit temporal parameter and requires a trajectory, flow, or evolution operator for its description.
TR_{OBJ} — Object	Emergent phenomenon understood as a persistent configuration stabilized by internal coherence and admissible constraints. An object exists as a distinguishable carrier that remains identifiable under small perturbations.
TR_{PAT} — Pattern	Emergent phenomenon understood as a reproducible regularity or motif recurring across components, time, or scales. A pattern is defined by form or relation rather than by a single specific carrier.
TR_{PROP} — Property	Emergent phenomenon understood as a system-level attribute that cannot be assigned to individual components but characterizes the system as a whole. A property functions as a derived macroscopic parameter.

Type role	Definition
TR_{EFF} — Effect	Emergent phenomenon understood as a fixed outcome or influence produced by system dynamics and propagated beyond the originating structure. An effect is defined primarily by its impact rather than by its internal organization.

Table 8: Conceptual identity of type roles.

Formal criteria and detection tests

The next table presents the formal criteria and indicative detection tests associated with each type role. These criteria provide a structured basis for assigning a phenomenon to a role by specifying the conditions under which its structural identity becomes analytically distinguishable. The table is intended as a diagnostic guide for classification across heterogeneous domains.

Type role	Formal criteria	Detection tests
TR_{PR} — Process	Explicit dependence on time or an ordering parameter t; existence defined by state transition rather than static configuration; absence of the phenomenon in frozen or time-collapsed description.	Removal of temporal evolution eliminates the phenomenon; different initial conditions yield distinct trajectories of the same process; process identity is preserved across realizations with varying substrates.
TR_{OBJ} — Object	Existence of a stable invariant, attractor, or bounded region in state space; persistence under perturbations within a defined tolerance; definable boundary separating the object from its environment.	The phenomenon can be referenced as a stable unit; removal of internal coherence destroys object identity; object persists across multiple realizations of the same constraints.
TR_{PAT} — Pattern	Repetition or invariance under transformation; independence from a single physical instance; statistical or structural regularity across realizations.	The same structure appears in multiple locations or instances; the pattern survives replacement of individual components; statistical measures reveal non-random regularity.
TR_{PROP} — Property	Defined as a function of many microstates; irreducible to any single component variable; well-defined only at a specific descriptive scale.	The property vanishes when the system is decomposed; different microconfigurations yield the same property value; the property governs effective behavior at higher levels.
TR_{EFF} — Effect	Existence as an output or consequence of system operation; ability to influence other systems or levels; fixation as an observable result.	The effect persists after internal dynamics are abstracted away; it can be measured without reconstructing the full system; it participates in new causal chains as an input.

Table 9: Formal criteria and detection tests for type roles.

Mathematical form and principal failure modes

The final table summarizes representative mathematical forms associated with each type role together with the principal failure modes that arise when roles are confused or misapplied. The equations should be read as schematic formalizations of structural role rather than as exhaustive models of all possible realizations.

Type role	**Mathematical form**	**Principal failure modes**
TR_{PR} — Process	Processes are represented by dynamical equations defining trajectories in state space: $$x_{t+1} = F(x_t), \quad (25)$$ $$\frac{dx}{dt} = f(x), \quad (26)$$ $$\Gamma = \{x(t) \mid t \in [t_0, t_1]\}. \quad (27)$$	Misclassification of stable dynamics as objects; reduction of a process to a single snapshot or isolated outcome.
TR_{OBJ} — Object	Objects correspond to invariant or weakly attracting structures: $$F(\mathcal{O}) = \mathcal{O}, \quad (28)$$ $$d(\mathcal{O}, \mathcal{O}') < \epsilon \quad \text{under perturbation.} \quad (29)$$	Confusing transient patterns with objects; overextending objecthood to loosely correlated assemblies.
TR_{PAT} — Pattern	Patterns are represented through symmetry, correlation, or equivalence classes: $$P(x) = P(gx), \qquad g \in G, \quad (30)$$ $$C(x_i, x_j) \approx \text{const}, \quad (31)$$ $$\rho(x,t) = \rho(x+\lambda, t+\tau). \quad (32)$$	Mistaking single instances for patterns; confusing objects with their repeated arrangements.
TR_{PROP} — Property	Properties are represented as macroscopic observables or order parameters: $$P = \Phi(x_1, \ldots, x_n), \quad (33)$$ $$P \notin \{x_i\}, \quad (34)$$ $$P = \langle \mathcal{O} \rangle. \quad (35)$$	Attributing system-level properties to individual elements; confusing properties with effects or outcomes.
TR_{EFF} — Effect	Effects are represented as output functionals or propagated impacts: $$E = \mathcal{O}[\psi], \quad (36)$$ $$R : (P, C) \to E, \quad (37)$$ $$E(t) \to \text{external domain.} \quad (38)$$	Confusing effects with properties; attributing causal structure to pure outcomes abstracted from their generating organization.

Type role	Mathematical form	Principal failure modes

Table 10: Mathematical form and principal failure modes of type roles.

4.8 Carrier groups as ontological classes of quantitative causality

The introduction of quantitative causality requires a precise distinction between what is counted and how it is structurally represented. In the causal graph of arisings, nodes denote units of arising, whereas edges denote typed causal relations. Not every causal relation, however, becomes active solely by virtue of its qualitative specification. Many relations depend on the accumulation of concrete carriers whose quantity determines whether the relation is inactive, linearly active, threshold-transformative, or saturated. The ontology of Carrier Groups formalizes the nature of such carriers prior to the application of any specific counting rule.

A Carrier Group is defined as a class of ontologically homogeneous quantitative bearers whose accumulation controls the activation or regime of a causal edge. Carrier Groups do not describe objects or processes in general. Rather, they specify the type of entity that must be numerically present for a given causal mechanism to operate. This distinction separates the structural topology of the graph from the ontological nature of the counted units.

Without this distinction, quantitative analysis collapses into pure graph theory, in which every carrier is reduced to a node or an edge. Such reduction erases critical differences between physical bodies, resonant modes, functional material units, structural motifs, informational tokens, and agents. Although all of these may be represented graphically, they obey fundamentally different quantitative regimes. The theory of Carrier Groups preserves these differences as ontological invariants.

Ontological criterion of a carrier group

A Carrier Group is characterized by four structural properties. First, it defines a unit of counting whose identity remains stable within a given descriptive level. Second, it defines a boundary rule specifying how a counted unit is individuated. Third, it defines a membership rule specifying when a unit participates in a given causal motif. Fourth, it defines a class of quantitative regimes admissible for that group.

Formally, let e be a causal edge and let N_x denote the number of carriers of type x participating in the motif of e. A Carrier Group specifies the ontological nature of x. The activation function $A_e(N_x)$ and the regime function $R_e(N_x)$ depend on the group to which x belongs. The same numerical value of N_x may therefore produce distinct structural effects depending on whether x belongs to a body-based, mode-based, functional, topological, informational, or agent-based group.

Field / mode carriers

Field / mode carriers denote excitations, coherent modes, or resonant components whose coupling constitutes the operative causal mechanism. Their counting ontology is defined by representational identity in mode space rather than by geometric separability. Quantitative thresholds in this group frequently arise from minimal closure in coupling space. Triadic closure is a common structural motif, in which three coherently coupled modes generate a self-sustaining configuration that is not reducible to pairwise interaction alone.

The quantitative logic of this group is primarily structural rather than statistical. In many cases, N_{crit} corresponds to the minimal number of modes required for closure under admissibility constraints. Beyond this point, additional modes may increase degrees of freedom or control parameters without necessarily introducing a qualitatively new regime.

Physical body carriers

Physical body carriers denote discrete, separable entities whose interactions form N-body configurations. Their individuation is geometric or operational, and membership is determined by dynamic coupling within a shared motif. The quantitative logic of this group is characterized by discrete structural thresholds. Two bodies define a minimal interaction. Three bodies introduce closure that often prevents full pairwise reduction. Four or more bodies typically add degrees of freedom that permit regime modulation rather than fundamental qualitative novelty.

Unlike collective-phase carriers, thresholds in this group are usually small and topological. Emergence in this context may therefore appear already at low N, especially where closure introduces irreducibility.

Functional material carriers

Functional material carriers include molecules, bonds, active sites, complexes, and cells treated as functionally individuated units. Their counting ontology depends on functional identity rather than purely geometric separability. Within this group, two distinct quantitative regimes coexist. Small-N thresholds correspond to cooperative or stoichiometric effects, in which a limited number of units generates nonlinear functional change. Large-N thresholds correspond to collective or phase transitions, in which bulk behavior emerges through statistical accumulation.

The quantitative thresholds in this group are often context-dependent. Temperature, boundary conditions, concentration, and geometric constraints may modify N_{crit} and N_{sat}. For this reason, this group often interacts strongly with contextual parameters in determining regime activation.

Topological / configurational carriers

Topological carriers denote structural elements of the causal graph itself, including edges, cycles, strongly connected components, cores, and percolating structures. Their counting ontology is purely structural. Units are defined by graph-theoretic predicates rather than by material individuation. This group is unique in being admissible across all levels of organization.

Quantitative thresholds in this group frequently correspond to motif closure, such as the formation of minimal cycles, or to connectivity transitions, such as the emergence of a giant component. The structural nature of this group makes it especially relevant for diagonal causation and feedback-dominated regimes.

Informational / symbolic carriers

Informational carriers are discrete representational units such as tokens, symbols, memory states, or effective parameters. Their individuation is representation-dependent rather than material. Quantitative thresholds

in this group often correspond to minimal expressivity, minimal memory capacity, or sufficient state-space diversity to stabilize a regime.

Unlike purely topological carriers, informational carriers require semantic or representational constraints in order to define unit identity. The same graph structure may therefore host different informational regimes depending on token diversity, coding rules, or state capacity.

Agent / population carriers

Agent / population carriers consist of autonomous participants, role instances, or interaction events at collective scales. Their individuation is defined by participation identity within a social or coordinated motif. Quantitative thresholds in this group often correspond to the stabilization of coordination regimes, norm formation, market formation, or institutional persistence.

Large-N behavior in this group frequently generates collective-phase regimes in which macro-level properties become stable and self-maintaining. At the same time, small-N thresholds may remain decisive where minimal coalition size or coordination number is required for regime activation.

Why carrier groups cannot be reduced to pure topology

Although all carriers can be represented within a graph, their quantitative semantics differ fundamentally. A triadic threshold in mode space is not equivalent to a triadic threshold in agent space. A phase transition in molecular populations is not equivalent to the appearance of a feedback cycle in a graph. Carrier Groups preserve these ontological distinctions by binding quantitative logic to the nature of the counted units.

Carrier Groups therefore function as an intermediate ontological layer between qualitative causation and quantitative activation. They ensure that identical numerical growth is interpreted according to the correct causal ontology. Without them, quantitative causality would collapse into undifferentiated node counting, eliminating the structural precision required for analyzing thresholds, emergence, and regime transitions.

4.9 Tabulated ontology of carrier groups

Because Carrier Groups combine ontological, quantitative, and diagnostic dimensions, their presentation is divided into three complementary tables. The first table establishes the conceptual identity of each group together with its counting ontology. The second specifies the levels at which each group is admissible, the typical quantitative regimes associated with it, and the principal quantitative remarks governing threshold interpretation. The third summarizes the characteristic failure modes that arise when carrier groups are confused, overextended, or detached from their proper ontological basis. Taken together, these tables provide a compact analytical reference for the role of Carrier Groups in quantitative causality.

Conceptual identity and counting ontology of carrier groups

The following table presents the conceptual identity of each Carrier Group. It specifies the formal code, the general definition, and the core ontology of counting: what kind of unit is counted, how the boundary of a

unit is established, and under what membership condition a unit belongs to the causal motif relevant to a given edge.

Carrier group	**Definition**	**Unit kind**	**Boundary rule**	**Membership rule**
CG_{FM} — Field / mode carriers	Carrier group in which the countable units are field excitations, coherent modes, or resonant components whose coupling constitutes the causal carrier of an edge.	`mode_instance`	Mode identity is determined by representation-specific labels, such as mode index, band, or resonance label, together with coherence or persistence within the motif window.	A unit is counted only if it participates in the same coupling motif relevant to the edge's mechanism, rather than merely coexisting in the same system.
CG_{PB} — Physical body carriers	Carrier group in which the countable units are discrete bodies or object-like participants whose mutual interactions constitute the causal carrier of an edge.	`body`	Body boundary is defined at the chosen descriptive scale through operational separability and persistence across the motif window.	A body is counted only if it is dynamically coupled into the motif relevant to the edge, rather than merely spatially present or nearby.
CG_{FMC} — Functional material carriers	Carrier group in which the countable units are functionally relevant material units such as molecules, bonds, active sites, complexes, or cells whose quantity controls the activation or regime of an edge.	`functional_material_unit`	Unit boundary is determined by functional individuation at the chosen level, for example chemical identity, bond instance, active-site identity, complex membership, or cell boundary.	A unit is counted only if it participates in the mechanism-relevant motif for the edge, such as bond instances contributing to a percolating network or active sites contributing to throughput.
CG_{TC} — Topological / configurational carriers	Carrier group in which the countable units are structural elements of the causal configuration, such as edges, cycles, cores, or components, whose abundance or presence controls activation or regime of an edge.	`graph_structural_element`	Element identity is defined by graph-theoretic criteria under the ontology's typing constraints, including interaction modes, causal directions, and role constraints.	Only elements satisfying the motif specification for the edge are counted, for example cycles containing specific interaction modes or strongly connected components above a required size.
CG_{IS} — Informational / symbolic carriers	Carrier group in which the countable units are informational or symbolic units, such as tokens, symbols, memory states, or effective parameters, whose quantity controls activation or regime of an edge.	`information_unit`	Unit identity is determined by representation-level distinctness, for example token type, symbol identity, state label, or parameter identity, within the motif window.	Only units that causally participate in the mechanism are counted, such as tokens entering a propagation loop or states reachable under the relevant constraint.

Carrier group	Definition	Unit kind	Boundary rule	Membership rule
CG_{AP} — Agent / population carriers	Carrier group in which the countable units are agents, population members, role instances, or interaction events whose quantity controls activation or regime of an edge at social or collective scales.	`agent_like_unit`	Unit identity is defined by participation identity at the relevant scale, for example agent identity, role-instance identity, or event identity within the motif window.	Only units participating in the relevant collective mechanism are counted, such as members within the same coordination regime or events within the same cascade.

Table 11: Conceptual identity and counting ontology of Carrier Groups.

Admissible levels, quantitative regimes, and threshold interpretation

The next table summarizes the domains of admissibility and the quantitative logic characteristic of each Carrier Group. It specifies the levels at which a group may be validly applied, the typical quantitative regimes associated with it, and the principal remarks needed for interpreting $N_{\min}$, N_{crit}, and N_{sat} within that ontological class.

Carrier group	Allowed levels	Typical quantitative regimes	Quantitative notes
CG_{FM} — Field / mode carriers	$L_0 - L_2$	*triad-closure*, *control-dof*, *scaling*	Typical structural thresholds arise from minimal closure motifs in mode-coupling space. N_{crit} is frequently small, often 3 in triadic closure, and is determined by admissibility constraints rather than bulk statistics. N_{sat} is usually model-dependent and may be omitted when additional modes scale amplitude without changing regime.
CG_{PB} — Physical body carriers	$L_1, L_2, L_3, L_9, L_{10}, L_{11}, L_{12}$	*pair-only*, *triad-closure*, *control-dof*, *scaling*	$N_{\min}$ is typically 2 for interaction-defined edges, but may be 1 for self-constraint edges on an already individuated body. N_{crit} is often structural or topological rather than statistical. N_{sat} may be absent when additional bodies mainly increase degrees of freedom without producing a new qualitative regime.
CG_{FMC} — Functional material carriers	$L_2 - L_5$	*pair-only*, *control-dof*, *collective-phase*, *scaling*	Two quantitative regimes frequently coexist: small-N cooperative or stoichiometric thresholds, and large-N collective or phase thresholds. N_{crit} is often context-dependent under changes in temperature, boundary, concentration, or geometry. N_{sat} becomes meaningful when additional units scale throughput or amplitude without changing regime.

Carrier group	Allowed levels	Typical quantitative regimes	Quantitative notes
CG_{TC} — Topological / configurational carriers	$L_0 - L_{13}$	*triad-closure*, *control-dof*, *collective-phase*, *scaling*	Small-N thresholds, typically in the range 3 to 7, often correspond to motif closure such as minimal cycles or feedback structures. Large-N thresholds often correspond to percolation or connectivity transitions. N_{crit} may be defined by a structural predicate threshold, for example the existence of at least one qualifying cycle or a k-core with $k \geq k_{\min}$.
CG_{IS} — Informational / symbolic carriers	$L_3 - L_8$	*pair-only*, *control-dof*, *collective-phase*, *scaling*	Thresholds may reflect minimal expressivity, minimal memory for control, or sufficient state-space diversity for stabilization. N_{crit} may represent minimal codebook size, minimal state-space coverage, or minimal parameter set for regime selection. N_{sat} may indicate diminishing structural novelty beyond a representational capacity threshold.
CG_{AP} — Agent / population carriers	$L_6 - L_{10}$	*control-dof*, *collective-phase*, *scaling*	Thresholds often correspond to stabilization of coordination regimes, norm formation, market formation, or institutional persistence. N_{crit} is commonly context-dependent and may be specified through structural predicates such as sustained participation or temporal stability rather than by a single universal number. N_{sat} may indicate participation saturation, where further growth affects scale more than causal regime.

Table 12: Admissible levels, quantitative regimes, and threshold interpretation of Carrier Groups.

Principal failure modes in the use of carrier groups

The final table summarizes the characteristic failure modes associated with each Carrier Group. These errors arise when counting is detached from the correct ontology of units, when motif participation is not properly enforced, or when one carrier group is incorrectly substituted for another. The table therefore functions as a diagnostic safeguard against category error in quantitative causal analysis.

Carrier group	Principal failure modes
CG_{FM} — Field / mode carriers	Counting energy quanta or intensity samples instead of mode instances; counting modes that do not coherently participate in the coupling motif; using field / mode carriers where object-body individuation is the correct ontology.
CG_{PB} — Physical body carriers	Counting parts or subcomponents as bodies; counting bodies not dynamically coupled into the motif; replacing body count with link count without explicitly declaring a topological carrier ontology.
CG_{FMC} — Functional material carriers	Counting bulk amount or concentration without a unit mapping; mixing different unit kinds in one edge without declaring a composite carrier rule; using functional material units to represent purely structural graph thresholds better treated as topological carriers.

Carrier group	Principal failure modes
CG_{TC} — Topological / configurational carriers	Counting all edges or cycles without enforcing motif typing constraints; confusing percolation in a causal graph with physical percolation without explicit mapping; using topological carriers where the operative mechanism is explicitly material or functional.
CG_{IS} — Informational / symbolic carriers	Counting raw data volume without defining symbolic units; mixing token count with agent count without declaring carrier ontology; treating informational carriers as purely topological and thereby erasing representation semantics.
CG_{AP} — Agent / population carriers	Counting passive exposure as participation; counting agents without specifying the mechanism, for example role, event, or member; projecting micro-level physical carriers onto agent-level edges without explicit mapping.

Table 13: Principal failure modes in the application of Carrier Groups.

4.10 Carrier types as quantitative operators of causality

In a graph of causal arisings, nodes represent units of arising and edges represent qualitatively typed causal relations. Qualitative typing alone, however, is insufficient to determine when a relation becomes structurally active. Many causal edges are conditional upon the presence of a sufficient number of concrete carriers. Carrier Types formalize this dependency by specifying what exactly must be counted in order for a causal mechanism to activate or to change its regime.

A Carrier Type is not an object, process, or property in itself. It is a quantitative operator attached to a causal edge. It specifies the ontological kind of unit whose accumulation determines the activation state of that edge. Formally, for a causal edge $e = (u \to v)$, let x denote the carrier associated with that edge. The quantity N_x denotes the number of carriers of type x participating in the relevant causal motif. The activation of the edge is governed by a function $A_e(N_x)$, while the qualitative regime of the edge is governed by a function $R_e(N_x)$.

The introduction of Carrier Types separates two dimensions of causality that would otherwise be conflated. The first dimension is qualitative: causal direction, interaction mode, dependency type, and the role relations between nodes. The second dimension is quantitative: whether the mechanism described by the edge is inactive, linearly active, threshold-transformative, or saturated. Carrier Types operate strictly within this second dimension while remaining ontologically grounded in the first.

Without Carrier Types, quantitative analysis degenerates into undifferentiated counting of nodes or edges. Such counting is structurally insufficient because different kinds of carriers obey different quantitative logics. The same numerical increase $N_x \to N_x + 1$ may correspond to closure, control, scaling, or phase transition depending on the ontological nature of x. Carrier Types preserve this distinction by binding numerical growth to the correct ontological substrate.

Carrier Types are therefore defined by three core elements. First, a unit-of-count definition determines what constitutes one carrier instance. Second, an individuation rule specifies how carriers are distinguished from one another within the chosen level of description. Third, a membership rule specifies which carriers are included in the count N_x for a given edge. Only carriers participating in the specific causal motif are admissible members of the counting set.

Given these definitions, each edge can be associated with threshold parameters such as $N_{\min}$, N_{crit}, and N_{sat}. The minimal threshold $N_{\min}$ determines when the causal mechanism is structurally defined. The critical threshold N_{crit} determines when a qualitative regime shift becomes admissible. The saturation threshold N_{sat} determines when further increases in N_x no longer alter the qualitative regime and instead only scale intensity or magnitude.

The regime function can be expressed schematically as a piecewise mapping $R_e(N_x)$, whose structure depends on the Carrier Type. In general form, if $N_x < N_{\min}$, the edge is inactive; if $N_{\min} \leq N_x < N_{\text{crit}}$, the edge operates in a linear or compositional regime; if $N_x \geq N_{\text{crit}}$, the edge may enter a threshold regime; and if $N_x \geq N_{\text{sat}}$, the regime becomes saturated. The specific interpretation of these regions depends on the ontological nature of the carrier.

Carrier Types thus function as quantitative mediators between structural possibility and realized activation. They determine not only whether a causal relation is operative, but also how the meaning of numerical growth is to be interpreted. In this sense, Carrier Types are necessary for distinguishing structural closure from statistical aggregation, control degrees of freedom from simple scaling, and collective-phase transitions from local motif activation.

By formalizing Carrier Types prior to enumerating their concrete instances, the ontology ensures that every quantitative parameter attached to an edge possesses a clear ontological referent. Numerical thresholds are therefore not arbitrary annotations but quantities grounded in the nature of the counted units. This grounding is essential for maintaining coherence between levels, preserving distinctions between emergence and composition, and enabling algorithmic evaluation of regime transitions across heterogeneous domains.

4.11 Tabulated ontology of carrier types

Because Carrier Types combine ontological identity, counting rules, and threshold logic, their presentation is divided into three complementary tables. The first table establishes the conceptual identity of each Carrier Type together with its counting unit and basic individuation logic. The second specifies the levels at which each type typically appears and the quantitative characteristics associated with its activation profile. The third groups Carrier Types by Carrier Group in order to clarify how local counting operators inherit their deeper ontological basis. Taken together, these tables provide the operational reference needed for attaching quantitative semantics to causal edges.

Conceptual identity and counting rules of carrier types

The following table presents the conceptual identity of each Carrier Type. It specifies the formal code, the general definition, the counting unit, the individuation rule, and the membership condition under which the unit belongs to the motif relevant to a causal edge.

Carrier type	**Definition**	**Counting unit**	**Individuation rule**	**Membership rule**
CT_{PB} — Physical body	Discrete material body treated as an individuated participant in an interaction configuration.	One separable body.	Geometric or operational separability with persistence across the motif window.	Counted only if dynamically coupled within the specific causal motif.

Carrier type	Definition	Counting unit	Individuation rule	Membership rule
CT_{FE} — Field excitation	Localized excitation instance within a field acting as a countable interaction participant.	One excitation.	Distinguished by mode index, localization, or coherence in representation space.	Counted only if participating in the same coupling motif.
CT_{RM} — Resonant mode	Coherently sustained oscillatory mode functioning as a causal carrier.	One coherent mode.	Defined by stable resonance parameters under admissibility constraints.	Included only if coherence is maintained within the motif.
CT_{AT} — Atom	Individual atom acting as a discrete participant in bonding or structural interaction.	One atom.	Chemical identity at the selected descriptive scale.	Counted only if participating in the bonding motif.
CT_{MO} — Molecule	Discrete molecule functioning as a causal unit in interactions or assemblies.	One molecule.	Stable molecular identity under the relevant conditions.	Counted only if participating in the relevant motif.
CT_{BD} — Bond instance	Single instantiated chemical bond contributing to structural coupling.	One bond.	Defined by bonded pair and bond type.	Counted only if contributing to the relevant structural motif.
CT_{AS} — Active site	Localized catalytic or functional region enabling selective interaction.	One functional site.	Defined by functional differentiation within a macromolecule.	Counted only if participating in catalytic throughput.
CT_{MC} — Macromolecular complex	Discrete multi-molecule assembly functioning as a unified causal unit.	One complex.	Defined by stable assembly identity.	Counted if functioning as an integrated module.
CT_{CE} — Cell	Autonomous living unit functioning as a causal participant.	One cell.	Defined by cellular boundary integrity and operational autonomy.	Counted if participating in the same physiological motif.
CT_{IE} — Interaction edge	Single instantiated causal link within a configuration.	One edge.	Defined by an ordered node pair with typed interaction.	Counted only if matching the motif constraints.
CT_{FL} — Feedback loop	Closed causal loop acting as a regulatory structural carrier.	One loop.	Defined as a cycle satisfying motif-typing constraints.	Counted if structurally closed and active.

Carrier type	Definition	Counting unit	Individuation rule	Membership rule
CT_{IT} — Information token	Discrete symbolic or informational unit participating in propagation or regulation.	One token or one state.	Defined by representation-level distinctness.	Counted if participating in the same informational motif.
CT_{AG} — Agent	Autonomous decision-capable participant in a collective configuration.	One agent.	Defined by participation identity at the selected scale.	Counted if participating in the same coordination regime.
CT_{NS} — Neural state / activity pattern	Discrete neurodynamic state or recurrent activity pattern participating in modulation, integration, or control across neural and social-neural interfaces.	One distinguishable neural state or activity pattern.	Defined by state-pattern identity at the selected resolution, that is, by a stable configuration of neural activation, synchronization, or dynamical regime that is causally distinguishable within the mechanism.	Counted if participating in the same neurodynamic motif, regulatory loop, or socially coupled neural mechanism relevant to the edge.

Table 14: Conceptual identity and counting rules of Carrier Types.

Typical levels and quantitative characteristics of carrier types

The next table summarizes the levels at which each Carrier Type typically appears together with its characteristic threshold profile. These values should not be interpreted as rigid universal constants. Rather, they function as default quantitative signatures indicating the typical scale at which a type begins to matter, reaches a structurally significant threshold, or enters saturation.

Carrier type	Typical levels	$N_{\min}$	N_{crit}	N_{sat}	Regime pattern
CT_{PB} — Physical body	$L_1, L_2, L_3, L_9, L_{10}, L_{11}, L_{12}$	2	3	—	Discrete structural thresholds: pair, closure, and control.
CT_{FE} — Field excitation	L_0, L_1, L_2	2	3	—	Closure in mode-coupling space.
CT_{RM} — Resonant mode	L_0, L_1	3	3	—	Triadic structural closure.
CT_{AT} — Atom	L_2	2	—	—	Stoichiometric thresholds.
CT_{MO} — Molecule	L_2, L_3	2	Context based	Context based	Cooperative and collective-phase behavior.
CT_{BD} — Bond instance	L_2	1	—	—	Connectivity scaling.
CT_{AS} — Active site	L_3	1	2	Context based	Cooperative activation.
CT_{MC} — Macromolecular complex	L_3, L_4	1	—	—	Module scaling.

Carrier type	**Typical levels**	$N_{\min}$	N_{crit}	N_{sat}	**Regime pattern**
CT_{CE} — Cell	L_4, L_5	1	Population-dependent	Context based	Population-level thresholds.
CT_{IE} — Interaction edge	$L_0 - L_7$	1	Structural	Structural	Connectivity thresholds.
CT_{FL} — Feedback loop	L_3, L_4, L_6, L_7	1	1	—	Closure-triggered activation.
CT_{IT} — Information token	L_3, L_6, L_7, L_8	1	Capacity-dependent	Capacity-dependent	Expressivity thresholds.
CT_{AG} — Agent	L_7, L_8, L_9	2	Coordination based	Context based	Collective stabilization.
CT_{NS} — Neural state / activity pattern	L_6, L_7	1	Regime-dependent	Capacity-dependent	State-space coverage and attractor/regime thresholds.

Table 15: Typical levels and quantitative characteristics of Carrier Types.

Carrier types within their ontological carrier groups

The final table groups Carrier Types according to the Carrier Group from which they inherit their ontological basis. This grouping is theoretically important because a Carrier Type is never a free-floating quantitative label: it is always a local specification of a broader ontological class of countable bearers. The table therefore clarifies how edge-level counting operators remain anchored in a deeper ontology of quantitative causality.

Carrier group	**Carrier types**	**Analytical significance**
CG_{FM} — Field / mode carriers	CT_{FE}, CT_{RM}	These types represent excitation-based and resonance-based counting ontologies in which threshold logic is primarily structural and closure-driven rather than statistical.
CG_{PB} — Physical body carriers	CT_{PB}	This type captures discrete body-based participation, where thresholds are typically topological and arise from N-body closure or control degrees of freedom.
CG_{FMC} — Functional material carriers	CT_{AT}, CT_{MO}, CT_{BD}, CT_{AS}, CT_{MC}, CT_{CE}	These types represent materially instantiated but functionally individuated units, allowing both small-N cooperative thresholds and large-N collective or phase-like regimes.
CG_{TC} — Topological / configurational carriers	CT_{IE}, CT_{FL}	These types formalize graph-structural carriers whose abundance or closure controls activation, especially in connectivity-, cycle-, and feedback-dependent regimes.
CG_{IS} — Informational / symbolic carriers	CT_{IT}, CT_{NS}	These types capture representational and state-based carriers whose thresholds are determined by expressivity, memory, state-space coverage, or regime selection.

Carrier group	Carrier types	Analytical significance
CG_{AP} — Agent / population carriers	CT_{AG}	This type captures participation-based collective counting, where thresholds typically correspond to coordination, stabilization, or emergence of collective regime.

Table 16: Carrier Types grouped by their ontological Carrier Groups.

4.12 Complexity levels as the hierarchical architecture of multi-scale organization

Complexity levels represent the major strata of organization through which causal arisings accumulate into progressively more complex structures. Each level defines a distinct regime of admissible interactions, characteristic components, boundary conditions, and typical transition mechanisms. Whereas interaction modes, causal directions, type roles, and level phases describe the internal mechanics of arising, levels describe the broader organizational architecture within which these mechanics operate.

Within the ontology of causal arisings, levels function as large-scale organizational domains. Each domain is characterized by its own forms of stability, dominant energetic and informational constraints, characteristic configurations, and the kinds of emergent phenomena that can arise within it. Progression across levels reflects the accumulation of structural potency: as internal regulation, closure, and complexity increase, new regimes of organization become admissible.

The classification spans a full hierarchy, extending from quantum fields and elementary interactions to molecular systems, biological organization, cognitive structures, social formations, technological ecosystems, planetary dynamics, and speculative meta-levels. Each level is defined not by scale alone but by a distinctive set of constraints and potentials determining which kinds of phenomena may arise and which causal pathways may become operative.

Every level contains its own phases of differentiation, integration, self-organization, and threshold transition. As these phases unfold, new components appear, novel interaction patterns emerge, and regulatory structures gain stability. Once these internal processes reach a critical limit, transition to a higher organizational level becomes admissible. This hierarchical logic provides continuity across the full range of physical, biological, cognitive, and social organization.

The following table provides a structured overview of the levels used in the framework. It summarizes their defining characteristics, key innovations, self-organizing principles, and the dominant aspects of causal arising within each domain. This overview serves as a conceptual scaffold for the theoretical system as a whole, establishing the multi-level background against which individual emergent phenomena are classified and analyzed.

4.13 Tabulated architecture of complexity levels

Because the ontology of complexity levels combines domain identity, emergent innovation, self-organizing logic, and internal phase articulation, its presentation is divided into two complementary tables. The first table provides a compact theoretical profile of each level as a large-scale domain of organization. The second table expands the internal phase sequence associated with each level. Taken together, these tables establish

the hierarchical scaffold within which causal arisings are situated and compared.

Theoretical profile of complexity levels

The following table presents the principal complexity levels as large-scale ontological domains. For each level it specifies the formal code, the domain name, the key innovation introduced at that level, the characteristic form of self-organization, and the dominant aspects of causal arising associated with it. This table serves as the primary reference for the hierarchical architecture of the framework.

Level	**Name**	**Key innovation**	**New self-organization**	**Key aspects**
L_0	Proto Field	Hypothesized substrates preceding known physics, admitting non-standard metrics and pre-causal regularities capable of seeding later physical law.	Organization, if present, is not constrained by standard spacetime; putative structures act as sources of randomness or hidden boundary data for lower levels.	Methodological limit; placeholder for theories beyond empirical reach; meta-constraints potentially shaping quantum vacuum structure and constants.
L_1	Quantum	Elementary fields and symmetry structures define conserved charges and interaction vertices; vacuum fluctuations and entanglement impose nonlocal constraints.	Color confinement forms hadrons, while symmetry breaking and running couplings determine effective low-energy behavior.	Field quanta and constants; renormalization logic; foundations for atomic binding and all higher organization.
L_2	Atomic	Bound electron states produce stable atoms; bonding taxonomies generate combinatorial chemical diversity.	Orbital energetics and symmetry produce periodicity and reaction pathways; supramolecular motifs prepare membranes and templating.	Electronic structure governs reactivity; phase behavior and lattice formation; substrate for materials and biochemistry.
L_3	Biochemical	Sequence-coded macromolecules enable replication and catalysis; folding yields functional micro-machines.	Autocatalytic sets and feedback-controlled metabolism stabilize production and energy transduction.	Genetic encoding, kinetic control, and compartment-ready chemistries; bridge to cellular autonomy.
L_4	Cellular	Boundary formation and compartmentalization localize chemistry; regulated transport and signaling maintain viable internal conditions.	Feedback circuits govern growth, division, and stress response; spatial intracellular organization supports stable function.	Autonomy, reproduction, and regulated information flow; platform for specialization and tissue formation.
L_5	Multicellular	Division of labor and morphogenetic programs generate tissues and organs with coordinated functions.	Mechanical and chemical gradients together with gene-regulatory networks orchestrate form, maintenance, and repair.	Morphogenesis, organogenesis, and systemic regulation; resilience through redundancy and regeneration.
L_6	Behavioral	Distributed neural circuits transform sensation into adaptive action; plasticity tunes performance to context.	Feedback across sensory–motor loops and internal models stabilizes behavior and supports abstraction.	Representation, prediction, and policy selection; conditions for subjective experience in some organisms.

Level	Name	Key innovation	New self-organization	Key aspects
L_7	Social	Symbolic communication and shared memory enable cumulative knowledge and complex coordination.	Norms, markets, and governance reduce friction and align incentives across populations.	Collective intelligence, cooperation–conflict dynamics, and cultural evolution; macro-patterns generated from local rules.
L_8	Technological	Artificial systems extend sensing, memory, and control; code becomes an operative agent of design and coordination.	Machine-to-machine protocols and data flows generate emergent capability and optimization regimes.	Automation, AI governance, and human–machine coupling; acceleration of discovery and deployment.
L_9	Planetary	Coupled atmosphere, ocean, land, and biosphere processes regulate habitability; anthropogenic forcing modifies global cycles.	Ecosystems and infrastructures co-adapt across basins and continents; cascading risks motivate planetary-scale response.	Earth-system resilience, tipping elements, and cross-border infrastructures; sustainability constraints.
L_{10}	Civilizational	Interlinked energy, materials, and information systems coordinate at continental scale; institutions manage externalities and long-range planning.	Standards, protocols, and incentive architectures align actors under uncertainty and systemic shock.	Long-horizon planning, circularity, and equity constraints; interface to orbital expansion.
L_{11}	Spacefaring infrastructure	Orbital and deep-space assets extend sensing, manufacturing, and habitation beyond Earth.	Multi-body dynamics, radiation environments, and sparse-resource economics shape robust extra-planetary architectures.	Launch-to-orbit value chains, on-orbit servicing, and in-situ resource utilization; foundations for diasporic expansion.
L_{12}	Cosmic	Large-scale structure emerges through gravity, dark matter, and dark energy; cosmological relics encode formation histories.	Resonances, star-formation feedback, active galactic nuclei, and mergers scaffold long-range order.	Cosmic-web topology, horizon limits, and astroengineering possibility space; terminal boundary conditions.
L_{13}	Upper speculative	All levels are treated as local modalities of a broader structural field in which causation and organization themselves become parameters.	Transitions operate on types of causality and admissibility rather than on carriers; pattern compatibility may span worlds.	Methodological closure of ontology; scope conditions for theory transfer and limits of reducibility.

Table 17: Theoretical profile of complexity levels.

Phase articulation within complexity levels

The following table expands the internal phase articulation of each complexity level. It shows how each level can be decomposed into a sequence of characteristic formations that prepare, integrate, stabilize, and transform the domain in question. The table should not be read as a rigid universal chronology, but as a structured phase grammar identifying the typical internal sequence through which each level becomes organized.

Level	Name	Phase 1	Phase 2	Phase 3	Phase 4
L_0	Proto Field	Pre-physical hyper-domains	Proto-metric fluctuations	Latent causal parameters	Vacuum-like substrate
L_1	Quantum	Quarks, leptons, and gauge bosons	Interactions and symmetry groups	Confinement and asymptotic freedom	Seeds of atomic structure
L_2	Atomic	Atomic orbitals and shells	Chemical bonds and molecules	Resonance and intermolecular forces	Proto-biochemical assemblies
L_3	Biochemical	Monomers and templates	Polymers and folding	Catalysis and cycles	Protocellular systems
L_4	Cellular	Membranes and compartments	Networks and cytoskeleton	Homeostasis and gene control	Multicellular readiness
L_5	Multicellular	Differentiated cell types	Tissues and extracellular matrix	Organs and physiological systems	Whole-organism integration
L_6	Behavioral	Neural ensembles and sensing	Synapses and rhythms	Learning and memory	Integrated cognition
L_7	Social	Agents and roles	Groups and norms	Institutions and culture	Large-scale coordination
L_8	Technological	Tools and computation	Networks and platforms	Autonomy and feedback algorithms	Cyber–physical ecosystems
L_9	Planetary	Biomes and geocycles	Teleconnections and transport	Global feedbacks	Planetary coordination
L_{10}	Civilizational	Infrastructure and supply webs	Integrated governance and finance	Resilience and optimization	Post-biospheric preparedness
L_{11}	Spacefaring infrastructure	Satellites and platforms	Cislunar and interplanetary logistics	Autonomous navigation and industry	Gateway to interstellar projects
L_{12}	Cosmic	Stars, galaxies, and webs	Hierarchical assembly	Gravitational geometry as record	Cosmic-scale constraints
L_{13}	Upper speculative	Meta-structural preconditions	Ontological phase grammars	Causal logic reconfiguration	Meta-distribution of realities

Table 18: Phase articulation within complexity levels.

5 Arising and causal configurations

A configuration is a structured set of nodes and causal connections forming a coherent functional arrangement. Although a configuration may change under the influence of internal interactions or external conditions, within the present framework it is treated as the fundamental basis for the arising of new levels of organization. A configuration determines how components are positioned, how their roles are related, and which causal pathways become admissible.

5.1 Configuration as a condition of arising

Arising does not occur at the level of isolated elements but at the level of their structural distribution. A configuration defines a system of connections in which the placement, function, and interaction modes of components create the conditions for transition to a new organizational state. It is not itself an object or a final product, but a functional structure determining which transformations are possible for the system.

A configuration establishes a field of potentials. It does not contain the transition in its realized form, but defines its admissibility. A new level arises when a configuration reaches the required threshold of coherence among components, links, and causal modes. This threshold is determined by the formal structure of assembly: which elements are present, how they are connected, what roles they perform, and which types of causal action are active.

A configuration comprises three principal structural dimensions. First, it includes the type roles of the nodes involved in arising, namely process, object, pattern, property, and effect. Second, it includes the causal directions and interaction modes linking these nodes, including upward, downward, horizontal, diagonal, decoupled, and meta-level causation, together with cooperation, constraint, feedback, initiation, and realization. Third, it includes the phase and level within which the configuration is embedded, together with the constraints and admissibilities characteristic of that level.

The same set of components may generate different systemic qualities depending on configuration, that is, depending on how those components are arranged, how they are linked, and which interaction modes are active. Conversely, very different substrates may realize structurally similar configurations and thereby give rise to analogous emergent effects. Configuration therefore becomes the primary unit of analysis: it marks the boundary between stable dynamics and the conditions under which transition to a higher level becomes admissible.

A configuration is not a template or a merely repetitive form. Each instance may be unique in content and history while remaining analytically tractable through its structural characteristics. Any specific structure may serve as a carrier of potentiality if it satisfies the necessary conditions for arising. Analysis of configurations thus makes it possible to identify minimal admissibility conditions, distinguish productive from non-productive combinations of elements and links, and recognize forms with a high probability of generating systemic novelty.

5.2 Formal definition of a configuration in code form

At the graph level, a configuration is a directed graph with annotations: a set of nodes V and edges E, together with functions assigning codes to nodes and edges. Each node $v \in V$ carries a type role, a level, and a phase. Each edge $e \in E$ carries interaction modes and causal directions, and may additionally be associated with a phase in which the relation is most relevant.

Node annotations. For each node $v \in V$,

$$\mathrm{lvl}(v) \in \{L_0, L_1, \ldots\}, \tag{39}$$

$$\mathrm{phase}(v) \in \{\mathrm{LP}_A, \mathrm{LP}_B, \mathrm{LP}_C, \mathrm{LP}_D\}, \tag{40}$$

$$\mathrm{role}(v) \in \{\mathrm{TR}_{PR}, \mathrm{TR}_{OBJ}, \mathrm{TR}_{PAT}, \mathrm{TR}_{PROP}, \mathrm{TR}_{EFF}\}. \tag{41}$$

Edge annotations. For each edge $e = (v \to w)$,

$$IM(e) \subseteq \{\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_I, \mathrm{IM}_R\}, \tag{42}$$

$$CD(e) \subseteq \{\mathrm{CD}_U, \mathrm{CD}_D, \mathrm{CD}_H, \mathrm{CD}_{DI}, \mathrm{CD}_{DC}, \mathrm{CD}_M\}, \tag{43}$$

$$OR(e) \in \{\textit{arising}, \textit{maintenance}, \textit{modulation}\}, \tag{44}$$

$$\mathrm{phase}(e) \in \{\mathrm{LP}_A, \mathrm{LP}_B, \mathrm{LP}_C, \mathrm{LP}_D\} \cup \{\varnothing\}. \tag{45}$$

For each emergent phenomenon described in the database, the schema assigns a level, a phase, a primary type role, and a list of parent relations. Each parent relation encodes a parent phenomenon together with its specific causal directions and interaction modes. The parent–child graph, together with these annotations, yields the configuration in the sense defined here.

For a phenomenon p with code $k.i$, where k is the level and i the local index,

$$L(p) = L_k, \tag{46}$$

$$\Phi(p) \in \{\mathrm{LP}_A, \mathrm{LP}_B, \mathrm{LP}_C, \mathrm{LP}_D\}, \tag{47}$$

$$TR(p) \in \{\mathrm{TR}_{PR}, \mathrm{TR}_{OBJ}, \mathrm{TR}_{PAT}, \mathrm{TR}_{PROP}, \mathrm{TR}_{EFF}\}, \tag{48}$$

$$\mathrm{Parents}(p) = \{r_1, \ldots, r_m\}. \tag{49}$$

For each parent relation r_j,

$$\mathrm{ParentCode}(r_j) = k_j.i_j, \tag{50}$$

$$IM(r_j) \subseteq \{\mathrm{IM}_*\}, \tag{51}$$

$$CD(r_j) \subseteq \{\mathrm{CD}_*\}, \tag{52}$$

$$OR(r_j) \in \{\textit{arising}, \textit{maintenance}, \textit{modulation}\}, \tag{53}$$

$$\mathrm{Phase}(r_j) \in \{\mathrm{LP}_*\} \cup \{\varnothing\}. \tag{54}$$

The configuration associated with a single phenomenon p may then be summarized by the union of codes over its parent relations. This union characterizes which structural resources are actually used to generate p from its parents.

Minimal configurational code. The minimal configurational code of p is defined as

$$IM(p) = \bigcup_{r \in \mathrm{Parents}(p)} IM(r), \tag{55}$$

$$CD(p) = \bigcup_{r \in \mathrm{Parents}(p)} CD(r), \tag{56}$$

$$\Gamma(p) = \big(L(p), \Phi(p), TR(p), IM(p), CD(p)\big). \tag{57}$$

Here $\Gamma(p)$ is a compressed representation of the configuration: it records which interaction modes, causal directions, and node roles are active at a given level and phase. For finer analysis, edge-level codes are used, preserving the pairing of parent and child roles.

Edge-level code. For a parent relation $r : q \to p$, define

$$\kappa(r) = \big(IM(r);\, TR(q), TR(p);\, CD(r);\, OR(r);\, L(p), \Phi(p)\big). \tag{58}$$

Edge-level codes $\kappa(r)$ make it possible to reconstruct configurational motifs, that is, typical combinations of interaction modes, causal directions, and node roles recurring across domains. The set of all $\kappa(r)$ belonging to a configuration constitutes its full configurational code.

5.3 Configurational codes as compositions of mechanisms

At an abstract level, a configurational code may be written as a composition of interaction modes, node roles, and causal directions. The earlier schematic form $M \to (T) \times (C)$ can now be made more precise in terms of the code alphabet introduced above.

Let

$$M \subseteq \{\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_I, \mathrm{IM}_R\}, \tag{59}$$

$$T \subseteq \{\mathrm{TR}_{PR}, \mathrm{TR}_{OBJ}, \mathrm{TR}_{PAT}, \mathrm{TR}_{PROP}, \mathrm{TR}_{EFF}\}, \tag{60}$$

$$C \subseteq \{\mathrm{CD}_U, \mathrm{CD}_D, \mathrm{CD}_H, \mathrm{CD}_{DI}, \mathrm{CD}_{DC}, \mathrm{CD}_M\}, \tag{61}$$

$$O \subseteq \{\textit{arising}, \textit{maintenance}, \textit{modulation}\}, \tag{62}$$

$$L \in \{L_0, L_1, \ldots\}, \qquad \Phi \in \{\mathrm{LP}_A, \mathrm{LP}_B, \mathrm{LP}_C, \mathrm{LP}_D\}. \tag{63}$$

The configurational code of a motif is then defined as

$$K = [M] \to (T) \times (C) \times (O)@(L, \Phi). \tag{64}$$

In this notation, $[M]$ denotes the set of interaction modes jointly participating in the transition; (T) denotes the set of node type roles involved; (C) denotes the set of causal directions through which influence flows; (O) denotes the set of ontological roles through which parent relations participate in the lifecycle of the child arising; L denotes the level at which the motif is realized; and Φ denotes the dominant phase. The arrow $\to$ denotes structural dependency: the combination of modes M, acting on nodes of types T, along directions C, with ontological roles O, at level L and phase Φ, defines a class of admissible transitions.

The same configuration may generate several codes of this form: global codes for the entire phenomenon, $\Gamma(p)$; edge-level codes, $\kappa(r)$, for each parent relation; and local motif codes, K, for typical subgraphs that recur across different phenomena and levels.

5.4 Example 1: DNA replication as a configurational code

DNA replication can be interpreted as an emergent configuration at the biochemical level. Without binding the example to a particular numbering scheme, the phenomenon may be placed at a molecular–biochemical level, for example L_3, and in the self-organization phase LP_C, where regulatory structures and stable cycles of transformation arise.

The configuration includes at least the following components. Object-nodes $\left(\mathrm{TR}_{OBJ}\right)$ represent DNA strands, polymerase complexes, and auxiliary proteins. Process-nodes $\left(\mathrm{TR}_{PR}\right)$ represent strand separation, template-directed synthesis, proofreading, and repair cycles. Pattern-nodes $\left(\mathrm{TR}_{PAT}\right)$ may represent recurrent topologies such as the replication fork or characteristic assembly motifs of multi-protein complexes.

The interaction modes active in this configuration include cooperation $\left(\mathrm{IM}_C\right)$, constraint $\left(\mathrm{IM}_K\right)$, feedback $\left(\mathrm{IM}_F\right)$, and initiation $\left(\mathrm{IM}_I\right)$. Cooperation integrates nucleotides and enzymes into stable assemblies. Constraint is imposed by base complementarity and geometric compatibility, which narrow the space of admissible molecular matching. Feedback appears in proofreading loops and in regulation of replication speed. Initiation is realized in the local triggers that start replication at specific origins once structural conditions are satisfied.

The dominant causal directions are upward and diagonal. Upward causation $\left(\mathrm{CD}_U\right)$ transforms local physicochemical interactions into meso-scale structures: base pairing and local binding events assemble

into stable replication complexes and reliable copying behavior. Diagonal causation $\big(\mathrm{CD}_{DI}\big)$ links these local events to cell-scale outcomes, since successful replication influences cell division, mutation rates, and evolutionary trajectories.

At the level of a global motif, replication may therefore be coded as

$$M_{\mathrm{DNA}} = \{\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_I\}, \tag{65}$$

$$T_{\mathrm{DNA}} = \{\mathrm{TR}_{OBJ}, \mathrm{TR}_{PR}, \mathrm{TR}_{PAT}\}, \tag{66}$$

$$C_{\mathrm{DNA}} = \{\mathrm{CD}_U, \mathrm{CD}_{DI}\}, \tag{67}$$

$$K_{\mathrm{DNA}} = [M_{\mathrm{DNA}}] \to (T_{\mathrm{DNA}}) \times (C_{\mathrm{DNA}})@(L_3, \mathrm{LP}_C). \tag{68}$$

In words, at level L_3 and phase LP_C, a combination of cooperation, constraint, feedback, and initiation acts between objects, processes, and patterns along upward and diagonal causal directions. This combination defines a class of configurations in which templated copying with high fidelity becomes admissible. Different biochemical implementations, for example different organisms, polymerases, or replication origins, instantiate the same structural code in different material forms.

Edge-level codes can be specified more precisely. For example, the relation from the spatial form of the DNA double helix (object-node, TR_{OBJ}) to the replication process (process-node, TR_{PR}) may be coded as

$$\kappa_1 = \big(\{\mathrm{IM}_K, \mathrm{IM}_I\};\ \mathrm{TR}_{OBJ}, \mathrm{TR}_{PR};\ \{\mathrm{CD}_U\};\ L_3, \mathrm{LP}_B\big). \tag{69}$$

This indicates that constraint and initiation jointly act from an object-node to a process-node via upward causation at the molecular level in an integration-like phase LP_B: the structural geometry of DNA both restricts and locally triggers replication processes. A different edge may encode feedback from the progression of replication to the regulation of origin firing:

$$\kappa_2 = \big(\{\mathrm{IM}_F\};\ \mathrm{TR}_{PR}, \mathrm{TR}_{PAT};\ \{\mathrm{CD}_{DI}\};\ L_3, \mathrm{LP}_C\big). \tag{70}$$

Here feedback acts from a process-node to a pattern-node via diagonal causation, stabilizing or modulating replication patterns in response to global cellular conditions. Together, κ_1, κ_2, and analogous edges form the detailed configurational code of DNA replication.

5.5 Example 2: Virality of digital content as a configurational code

Virality of digital content may be described at a coupled technological–social level, for instance at L_9 for technological systems or across an L_8–L_9 interface for coupled social–technological configurations, in a late self-organization or threshold phase $\big(\mathrm{LP}_C\text{–}\mathrm{LP}_D\big)$. The phenomenon emerges from the interaction between user behavior and platform algorithms.

The dominant node roles in this case are patterns and properties. Pattern-nodes $\big(\mathrm{TR}_{PAT}\big)$ represent recurrent interaction schemes such as recommendation loops, sharing cascades, and propagation motifs in the network. Property-nodes $\big(\mathrm{TR}_{PROP}\big)$ represent systemic attributes such as effective reach or virality index, which characterize the behavior of the platform–user ensemble as a whole.

The relevant interaction modes include feedback $\big(\mathrm{IM}_F\big)$, cooperation $\big(\mathrm{IM}_C\big)$, and realization $\big(\mathrm{IM}_R\big)$. Feedback is expressed in the circular relation between user actions and algorithmic ranking. Cooperation aggregates the actions of many users into amplified visibility for specific content. Realization corresponds to stabilization

of the viral effect: the joint configuration of platform parameters and user responses yields an observable macro-outcome in which content dominates attention over a limited period.

The dominant causal directions are horizontal and diagonal. Horizontal causation $\left(\mathrm{CD}_H\right)$ captures user–user interactions at the same level, including direct sharing, commenting, and mutual exposure. Diagonal causation $\left(\mathrm{CD}_{DI}\right)$ links micro-level actions to macro-level algorithmic behavior and back: local clicks update global models, which reshape the visibility landscape that in turn affects subsequent local action.

A global motif code for virality may therefore be written as

$$M_V = \{\mathrm{IM}_F, \mathrm{IM}_C, \mathrm{IM}_R\}, \tag{71}$$

$$T_V = \{\mathrm{TR}_{PAT}, \mathrm{TR}_{PROP}\}, \tag{72}$$

$$C_V = \{\mathrm{CD}_H, \mathrm{CD}_{DI}\}, \tag{73}$$

$$K_V = [M_V] \to (T_V) \times (C_V)@(L_9, \mathrm{LP}_C\text{–}\mathrm{LP}_D). \tag{74}$$

This indicates that at technological level L_9, in late self-organization and threshold phases, a combination of feedback, cooperation, and realization acts between patterns and properties along horizontal and diagonal causal directions, thereby defining a class of configurations in which viral dissemination becomes admissible. Specific platforms and content types instantiate this code through concrete architectures of feeds, ranking models, and user interaction affordances.

5.6 Example 3: Simple protein-based causal map in code form

Consider the earlier minimal map describing a transition from a physicochemical process to a functional effect: protein folding (node A), stable structural pattern (node B), catalytic activity as a property (node C), and reaction-rate acceleration as an effect (node D). The same configuration can be expressed entirely in code form.

At the biochemical level L_3, across phases LP_B–LP_C, assign the following roles: node A carries TR_{PR} (process), node B carries TR_{PAT} (pattern), node C carries TR_{PROP} (property), and node D carries TR_{EFF} (effect). The edge from A to B uses cooperation and constraint, IM_C and IM_K, together with upward causation CD_U: local interactions among amino acids generate a stable folded pattern. The edge from B to C uses constraint and realization, IM_K and IM_R, together with downward causation CD_D: the spatial form restricts admissible modes of vibration and electron distribution, thereby enabling catalytic property. The edge from C to D uses realization together with diagonal causation, IM_R and CD_{DI}: the localized property becomes an effect at the level of the reaction medium.

The three edge-level codes may therefore be written as

$$\kappa_{A\to B} = \big(\{\mathrm{IM}_C, \mathrm{IM}_K\};\ \mathrm{TR}_{PR}, \mathrm{TR}_{PAT};\ \{\mathrm{CD}_U\};\ L_3, \mathrm{LP}_B\big), \tag{75}$$

$$\kappa_{B\to C} = \big(\{\mathrm{IM}_K, \mathrm{IM}_R\};\ \mathrm{TR}_{PAT}, \mathrm{TR}_{PROP};\ \{\mathrm{CD}_D\};\ L_3, \mathrm{LP}_C\big), \tag{76}$$

$$\kappa_{C\to D} = \big(\{\mathrm{IM}_R\};\ \mathrm{TR}_{PROP}, \mathrm{TR}_{EFF};\ \{\mathrm{CD}_{DI}\};\ L_3, \mathrm{LP}_D\big). \tag{77}$$

The union code for the configuration is then

$$IM(\text{protein}) = \{\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_R\}, \tag{78}$$

$$CD(\text{protein}) = \{\mathrm{CD}_U, \mathrm{CD}_D, \mathrm{CD}_{DI}\}, \tag{79}$$

$$\Gamma(\text{protein}) = \begin{pmatrix} L_3, \{\mathrm{LP}_B, \mathrm{LP}_C, \mathrm{LP}_D\}, \\ \{\mathrm{TR}_{PR}, \mathrm{TR}_{PAT}, \mathrm{TR}_{PROP}, \mathrm{TR}_{EFF}\}, \\ IM(\text{protein}),\ CD(\text{protein}) \end{pmatrix}. \tag{80}$$

This compact code preserves the essential structure of the example: a sequence of upward, downward, and diagonal causal steps, mediated by cooperation, constraint, and realization, and linking process, pattern, property, and effect at the biochemical level.

5.7 How to compute configurational codes from data

The emergent-phenomena database supports direct computation of configurational codes. For each phenomenon p, the schema provides the necessary fields: *Code* ($k.i$), *Level* (integer k), *Phase* (0–3), *TypeRole* (0–4), and a list of parent relations with *InteractionModes* and *CausalDirections* identifiers. The computation of $\Gamma(p)$ and $\kappa(r)$ then proceeds through a finite sequence of mechanical steps.

1. Read the basic attributes of p: *Code*, *Level*, *Phase*, and *TypeRole*. Map *Level* $= k$ to L_k, *Phase* $\in \{0, 1, 2, 3\}$ to $\{\mathrm{LP}_A, \mathrm{LP}_B, \mathrm{LP}_C, \mathrm{LP}_D\}$, and *TypeRole* to one of TR_{PR}, TR_{OBJ}, TR_{PAT}, TR_{PROP}, or TR_{EFF}.
2. For each parent relation $r \in \mathrm{Parents}(p)$, map *InteractionModes* identifiers (0–4) to $\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_I, \mathrm{IM}_R$, and *CausalDirections* identifiers (0–5) to $\mathrm{CD}_U, \mathrm{CD}_D, \mathrm{CD}_H, \mathrm{CD}_{DI}, \mathrm{CD}_{DC}, \mathrm{CD}_M$. Record the parent role $TR(q)$ from the parent phenomenon q, if required.
3. Construct edge-level codes of the form
$$\kappa(r) = \big(IM(r);\ TR(q), TR(p);\ CD(r);\ L(p), \mathrm{Phase}(r)\big), \tag{81}$$
where $\mathrm{Phase}(r)$ is either explicitly specified in the relation or inherited from p.
4. Compute the union sets $IM(p)$ and $CD(p)$ by taking the union of $IM(r)$ and $CD(r)$ across all parent relations r. If a broader $\Gamma(p)$ is required, collect all type roles encountered in p and in its immediate parent set.
5. Form the global code
$$\Gamma(p) = \big(L(p), \Phi(p), TR(p), IM(p), CD(p)\big). \tag{82}$$
Optionally, attach the full set $\{\kappa(r)\}$ for edge-level configurational detail.

For a set of phenomena S within the same level, the same procedure yields level-wide statistics: the relative frequency of each interaction mode, the dominant causal directions, and the distribution of type roles. This makes it possible to compute quantitative indicators of structural potential and to compare different regions of the ontology.

As a simple example, for a configuration $\mathrm{Conf} = (V, E)$, define

$$w_{\mathrm{IM}_K}(\mathrm{Conf}) = \big|\{\, e \in E \mid \mathrm{IM}_K \in IM(e) \,\}\big|, \tag{83}$$

$$\rho_K(\mathrm{Conf}) = \frac{w_{\mathrm{IM}_K}(\mathrm{Conf})}{|E|}. \tag{84}$$

Here $\rho_K(\mathrm{Conf})$ measures the fraction of edges in the configuration involving constraint as an interaction mode. Analogous metrics may be defined for cooperation, feedback, realization, initiation, or for particular causal directions.

By computing such metrics across levels and codes, it becomes possible to identify configurations with high densities of specific mechanisms, for example regions dominated by feedback and diagonal causation, and to correlate these with phases of emergent transition such as LP_C and LP_D.

5.8 Configuration $\neq$ result

A configuration describes a structure of admissibility, not realization. It defines the conditions under which a transition is possible, but it does not determine whether the transition will in fact occur. Arising is registered only through actual activation in the dynamics of the system.

A fully specified code $\Gamma(p)$ indicates that, at a given level and phase, nodes with certain type roles are connected through particular interaction modes and causal directions. This does not guarantee that the corresponding emergent effect will occur in a concrete system. The same structural code may remain latent, may be only partially active, or may be realized only under additional contextual conditions, such as triggering events, threshold crossings, or external inputs.

A clear distinction between potentiality and realization is therefore essential. A configuration may satisfy all structural conditions for arising and yet remain inactive. Arising requires not only structural coherence, as encoded in Γ and κ, but also activation through specific dynamic or contextual parameters: sufficient density of interactions, adequate energy flow, crossing of fluctuation thresholds, or the presence of initiation modes IM_I at the relevant points.

Analytically, a configuration functions as an ontological landscape. The codes L, LP, TR, IM, and CD define the boundaries of what is possible, determine the directions of causal linkage, and specify which forms of coordination are admissible. Transitions within this landscape depend on dynamic parameters interacting with it, including flows, delays, noise, and control signals. Without a strict distinction between code-level admissibility and dynamical realization, analysis degenerates into retrospective fitting of configurations to any observed novelty.

Emergence in this framework does not suspend regularity. It expresses the realization of structural potency within a configuration whose code can be made explicit. Configuration functions as a boundary structure that permits transitions, whereas the actual occurrence of an emergent event corresponds to the passage of a system trajectory through a region of code-space in which all necessary structural and dynamical conditions are simultaneously satisfied.

5.9 Role of configurational codes in the overall theory

The preceding sections introduced the conceptual architecture of arising: ontological node types, causal directions, interaction modes, phases, levels, and the associated causal matrix. The present chapter refines that architecture by introducing configurational codes, that is, symbolic representations describing how these elements are assembled into concrete patterns of admissibility.

Configurational codes perform four principal functions. First, they provide *compression*: complex processes can be represented as combinations of a limited set of code modules, thereby rendering structural logic explicit. Second, they provide *comparison*: heterogeneous phenomena drawn from physics, biology, cognition, and social organization can be compared according to code structure, revealing families of structurally equivalent transitions. Third, they provide *recognition*: new or only partially understood processes can be analyzed through their codes, allowing localization of tension nodes, unstable motifs, and latent structural potentials. Fourth, they provide *forecasting*: given a current code, the classes of admissible transitions and the structural

modifications required for activation of new organizational levels can be explored in a systematic way.

In this form, configuration becomes an operational object. It can be stored in databases, processed algorithmically, matched against known motifs, and used as a basis for simulation, inference, and hypothesis generation. The next step of the theory is therefore a meta-model of arising describing how configurations with different codes unfold, interact, and generate cascades of formation across time, levels, and systemic embeddings.

6 Principle of structural potency and ontology of possibility

> Structural potency denotes a state in which a system does not yet realize a new quality but already contains within its configuration the conditions that make such an arising admissible. Potency refers to a structured allowance embedded in the distribution of components, connections, and causal mechanisms. This allowance expresses which transformations are possible even before they manifest.

6.1 The problem of possibility as a fundamental lacuna in science

Scientific practice has historically centered on the actual: on what can be observed, measured, reproduced, and verified. Models are typically built around realized states, completed processes, and empirically established relations. Even dynamic and probabilistic descriptions tend to focus either on trajectories that have already unfolded or on distributions inferred from realized data.

Within such an orientation, possibility rarely becomes an autonomous analytical category. States and transitions not yet realized are usually treated either as speculative constructions or as auxiliary hypotheses lacking full ontological status. As a result, the structural conditions that make arising admissible prior to realization remain outside the scope of formal analysis.

This gap is especially pronounced in the study of emergence. The arising of new levels of organization is not reducible to the sum of components and cannot be derived from element-level parameters through straightforward extrapolation. It occurs within what may be called a *configurational field*: a structured space of nodes, links, constraints, and interaction modes in which some transitions are allowed, others prohibited, and many remain dynamically improbable yet structurally admissible.

Within the present framework, possibility is treated as a structural characteristic of this configurational field. For a given configuration cfg, the set of admissible transitions is determined by the architecture of connections, causal directions, and interaction modes actually present. Possibility thereby becomes a property of system topology rather than a function of epistemic uncertainty alone.

Let C be the class of all configurations, understood as typed causal graphs. For each configuration $cfg \in C$, define a set of structurally admissible transitions

$$P(cfg) \subseteq T, \tag{85}$$

where T is the space of conceivable transitions, including changes of level, phase, or code. Formally,

$$P(cfg) := \{\, t \in T \mid A(cfg, t) = 1 \,\}, \tag{86}$$

where $A(cfg, t)$ is a predicate evaluating the structural requirements of transition t against the typed resources present in cfg. A transition $t \in T$ belongs to $P(cfg)$ if and only if the current configuration provides all necessary structural resources—nodes, edges, interaction modes, causal directions, and level conditions—for t to be realizable under some admissible dynamics.

Law of structural admissibility. A transition can be realized only if it belongs to the admissible transition set generated by the current configuration:

$$\text{Realizable}(t, cfg) \Rightarrow t \in P(cfg). \tag{87}$$

This law states that realization is subordinate to structural admissibility. A transition may occur only if it is contained in the set of structurally admissible transitions determined by the configuration.

Strong exclusion law. If a transition is not contained in the admissible transition set of a configuration, then it is structurally impossible for that configuration:

$$t \notin P(cfg) \Rightarrow \neg\text{Realizable}(t, cfg). \tag{88}$$

This law provides the negative form of admissibility. Structural impossibility is defined as exclusion from the admissible transition set associated with the configuration.

Neglecting this layer produces an explanatory void: novel organizational qualities appear as exceptions disconnected from prior structure. Without explicit sets $P(cfg)$, analysis is reduced to retrospective fitting of explanations after the fact. A systematic ontology of possibility therefore requires that $P(cfg)$ be treated as a first-class object: something that can be computed, compared, and modified.

Introducing possibility as a structural category makes it possible to treat several tasks in a unified way:

- identify allowances in complex systems by computing $P(cfg)$;
- document configurations that contain structural potential for arising, that is, configurations with non-empty and non-trivial $P(cfg)$;
- describe the boundaries within which transitions are possible by distinguishing $P(cfg)$ from its complement;
- model emergent processes at the level of preliminary structural conditions rather than only at the level of realized effects.

6.2 What is structural potency

Structural potency is the configurational state of a system in which the conditions permitting the arising of a new level of organization are already embedded in its internal architecture. It is a concrete structured allowance generated by the arrangement of nodes, links, constraints, and causal directions. In code terms, potency corresponds to a configuration cfg for which the set $P(cfg)$ is non-empty and contains transitions that lift the system into a higher organizational regime.

Given a configuration cfg, define its structural potency set by

$$\Pi(cfg) = \{\, t \in P(cfg) \mid \Delta L(t) > 0 \,\}, \tag{89}$$

where

$$\Delta L(t) := L_{\text{tgt}}(t) - L_{\text{src}}(t). \tag{90}$$

A transition t raises the system's organizational level if its target code introduces either

1. a level L_k not present in the current dominant level set, or

2. a phase upgrade, for example $\mathrm{LP}_C \to \mathrm{LP}_D$, associated with threshold transition.

Structural potency is present if and only if

$$\Pi(cfg) \neq \varnothing. \tag{91}$$

Law of potency-restricted emergence. A higher-level arising is possible only through the structural potency set of the configuration:

$$\mathrm{Emergence}(t, cfg) \Rightarrow t \in \Pi(cfg) \subseteq P(cfg). \tag{92}$$

This law states that emergence is a restricted case of admissibility. Every emergent transition must belong to the structurally potent subset of admissible transitions.

Unlike abstract potentiality, structural potency is anchored in the actual topology of the system. It can be localized, analyzed, and described using formal criteria derived from node and edge codes. Possibility here takes the form of predicative compatibility: the current configuration satisfies a set of structural predicates making one or more transitions in $\Pi(cfg)$ admissible, even if none of them has yet been realized.

Three characteristics of structural potency

Structural potency is characterized by three principal features: localizability, multiplicity of admissible transitions, and irreversibility after activation. Each of these properties can be expressed directly in configurational terms.

Localizability. Structural potency is not diffusely distributed throughout the system. It is concentrated in specific subconfigurations, typically around tension nodes and multi-edge motifs where interaction modes, causal directions, and level couplings align. Formally, potency is carried by subgraphs whose local admissible-transition sets are non-empty.

For a configuration

$$cfg = (V, E, \ldots), \tag{93}$$

let

$$G(cfg) \tag{94}$$

denote the set of its connected subgraphs g. Define local potency by

$$\Pi_{\mathrm{loc}}(g) = P(g) \cap \Pi(cfg). \tag{95}$$

A subgraph g is a carrier of potency if

$$\Pi_{\mathrm{loc}}(g) \neq \varnothing. \tag{96}$$

Localizability means that there exists a proper subgraph $g \subset cfg$ such that

$$\Pi_{\mathrm{loc}}(g) \neq \varnothing. \tag{97}$$

Law of localized potency. Structural potency is concentrated in specific subconfigurations rather than uniformly distributed across the whole configuration:

$$\Pi(cfg) \neq \varnothing \Rightarrow \exists g \subseteq cfg : \Pi_{\mathrm{loc}}(g) \neq \varnothing. \tag{98}$$

This law states that if a configuration possesses structural potency, then there exists at least one subgraph that locally carries this potency.

This formulation makes it possible to identify the nodes and edges contributing to potency, namely those belonging to subgraphs with non-empty $\Pi_{\mathrm{loc}}(g)$. Such subgraphs may be interpreted as local pockets of admissible transition within a larger configuration.

Multiplicity of admissible transitions. Structural potency does not encode a single predetermined outcome. It defines a set of transitions in $\Pi(cfg)$ compatible with the same configuration. These transitions may differ in target level L_k, phase Φ, or detailed code, while sharing a common structural basis.

Multiplicity can be expressed by the condition

$$|\Pi(cfg)| > 1. \tag{99}$$

If $\Pi(cfg)$ contains more than one element, then the same configuration admits several distinct organizational trajectories. Structural constraints restrict $\Pi(cfg) \subseteq T$, but do not collapse it to a unique t.

Law of multiple admissibility. A single configuration may structurally admit more than one transition:

$$|\Pi(cfg)| > 1. \tag{100}$$

This law states that the same configuration can contain more than one structurally admissible emergent trajectory. Admissibility does not imply uniqueness of outcome.

In this sense, development is not arbitrary; it is a selection within $\Pi(cfg)$. The configuration filters out impermissible trajectories and retains only those transitions whose codes are structurally compatible with the current arrangement of nodes and edges.

Irreversibility after activation. Once a transition $t \in \Pi(cfg)$ is realized, the configuration of the system changes in such a way that return to the initial potency state without loss of integrity becomes impossible. This condition can be expressed by comparing the potency sets before and after realization.

Let cfg_0 be the initial configuration and let

$$t \in \Pi(cfg_0) \tag{101}$$

be a realized transition. Let cfg_1 denote the configuration after realization of t. Irreversibility may then be expressed as

$$cfg_1 \ncong cfg_0 \quad \wedge \quad \Pi(cfg_1) \neq \Pi(cfg_0), \tag{102}$$

where $cfg_1 \ncong cfg_0$ denotes non-isomorphism under the relevant code mapping of levels, phases, node roles, and edge modes, and $\Pi(cfg_1)$ is computed relative to the transformed configuration cfg_1. Frequently there is also a strict loss of return potency:

$$t^{-1} \notin \Pi(cfg_1), \tag{103}$$

meaning that the exact inverse transition is no longer structurally admissible.

Law of configurational irreversibility. Once a structurally potent transition is realized, the configuration changes and the potency landscape is no longer identical to the previous one:

$$cfg_1 \ncong cfg_0 \ \wedge \ \Pi(cfg_1) \neq \Pi(cfg_0). \tag{104}$$

This law states that realization transforms the configuration itself. After activation, the system no longer possesses exactly the same structural identity or the same set of potent transitions.

The new configuration defines a different landscape of admissible transitions: the previous potency is exhausted, and a new set $\Pi(cfg_1)$ becomes relevant, embedded in an altered architecture of levels, roles, and causal directions.

In summary, structural potency may be defined in the following way. A configuration cfg possesses structural potency for emergence if and only if

$$\Pi(cfg) \neq \varnothing, \tag{105}$$

local carriers of potency exist, that is,

$$\exists g \subset cfg : \Pi_{\text{loc}}(g) \neq \varnothing, \tag{106}$$

and every realized transition $t \in \Pi(cfg_0)$ leads to a transformed configuration cfg_1 such that

$$\Pi(cfg_1) \neq \Pi(cfg_0). \tag{107}$$

Theorem of minimal structural potency. A configuration possesses structural potency only if admissible emergence exists, local carriers of potency exist, and realization transforms the potency landscape:

$$\Pi(cfg) \neq \varnothing \ \wedge \ \exists g \subseteq cfg : \Pi_{\text{loc}}(g) \neq \varnothing \ \wedge \ \forall t \in \Pi(cfg_0), \ \text{Realized}(t, cfg_0) \Rightarrow \Pi(cfg_1) \neq \Pi(cfg_0). \tag{108}$$

This theorem summarizes the minimal structural conditions under which potency is not merely an abstract possibility but a genuine configurational property.

6.3 Emergence as the realization of structural potency

Emergence is interpreted here as the realization of an internal allowance encoded in $\Pi(cfg)$. It occurs when the dynamics of the system follow a trajectory that implements one of the structurally admissible transitions, thereby transforming potency into realized structure. Emergence is therefore neither a random fluctuation nor an externally imposed effect; it is a configuration-dependent transition.

Given an initial configuration cfg_0 and a structurally admissible transition $t \in \Pi(cfg_0)$, the passage

$$cfg_0 \to cfg_1 \tag{109}$$

with t realized is called an *emergent transition* if both of the following conditions hold:

1. the code of cfg_1 contains a new level or phase, for example entry into LP_D at some L_k;
2. there exists no sequence of elementary rearrangements internal to the state space of cfg_0 that reproduces cfg_1 without crossing the structural boundary defined by t.

Theorem of realized emergence. A transition counts as realized emergence only if it belongs to the structural potency set and produces a new organizational regime:

$$t \in \Pi(cfg_0) \ \wedge \ (cfg_0 \to cfg_1) \Rightarrow \Delta L(t) > 0. \tag{110}$$

This theorem states that emergence is not identical with any realized change. It must be a realized transition that produces an increase in organizational level.

At a more concrete level, activation of structural potency requires a coordinated set of conditions. These conditions can be expressed as constraints on both the configuration and its dynamics.

Conditions for activation of structural potency

1. Interaction density. The configuration must possess sufficient density of connections to support global propagation of local change.

Let $|E|$ be the number of directed edges and $|V|$ the number of nodes. Define the average in-degree and out-degree by

$$\bar{d}_{\text{in}}(cfg) = \frac{|E|}{|V|}, \qquad \bar{d}_{\text{out}}(cfg) = \frac{|E|}{|V|}. \tag{111}$$

If one instead uses the undirected projection of the directed graph, define the total average degree by

$$\bar{d}_{\text{tot}}(cfg) = \frac{2|E|}{|V|}. \tag{112}$$

The interaction-density requirement is then

$$\bar{d}_{\text{tot}}(cfg) \geq d_{\min}, \tag{113}$$

or equivalently

$$\bar{d}_{\text{in}}(cfg) \geq \frac{d_{\min}}{2}, \qquad \bar{d}_{\text{out}}(cfg) \geq \frac{d_{\min}}{2}, \tag{114}$$

for some threshold $d_{\min}$ depending on level L_k and phase Φ. Above this threshold, local variations can percolate through the graph rather than remaining isolated.

2. Systemic sensitivity. The system must amplify relevant fluctuations rather than merely damping them. At code level, this condition is captured by the presence and arrangement of feedback modes and diagonal causal directions.

Let

$$E_F = \{\, e \in E \mid \text{IM}_F \in IM(e) \,\}, \qquad E_G = \{\, e \in E \mid \text{CD}_{DI} \in CD(e) \,\}. \tag{115}$$

Define

$$\rho_F(cfg) = \frac{|E_F|}{|E|}, \qquad \rho_G(cfg) = \frac{|E_G|}{|E|}. \tag{116}$$

A configuration has high systemic sensitivity if

$$\rho_F(cfg) \geq \rho_{F,\min} \qquad \text{and} \qquad \rho_G(cfg) \geq \rho_{G,\min}, \tag{117}$$

indicating a sufficient density of feedback edges and diagonal, that is, cross-level, couplings.

3. Presence of constraints. Constraints must shape admissible trajectories without eliminating potency. At code level, the constraint mode IM_K must be present, but not dominant to the point of over-constraining the graph.

Let

$$E_K = \{\, e \in E \mid \text{IM}_K \in IM(e) \,\}. \tag{118}$$

Define

$$\rho_K(cfg) = \frac{|E_K|}{|E|}. \tag{119}$$

The configuration must then satisfy

$$\rho_{K,\min} \leq \rho_K(cfg) \leq \rho_{K,\max}, \tag{120}$$

where the lower bound ensures sufficient shaping of trajectories, and the upper bound prevents rigidification that would block emergence.

4. Cooperative assembly. There must exist motifs in which cooperative modes connect several nodes into a coherent unit. Such motifs often take the form of small subgraphs with high clustering coefficient.

Let Tri_C denote the set of triangles, or 3-cycles, in cfg such that every edge e in the triangle satisfies

$$\mathrm{IM}_C \in IM(e). \tag{121}$$

Define cooperative clustering by

$$\kappa_C(cfg) = \frac{|Tri_C|}{|Tri_{\text{total}}|}, \tag{122}$$

where Tri_{total} is the set of all triangles in cfg. A non-zero $\kappa_C(cfg)$ indicates the presence of cooperative assemblies. Above some threshold $\kappa_{C,\min}$, these assemblies may function as seeds of new levels, for example modules, cores, or protocellular clusters.

Collectively, these four conditions define a region in configuration space within which activation of $\Pi(cfg)$ becomes dynamically plausible. Emergence is then interpreted as a trajectory that moves cfg into this region and realizes one of the transitions contained in $\Pi(cfg)$.

Theorem of threshold activation. Structural admissibility alone is insufficient for realization; activation requires that the configuration enter a threshold region in which realization becomes dynamically possible:

$$t \in \Pi(cfg) \ \wedge \ \sigma(cfg, t) \geq \theta_t \Rightarrow \text{Activatable}(t, cfg). \tag{123}$$

This theorem states that a structurally potent transition becomes activatable only when a threshold condition is satisfied. Potency therefore precedes activation but does not guarantee it.

6.4 How is emergence different from a random fluctuation?

Fluctuation and arising belong to different dynamical regimes. A fluctuation is a deviation of state variables within a fixed configurational code, whereas emergence is a change of the code itself. This distinction can be expressed in terms of how a trajectory in state space interacts with the underlying configuration.

Let X be the state space over a fixed configuration cfg. A fluctuation is a trajectory

$$\gamma_{\text{fluct}} : [0, T] \to X \tag{124}$$

such that the underlying configuration code $\Gamma(cfg)$ remains constant. An emergent transition is a trajectory

$$\gamma_{\text{em}} : [0, T] \to X \times C \tag{125}$$

such that the projection to configuration space changes:

$$cfg(t) = \pi_C(\gamma_{\text{em}}(t)), \qquad \Gamma\big(cfg(0)\big) \neq \Gamma\big(cfg(T)\big). \tag{126}$$

Random fluctuations preserve the code $\Gamma(cfg)$: node roles, level assignments, interaction modes, and causal directions remain unchanged, even if individual state values vary strongly. After the fluctuation, the system returns to the same region of admissible trajectories defined by $\Pi(cfg)$.

Emergent transitions, by contrast, alter the code itself. New edges with modes IM_C, IM_F, or IM_R may appear; causal directions such as CD_U, CD_D, or CD_{DI} may be reorganized; node roles may be reclassified, for

example a pattern-node becoming an effect-node at a higher level; and new levels L_k may enter the dominant level set. The state space is thereby expanded or structurally redefined.

Two consequences follow:

- **Element-level parameters are insufficient.** Knowledge of local parameters without the code $\Gamma(cfg)$ does not determine $\Pi(cfg)$. Emergence depends on the configuration of connections and modes, not only on the intrinsic properties of elements.
- **Superficial reproducibility is insufficient.** Reproducing local conditions, for example temperature or local density, without reconstructing the full configuration of nodes, edges, modes, and directions does not guarantee repeatability of the emergent transition.

A fluctuation is motion within a fixed field; emergence is transformation of the field itself, expressible as a change in $\Gamma(cfg)$ and $\Pi(cfg)$. Without this distinction, analysis collapses into narrative description rather than structural explanation.

Law of code transformation. Emergence differs from fluctuation because emergence changes the configurational code, whereas fluctuation preserves it:

$$\Gamma(cfg_1) \neq \Gamma(cfg_0) \Rightarrow \text{Emergence}, \qquad \Gamma(cfg_1) = \Gamma(cfg_0) \Rightarrow \text{Fluctuation}. \tag{127}$$

This law states that the distinction between emergence and fluctuation is determined by whether the structural code of the configuration changes.

6.5 Potency and multilevel causality

Structural potency is distributed across multiple levels of organization. Micro-level interactions, meso-scale structures, and macro-level contexts jointly determine the set $\Pi(cfg)$. In code terms, potent configurations are those for which the level set $L_\star$ contains at least two levels and whose causal directions include diagonal or downward components linking those levels into a coherent architecture.

Let

$$\Gamma(cfg) = (M_\star, T_\star, C_\star, L_\star, \Phi_\star). \tag{128}$$

Multilevel potency requires

$$|L_\star| \geq 2 \tag{129}$$

and

$$\mathrm{CD}_{DI} \in C_\star, \tag{130}$$

that is, the presence of diagonal causality, with additional contributions where

$$\mathrm{CD}_D \in C_\star \qquad \text{or} \qquad \mathrm{CD}_{DC} \in C_\star. \tag{131}$$

Law of downward admissibility control. Once a higher-order regime is realized, it reshapes the admissible state space of lower-level components:

$$\text{Realized}(L_{k+1}) \Rightarrow \exists x \in L_k : \dot{x} = f(x; L_{k+1}). \tag{132}$$

This law states that realized higher-level organization does not merely coexist with lower-level dynamics. It constrains and modulates them by changing the admissible micro-dynamical regime as a function of the higher-level state.

Thermal and energetic conditions constrain which level codes L_k may appear in $L_\star$. High temperatures restrict configurations to levels at which only simple bonds and fields remain stable; low temperatures may reduce reaction rates and suppress dynamical assembly. For potency to exist, for example, at biochemical level L_4 and cellular level L_5, environmental parameters must preserve atomic and molecular stability while still allowing sufficient reaction rates and free-energy throughput.

Let

$$L_{\text{set}}(T) \subseteq \{L_0, \ldots, L_{13}\} \tag{133}$$

denote the set of levels physically admissible at temperature T. Any configuration $cfg(T)$ must then satisfy

$$L_\star \subseteq L_{\text{set}}(T). \tag{134}$$

For biochemical–cellular potency one requires

$$\{L_4, L_5\} \subseteq L_{\text{set}}(T). \tag{135}$$

Energy flow imposes an additional constraint. Even if a level L_k is thermodynamically admissible, insufficient free-energy flux may prevent the maintenance of structures characteristic of that level. Let J denote free-energy flux density through the system, and let $J_{\min(L_k)}$ denote the minimal flux required to sustain structures characteristic of L_k.

The energetic condition for level L_k is

$$J \geq J_{\min(L_k)}. \tag{136}$$

For multilevel potency involving both L_4 and L_5, one requires

$$J \geq \max\bigl(J_{\min(L_4)}, J_{\min(L_5)}\bigr). \tag{137}$$

Multilevel structural potency arises when these physical constraints are satisfied and the configuration contains coherent cross-level couplings. Diagonal causality then appears directly in the code: micro-level nodes at L_4 are linked by edges that simultaneously participate in meso- or macro-scale structures, while macro-contexts constrain or orient micro-dynamics.

Such couplings can be represented by edge-level codes of the form

$$\kappa = \bigl(M_e;\, T_{\text{src}}, T_{\text{tgt}};\, C_e;\, L_{\text{src}} \to L_{\text{tgt}}\bigr), \tag{138}$$

where

$$M_e \subseteq M, \qquad C_e \subseteq C, \qquad L_{\text{src}} \neq L_{\text{tgt}}. \tag{139}$$

Diagonal potency is present when there exists a set of edges $\{e_i\}$ such that

$$\text{CD}_{DI} \in C_{e_i} \tag{140}$$

and the set

$$\{L^{(i)}_{\text{src}}, L^{(i)}_{\text{tgt}}\} \tag{141}$$

collectively spans the levels contained in $L_\star$.

In this form, potency is not a hidden parameter but a structurally expressed state: a definite pattern of level codes, causal directions, and interaction modes that jointly determine $\Pi(cfg)$. Only configurations in which allowances and flows are coordinated across levels can support emergent transitions that genuinely elevate the organizational regime of the system.

6.6 From theory of possibility to engineering of arising

Once possibility is treated as a structurally expressed property, the sets $P(cfg)$ and $\Pi(cfg)$ become objects of computation and design. The focus thereby shifts from post hoc explanation of realized transitions to anticipatory modeling of admissible transitions. In this sense, the framework moves from descriptive ontology toward an engineering of arising.

Given a family of configurations C and a desired class of transitions T_{desired}, the engineering of arising may be formulated as the following problem: find

$$cfg \in C \tag{142}$$

such that

$$T_{\text{desired}} \subseteq \Pi(cfg). \tag{143}$$

Subject to physical and resource constraints, for example temperature, flux, or size, this becomes a constrained search within the space of codes $\Gamma(cfg)$.

In this setting, potency is not a byproduct of the actual state but a design target. Codes $\Gamma(cfg)$ can be compared, measured, and modified. Structural interventions—such as adding or removing edges with specific interaction modes, reclassifying node roles, or altering level couplings—must then be evaluated not only by their effect on current dynamics but by the way in which they expand, contract, or redirect $\Pi(cfg)$.

The scientific framework thereby acquires an additional descriptive layer: the level of configurational possibility. Empirical data constrain which configurations are realized, but codes and potency sets explain why certain emergent transitions are possible in one region of configuration space and not in another. Analytical and constructive tasks therefore include:

- computing $\Gamma(cfg)$ from observed or designed structures;
- deriving $P(cfg)$ and $\Pi(cfg)$ from $\Gamma(cfg)$ together with physical constraints;
- identifying critical subgraphs g such that $\Pi_{\text{loc}}(g) \neq \varnothing$;
- modifying cfg so as to expand or restrict $\Pi(cfg)$ in desired directions.

The theory of possibility thus becomes a practice of configurational modeling. Configurations are not merely recorded; they are actively manipulated with respect to their potency sets. This prepares the ground for a subsequent metamodeling framework in which configurations, their codes, and their potency sets evolve, intersect, and generate cascades of arising across time and across levels L_k.

7 Causal maps of arising: structure, tensions, and prediction

> A causal map is not a diagram of facts but a computable field of permissions: an abstract structural model in which the admissibility of arising becomes localizable, comparable, and designable. Its function is not to depict events but to formalize the configurational conditions under which activation of structural potency becomes possible.

7.1 Why mapping is necessary

Emergence is not an observable object and cannot be recorded as a discrete outcome. It is a structural transition in which a configuration reaches admissibility for a new organizational regime. This transition is

not reducible to a single element, instant, or event. It forms within a field of typed causal relations whose arrangement determines whether structural potency can activate. The analytical task is therefore not event detection but identification of admissibility conditions embedded in configuration.

A causal map of arising is a formal representation of this admissibility field. It encodes ontological positions as nodes, typed causal operators as edges, and their embedding in level and phase. Its function is to compute where structural potency accumulates and which transitions are admissible prior to realization.

Formally, a causal map is a directed typed graph

$$M = (V, E, \tau_V, \tau_E). \tag{144}$$

The set V is finite and represents arisings. The set $E \subseteq V \times V$ represents directed parent relations. The node-typing function τ_V assigns to each $v \in V$ a tuple

$$\tau_V(v) = (\mathrm{Level}(v), \mathrm{Phase}(v), \mathrm{TypeRole}(v)). \tag{145}$$

The edge-typing function τ_E assigns to each $e \in E$ a tuple

$$\tau_E(e) = \big(\mathrm{InteractionModes}(e), \mathrm{CausalDirections}(e), \mathrm{DependencyType}(e), \mathrm{Weight}(e), \mathrm{OntologicalRole}(e)\big). \tag{146}$$

This structure is compatible with the arising-schema specification and makes explicit not only how a relation operates, but also whether it contributes to first admissibility, persistence, or modulation of the child arising.

Nodes do not denote empirical objects but structural roles. Each node occupies exactly one primary TypeRole—Process, Object, Pattern, Property, or Effect—and is embedded at a defined organizational Level and Phase. Node identity is therefore functional and relational. It is determined by incoming and outgoing typed edges rather than by intrinsic material content.

Edges are operators of admissible transformation. The presence of a directed edge

$$e = (u, v) \tag{147}$$

indicates that node u contributes to the admissibility, persistence, or modulation of node v under the operator set encoded in $\tau_E(e)$. InteractionModes specify mechanism, CausalDirections specify structural orientation, DependencyType specifies the category of condition contributed by the relation, and OntologicalRole specifies whether the relation belongs to first admissibility, maintenance, or modulation of the child. Edges therefore formalize not only propagation of structural allowance across the configuration but also the lifecycle function of that allowance.

Given a configuration M, define the admissible transition set $P(M)$ as the set of higher-level or phase transitions whose structural predicates are satisfied by the current distribution of typed nodes and edges. A transition t is admissible if the required dependency categories are covered, the required interaction modes are present in supporting subgraphs, and cross-level directional constraints are satisfied. Mapping is necessary because $P(M)$ cannot be inferred from isolated inspection of nodes; it is a property of global typed structure.

Pre-transition zones correspond to subgraphs $g \subseteq M$ in which three structural indicators coincide: elevated feedback density, cross-level connectivity, and a balanced constraint–cooperation structure. Let E_F denote the set of edges containing Feedback mode, E_G the set of edges containing Diagonal or Downward causal directions, and E_K the set of edges containing Constraint mode. Define

$$\rho_F(g) = \frac{|E_F(g)|}{|E(g)|}, \qquad \rho_G(g) = \frac{|E_G(g)|}{|E(g)|}, \qquad \rho_K(g) = \frac{|E_K(g)|}{|E(g)|}. \tag{148}$$

A subgraph g is a candidate tension zone if

$$\rho_F(g) \geq \theta_F, \qquad \rho_G(g) \geq \theta_G, \qquad \theta_{K,\min} \leq \rho_K(g) \leq \theta_{K,\max}. \tag{149}$$

Such regions concentrate structural potency.

The causal map is analytic rather than merely representational. Its function is the computation of structural quantities: degree distributions $\deg(v)$, motif-closure counts, feedback-loop detection, dependency-coverage measures, and causal-completeness indices. Emergence becomes predictable only when these quantities are computable and comparable across configurations.

Without a causal map, analysis remains post hoc: emergence is described only after realization. With a causal map, emergence becomes an evaluation of structural admissibility. The map enables computation of configurational readiness, localization of tension nodes, and projection of admissible trajectories. Emergence is thereby transformed from a narrative category into a computable property of typed ontological graphs.

7.2 The node as an ontological unit

Within a causal map of arising, a node does not denote an empirical object or event. It is a formally defined ontological unit representing a structural position in a typed causal configuration. A node localizes a role through which causal operators are transmitted, transformed, constrained, or realized. Its identity is relational and functional rather than material.

Formally, for a causal map

$$M = (V, E, \tau_V, \tau_E), \tag{150}$$

each edge $e \in E$ is a directed ordered pair

$$e = (u, v) \tag{151}$$

equipped with a typing function

$$\tau_E(e) = \begin{pmatrix} \text{InteractionModes}(e), & \text{CausalDirections}(e), \\ \text{DependencyType}(e), & \text{Weight}(e), \\ \text{OntologicalRole}(e) & \end{pmatrix}. \tag{152}$$

The value

$$\text{Weight}(e) \in (0, 1] \tag{153}$$

represents a normalized structural contribution. The edge therefore encodes mechanism, structural orientation, ontological dependency category, relative influence, and lifecycle function with respect to the child arising.

Each TypeRole encodes a distinct structural function. A process-node represents temporally extended transformation. An object-node represents stabilized configurational persistence. A pattern-node represents reproducible relational topology. A property-node represents a system-level functional parameter derived from configuration. An effect-node represents a realized structural outcome that participates in subsequent causal chains. These roles are mutually exclusive at the primary typing level and prevent ontological mixing within a single node.

The ontological status of a node is independent of physical scale or substrate. A biochemical structure, a neural-circuit motif, or a social coordination schema may each instantiate the same TypeRole at different levels. What defines the node is not its substance but its structural function within the typed graph: whether it transforms states, stabilizes form, encodes topology, expresses collective parameters, or fixes outcomes.

A node has no standalone ontological meaning outside its relational embedding. Its effective identity is determined by its incoming and outgoing edges. Define

$$\text{In}(v) = \{\, e \in E \mid e = (u, v) \,\}, \qquad \text{Out}(v) = \{\, e \in E \mid e = (v, w) \,\}. \tag{154}$$

The structural function of v depends on the composition of InteractionModes and CausalDirections across $\mathrm{In}(v)$ and $\mathrm{Out}(v)$. Node identity is therefore emergent from adjacency structure and edge typing.

Connections between nodes are typed causal operators. Each edge $e \in E$ carries attributes

$$\tau_E(e) = \big(\mathrm{InteractionModes}(e), \mathrm{CausalDirections}(e), \mathrm{DependencyType}(e), \mathrm{Weight}(e)\big). \tag{155}$$

These attributes specify how the source node contributes to the admissibility of the target node. Upward causation aggregates lower-level interactions into higher-level structure. Downward causation constrains lower-level dynamics from higher-level configuration. Horizontal causation coordinates peers within a level. Diagonal causation links levels through feedback-mediated coupling. Decoupled causation encodes effective macro-parameter influence. Metacausation modifies admissibility rules themselves.

From the perspective of admissibility, a node v becomes structurally significant when its local configuration supports a non-empty admissible transition set. Let $P_{\mathrm{loc}}(v)$ denote the set of transitions supported by the subgraph induced by v and its neighbors. A node functions as a locus of structural potency when

$$P_{\mathrm{loc}}(v) \neq \varnothing \tag{156}$$

and its adjacency metrics satisfy threshold predicates for feedback density, cross-level connectivity, and balanced constraint–cooperation composition.

The causal map is therefore a network of functional ontological positions rather than a catalogue of empirical entities. A node marks a locus at which configuration can accumulate structural tension, permissions can be activated, and new organizational regimes may become admissible. Analysis of nodes thereby enables localization of potential transition zones and formal comparison of structurally analogous positions across heterogeneous domains.

7.3 The edge as an operator of arising

Within a causal map of arising, the node defines an ontological position, whereas the edge defines the admissible transformation between positions. An edge is not a passive link but a typed causal operator. It specifies how the structural state associated with a source node contributes to the admissibility conditions of a target node. Through edges, structural potency propagates, constraints are redistributed, and transitions become conditionally realizable.

Formally, for a causal map

$$M = (V, E, \tau_V, \tau_E), \tag{157}$$

each edge $e \in E$ is a directed ordered pair

$$e = (u, v) \tag{158}$$

equipped with a typing function

$$\tau_E(e) = \big(\mathrm{InteractionModes}(e), \mathrm{CausalDirections}(e), \mathrm{DependencyType}(e), \mathrm{Weight}(e)\big). \tag{159}$$

The value

$$\mathrm{Weight}(e) \in (0, 1] \tag{160}$$

represents a normalized structural contribution. The edge therefore encodes mechanism, structural orientation, ontological dependency category, and relative influence.

Operationally, an edge functions as a transformation constraint between admissibility sets. Let $S(u)$ denote the admissible state-space region associated with node u, and let $S(v)$ denote the admissible region associated with node v. The edge e defines a mapping operator

$$\mathrm{Op}_e : S(u) \to \mathrm{Conditions}(v), \tag{161}$$

where Conditions(v) specifies the structural predicates under which v becomes admissible. The operator does not directly produce v; rather, it modifies the admissibility landscape of v.

InteractionModes(e) determine the internal logic of Op_e. Cooperation contributes constructive alignment of states. Constraint restricts admissible trajectories. Feedback introduces recursive dependence and potential amplification. Initiation lowers activation barriers. Realization fixes a transition into an effect-node. The composition of modes within InteractionModes(e) therefore defines the qualitative class of transformation enacted by the edge.

CausalDirections(e) determine structural orientation across levels. Upward direction aggregates lower-level structure into higher-level admissibility. Downward direction parameterizes lower-level dynamics through higher-level configuration. Horizontal direction coordinates peers within a level. Diagonal direction couples levels through cross-scale loops. Decoupled direction encodes effective macro-relations independent of detailed microstructure. Meta-causal direction modifies the admissibility rules governing edges themselves.

The organizational scale of the edge is determined by level relations. Let Level(u) and Level(v) denote the levels of nodes u and v. If

$$\mathrm{Level}(u) = \mathrm{Level}(v), \tag{162}$$

the edge is intra-level. If

$$\mathrm{Level}(u) < \mathrm{Level}(v), \tag{163}$$

the edge participates in upward structural assembly. If

$$\mathrm{Level}(u) > \mathrm{Level}(v), \tag{164}$$

it participates in downward constraint. If CausalDirections(e) contains CD_{DI}, the edge contributes to multi-level loop closure. Scale classification is therefore derived from typing rather than declared independently.

Directionality encodes a vector of admissible transformation. The edge does not represent temporal succession but structural possibility. Through Op_e, the system may accumulate coherence, redistribute control parameters, cross threshold conditions, or stabilize new regimes. The edge is thus an ontological trajectory: a structured pathway along which potency can be converted into realization.

Edges contribute to admissible transition sets collectively. Let $P(M)$ denote the admissible transition set of configuration M. A transition $t \in P(M)$ is structurally supported if there exists a subgraph

$$g \subseteq M \tag{165}$$

such that the composition of operators

$$\{\mathrm{Op}_e \mid e \in g\} \tag{166}$$

satisfies the dependency-coverage and mode–direction predicates required by t. Emergence therefore depends on operator composition rather than on isolated edges.

Define the edge activation contribution A_e as a function of structural density and compatibility. A simple structural proxy is

$$A_e = \mathrm{Weight}(e) \cdot f\big(\mathrm{InteractionModes}(e), \mathrm{CausalDirections}(e)\big), \tag{167}$$

where f evaluates compatibility with the predicates required by a transition, for example the presence of feedback and diagonal coupling. Edges with higher A_e contribute more strongly to tension accumulation around their incident nodes.

Without typed operators, the causal map would collapse into an untyped adjacency graph incapable of distinguishing stabilization from transformation. By encoding mechanism, orientation, dependency category, and weight, each edge becomes a computable operator in the algebra of arising. The map is thereby transformed from a descriptive inventory into an operational system capable of identifying transition-supporting motifs, modeling admissible trajectories, and engineering configurational modification.

7.4 Ontological stratification of parent relations

> A parent relation does not merely indicate that one arising contributes to another. It also specifies what ontological function this contribution performs in the lifecycle of the child arising: whether it makes the child admissible for the first time, sustains an already admissible regime, or modulates that regime without producing a new ontological unit.

The current ontology of causal arisings already distinguishes several dimensions of a parent relation. Interaction modes describe how a relation operates. Causal directions describe how influence is oriented across levels. Dependency types describe what category of condition the relation contributes. Weight expresses relative structural contribution. Taken together, these dimensions define the mechanism, orientation, category, and strength of a causal relation. They do not yet determine its ontological function with respect to the lifecycle of the child arising.

This missing dimension becomes critical as soon as the graph begins to represent cyclic, self-maintaining, and regulatory systems. In such systems, the same parent may participate in the first admissibility of an arising, in its later stabilization, or in subsequent parameter shifts. If these roles are not distinguished, genesis, persistence, and regulation collapse into a single undifferentiated parent structure. Cycles then become ontologically ambiguous: they may describe genuine maintenance loops while falsely appearing as loops of first arising.

To prevent this collapse, each parent relation is assigned an ontological role. Ontological role does not replace interaction modes, causal directions, or dependency types. It introduces an additional axis specifying what the relation does relative to the child node as an arising. In the schema, this axis is encoded as

$$\text{OntologicalRole} \in \{\textit{arising}, \textit{maintenance}, \textit{modulation}\}. \tag{168}$$

Formal role function of a parent relation

Let N be an arising, and let

$$e = (p \to N) \tag{169}$$

denote a parent relation from parent p to child N. Define the ontological role function of the relation by

$$\lambda(e, N) \in \{A, M, U\}, \tag{170}$$

where A denotes *arising*, M denotes *maintenance*, and U denotes *modulation*. The full typing of a parent relation may then be written as

$$\tau_E(e) = \big(\text{InteractionModes}(e), \text{CausalDirections}(e), \text{DependencyType}(e), \text{Weight}(e), \text{OntologicalRole}(e)\big). \tag{171}$$

This extension preserves the existing logic of the causal map while making the lifecycle function of the relation explicit.

Arising relations

A parent relation has the ontological role of *arising* if it belongs to the minimal or structurally necessary set of conditions under which the child phenomenon becomes admissible for the first time. Such a relation

contributes to first admissibility rather than merely improving, stabilizing, or regulating an already existing regime.

Let

$$\mathrm{Adm}(N \mid C) \tag{172}$$

denote that the arising N is structurally admissible in configuration C. Let

$$C \setminus e \tag{173}$$

denote the same configuration with parent relation e removed. Then a relation is classified as *arising* when removal of that relation destroys first admissibility:

$$\begin{aligned} \lambda(e, N) = A \iff & \ \mathrm{Adm}(N \mid C) = 1 \\ & \wedge \ \mathrm{Adm}(N \mid C \setminus e) = 0. \end{aligned} \tag{174}$$

In this role, the parent relation functions as part of the ontological genesis of the child. It belongs to the configuration that makes the child possible as a new arising rather than to a later regime that merely supports its continuation.

Maintenance relations

A parent relation has the ontological role of *maintenance* if the child arising is already admissible, but the relation is required for persistence, stability, reproducibility, or autonomous continuation of the regime. Maintenance relations do not generate first admissibility. They preserve an arising that has already crossed the threshold of formation.

Let

$$\mathrm{Persist}(N \mid C) \tag{175}$$

denote that the arising N remains stably sustained in configuration C. A relation is classified as *maintenance* when removal of that relation does not erase first admissibility in principle, but destroys persistence of the already formed regime:

$$\begin{aligned} \lambda(e, N) = M \iff & \ \mathrm{Adm}(N \mid C) = 1 \\ & \wedge \ \mathrm{Persist}(N \mid C) = 1 \\ & \wedge \ \mathrm{Persist}(N \mid C \setminus e) = 0. \end{aligned} \tag{176}$$

Maintenance therefore concerns closure, retention, and stability. It marks the causal architecture through which a formed arising remains viable across time, perturbation, or internal fluctuation.

Modulation relations

A parent relation has the ontological role of *modulation* if removal or alteration of that relation does not destroy first admissibility and does not abolish persistence, but changes the operating regime of the child arising. Such changes may include intensity, sensitivity, gain, range, timing, amplitude, or parameterization of the already existing regime.

Let

$$\mathrm{Param}(N \mid C) \tag{177}$$

denote the parameter-state vector of the arising N in configuration C. A relation is classified as *modulation* when admissibility and persistence remain intact after its removal, but the operating regime changes:

$$\begin{aligned}\lambda(e, N) = U \iff & \ \mathrm{Adm}(N \mid C) = 1 \\ & \wedge\ \mathrm{Adm}(N \mid C \setminus e) = 1 \\ & \wedge\ \mathrm{Persist}(N \mid C \setminus e) = 1 \\ & \wedge\ \mathrm{Param}(N \mid C) \neq \mathrm{Param}(N \mid C \setminus e).\end{aligned} \tag{178}$$

Modulation therefore concerns controlled variation without ontological birth of a new unit. It belongs to regime regulation rather than to the origin of the child arising itself.

Three ontological questions answered by parent relations

The distinction between *arising*, *maintenance*, and *modulation* separates three questions that must not be conflated in a causal ontology. First, what makes a phenomenon admissible for the first time. Second, what keeps an already admissible phenomenon in existence. Third, what changes its mode of operation without generating a new ontological unit. Without this distinction, the graph cannot cleanly separate genesis, persistence, and regulation.

Relation to existing edge dimensions

Ontological role is independent of interaction mode, causal direction, and dependency type. The same interaction mode may appear in different ontological roles. Constraint may participate in arising by filtering initial admissibility, in maintenance by preserving a viable regime, or in modulation by reshaping a parameter window. Feedback may stabilize a formed regime as maintenance or tune its gain as modulation. Ontological role is therefore not reducible to mechanism.

Ontological role is also independent of necessity. Necessity specifies how essential a relation is within the represented description, using values such as *necessary*, *enabling*, *contextual*, or *optional*. Ontological role specifies the lifecycle function performed by that relation with respect to the child arising. A relation may therefore be necessary and arising, necessary and maintenance, or contextual and modulation. These are orthogonal dimensions.

Priority rule for ambiguous cases

In some systems, the same parent relation appears to contribute both to first admissibility and to later persistence. To preserve ontological clarity, a dominant-role priority rule is introduced. If a relation qualifies as *arising*, it is classified as arising even if it also supports maintenance. If it does not qualify as arising but does qualify as maintenance, it is classified as maintenance rather than modulation. The priority order is therefore

$$\textit{arising} \succ \textit{maintenance} \succ \textit{modulation}. \tag{179}$$

This rule prevents the graph from collapsing first genesis into later regulation and ensures that each relation receives one dominant ontological role relative to the child.

Connection to level phases

The ontological role of parent relations also clarifies the internal logic of level phases. During differentiation, *arising* relations dominate, because new basic units are being made admissible. During integration, *arising* relations remain important, but *maintenance* relations increase as assembled structures must retain coherence. During self-organization, *maintenance* and *modulation* relations become dominant, because the system increasingly stabilizes and tunes its internal regime. During threshold transition, a new configuration of *arising* relations appears, making the next organizational level admissible.

This interpretation strengthens the status of phases as structural stages rather than chronological labels. The phase of a node is reflected not only in the node itself, but also in the distribution of ontological roles across its incoming parent relations.

Ontological significance for cyclic systems

The distinction becomes especially important in cyclic and feedback-dominated domains such as metabolism, immune regulation, neural modulation, and social stabilization. In such domains, loops are real and necessary, but not every loop is a loop of first arising. Many cycles are instead loops of maintenance or modulation. By assigning ontological role explicitly, the map can preserve cyclic organization without falsely converting every stabilizing loop into a genealogical loop of origin.

OntologicalRole therefore acts as a stratifying principle for parent relations. It allows the causal map to distinguish the architecture of first admissibility from the architecture of persistence and from the architecture of regime control. In this form, parent structure becomes significantly more precise, and the graph becomes capable of representing complex self-maintaining systems without loss of ontological clarity.

7.5 Tension nodes and points of potential

Within a causal map

$$M = (V, E, \tau_V, \tau_E), \tag{180}$$

nodes do not contribute uniformly to structural admissibility. Certain nodes occupy positions at which multiple causal directions intersect, feedback loops accumulate, and cross-level dependencies converge. Such nodes are defined as *tension nodes*. A tension node is a structural position at which the local configuration supports a non-trivial admissible transition set despite the absence of realized transition.

Formally, for $v \in V$, define its local subgraph $g(v)$ as the induced subgraph containing v and its first-order neighbors. Let

$$P_{\mathrm{loc}}(v) \tag{181}$$

denote the admissible transition set computed over $g(v)$. A node v is a candidate tension node if

$$P_{\mathrm{loc}}(v) \neq \varnothing \tag{182}$$

and its structural metrics exceed the specified thresholds for feedback density, cross-level connectivity, and balanced constraint–cooperation composition.

These indicators are computed locally. Let $E_F(v)$ denote the set of edges incident to v containing Feedback mode, let $E_G(v)$ denote the set of edges containing Diagonal or cross-level directions, and let $E_K(v)$ denote

the set of edges containing Constraint mode. Define local densities by

$$\rho_F(v) = \frac{|E_F(v)|}{\deg(v)}, \qquad \rho_G(v) = \frac{|E_G(v)|}{\deg(v)}, \qquad \rho_K(v) = \frac{|E_K(v)|}{\deg(v)}, \tag{183}$$

where $\deg(v)$ is the total degree of v. A tension node requires

$$\rho_F(v) \geq \theta_F, \qquad \rho_G(v) \geq \theta_G, \qquad \theta_{K,\min} \leq \rho_K(v) \leq \theta_{K,\max}. \tag{184}$$

Additionally, let

$$\lambda_{\max}(v) \tag{185}$$

denote the spectral radius of the adjacency matrix of $g(v)$, interpreted as a local amplification coefficient. Elevated $\lambda_{\max}(v)$ indicates increased capacity for recursive reinforcement. A tension condition may therefore include

$$\lambda_{\max}(v) \geq \theta_\lambda, \tag{186}$$

reflecting the potential for amplification of structural perturbation.

The ontological type role of a tension node is not fixed. Any node type—Process, Object, Pattern, Property, or Effect—may become a tension node if its adjacency structure satisfies the relevant conditions. Tension is therefore positional and relational rather than categorical.

Three operational modes of a tension node may be distinguished according to activation state. In *stable mode*,

$$P_{\text{loc}}(v) = \varnothing \tag{187}$$

or the structural indicators remain below threshold; the node participates mainly in regime stabilization and exhibits low amplification capacity. In *latent mode*,

$$P_{\text{loc}}(v) \neq \varnothing \tag{188}$$

and the thresholds are satisfied, but no transition has yet been realized; the node functions as a concentrated reservoir of structural potency. In *active mode*, some transition

$$t \in P_{\text{loc}}(v) \tag{189}$$

is being realized, and the adjacency structure of v changes, typically through addition or re-typing of edges, indicating reconfiguration of admissibility.

The transition from latent to active mode may be modeled as threshold crossing in a composite tension index $T(v)$. Define

$$T(v) = a\,\rho_F(v) + b\,\rho_G(v) + c\,C_{\text{closure}}(v) + d\,\widehat{\lambda}_{\max}(v), \tag{190}$$

where $C_{\text{closure}}(v)$ measures motif closure, for example participation in feedback loops or strongly connected components, and $\widehat{\lambda}_{\max}(v)$ is a normalized spectral term, for example rescaled to $[0, 1]$ over the evaluated configuration family. When

$$T(v) \geq T_{\text{crit}}, \tag{191}$$

node v becomes dynamically capable of supporting structural reconfiguration under minimal perturbation.

Tension nodes are therefore computable loci of structural sensitivity. They indicate positions at which the configuration is locally prepared for regime shift. A high density of nodes satisfying

$$T(v) \geq T_{\text{crit}} \tag{192}$$

implies elevated global configurational readiness. This does not guarantee transition, but it reduces structural barriers to activation.

Let

$$N_T = |\{\, v \in V \mid T(v) \geq T_{\text{crit}} \,\}| \tag{193}$$

denote the number of tension nodes. Define configurational readiness by

$$R(M) = \frac{N_T}{|V|}. \tag{194}$$

Higher $R(M)$ corresponds to greater systemic permissibility for emergence. Emergent transitions occur when local tension nodes couple through cross-level edges, thereby forming a transition-supporting subgraph whose combined admissibility exceeds the global structural constraints.

Tension nodes therefore function as operational markers of potential rather than as descriptive anomalies. They are primary analytical targets for predictive mapping, because admissible trajectories originate in their vicinity. By identifying nodes with elevated $T(v)$, one localizes structural points at which a configuration may pass from potency to realization.

Law of local admissibility concentration. Admissible transitions originate preferentially in structurally tense local regions of the graph:

$$T(v) \geq T_{\text{crit}} \Rightarrow \Pi_{\text{loc}}(g(v)) \neq \varnothing. \tag{195}$$

This law states that sufficiently high local tension indicates the presence of local structural potency and therefore marks a region from which admissible transitions may originate.

7.6 Example: a simple causal map

Consider a minimal configuration illustrating transition from physicochemical dynamics to functional biological effect. The causal map

$$M = (V, E, \tau_V, \tau_E) \tag{196}$$

contains four nodes and three directed edges. The configuration represents not a temporal sequence but a structural chain of admissibility culminating in a functional outcome.

Let

$$V = \{A, B, C, D\}. \tag{197}$$

Node A is a process-node representing protein-folding dynamics. Node B is a pattern-node representing stabilized three-dimensional configuration. Node C is a property-node representing catalytic capability. Node D is an effect-node representing acceleration of a chemical reaction. All nodes are embedded at biochemical level L_3, with phases progressing from LP_B to LP_D.

Edge

$$e_1 = (A, B) \tag{198}$$

encodes upward causality. Its typing is

$$\tau_E(e_1) = \big(IM = \{\text{IM}_C, \text{IM}_K\}, CD = \{\text{CD}_U\}, DT = \text{DT}_{SUB}, \text{Weight} = w_1\big). \tag{199}$$

The operator Op_{e_1} aggregates micro-level amino-acid interactions into a stable structural topology. The admissibility of B depends on cooperative alignment and geometric filtering imposed by the folding process.

Edge

$$e_2 = (B, C) \tag{200}$$

encodes downward causality. Its typing is

$$\tau_E(e_2) = \big(IM = \{\mathrm{IM}_K, \mathrm{IM}_R\}, CD = \{\mathrm{CD}_D\}, DT = \mathrm{DT}_{CON}, \mathrm{Weight} = w_2\big). \tag{201}$$

The operator Op_{e_2} restricts admissible vibrational and electronic configurations to those compatible with catalytic activity. The spatial pattern B parameterizes the functional space of C.

Edge

$$e_3 = (C, D) \tag{202}$$

encodes diagonal causality. Its typing is

$$\tau_E(e_3) = \big(IM = \{\mathrm{IM}_R, \mathrm{IM}_I\}, CD = \{\mathrm{CD}_{DI}\}, DT = \mathrm{DT}_{CAT}, \mathrm{Weight} = w_3\big). \tag{203}$$

The operator Op_{e_3} transforms a molecular property into a cross-level kinetic effect. Catalytic capability C modifies reaction rates in the surrounding medium, thereby producing D as a realized systemic effect.

The local admissible transition set becomes non-empty once e_1 and e_2 are present and the relevant structural predicates are satisfied. Let

$$P_{\mathrm{loc}}(C) \tag{204}$$

denote the admissible transition set over the subgraph induced by C. In the minimal configuration without feedback loops,

$$P_{\mathrm{loc}}(C) \neq \varnothing, \tag{205}$$

but amplification remains limited.

Tension metrics may also be evaluated locally. For node C, the cross-level density satisfies

$$\rho_G(C) > 0 \tag{206}$$

because of diagonal edge e_3. If additional feedback edges were introduced, for example if reaction products stabilized folding, then $\rho_F(C)$ would increase and the local amplification coefficient $\lambda_{\max}(C)$ could also rise. If

$$T(C) \geq T_{\mathrm{crit}}, \tag{207}$$

catalytic activity would participate in higher-order regime shifts, such as autocatalytic cycles.

The compositional structure of operators

$$\{\mathrm{Op}_{e_1}, \mathrm{Op}_{e_2}, \mathrm{Op}_{e_3}\} \tag{208}$$

defines a transformation chain:

$$\text{Process} \to \text{Pattern} \to \text{Property} \to \text{Effect}. \tag{209}$$

Catalytic function is absent in isolated amino acids because Op_{e_1} and Op_{e_2} are not jointly satisfied. Only when structural closure at B exists can Op_{e_2} produce C, and only then can Op_{e_3} propagate C into D.

This example demonstrates that emergence is configurational rather than additive. The catalytic property is not contained independently in the components, but arises only when typed operators collectively satisfy admissibility predicates. The map therefore captures a sequential transformation of structural permissions rather than a chronological narrative of events.

In graph-theoretic terms, the minimal motif is a directed path of length three with heterogeneous type roles and heterogeneous causal directions. In ontological terms, it represents a minimal admissibility cascade across role transitions. The example illustrates how typed edges function as operators in the algebra of arising, thereby enabling systematic identification of transition-supporting motifs across domains.

7.7 How the map becomes predictive

A causal map becomes predictive when it operates on structural admissibility rather than on realized events. Let

$$M = (V, E, \tau_V, \tau_E) \tag{210}$$

be a typed causal graph. Prediction in this framework does not attempt to infer future states directly. Instead, it evaluates the admissible transition set $P(M)$ derived from the current configuration. The predictive question is therefore whether $P(M)$ contains higher-level or phase-shift transitions that are structurally supported but not yet realized.

Formally, define an admissibility function

$$A(M, t) \in \{0, 1\}, \tag{211}$$

where $A(M, t) = 1$ if transition t satisfies the relevant dependency-coverage, operator-composition, and level-direction constraints within M. The predictive capacity of the map is then the capacity to compute

$$P(M) = \{\, t \in T \mid A(M, t) = 1 \,\}. \tag{212}$$

A non-empty $P(M)$ indicates structural readiness for transformation.

Law of predictive admissibility. Prediction in a causal map concerns the admissibility of transitions rather than the direct forecasting of realized events:

$$A(M, t) = 1 \Rightarrow t \in P(M). \tag{213}$$

This law states that a transition is predictive in the structural sense when the map evaluates it as admissible. The predictive target is permissibility rather than event occurrence.

Predictiveness becomes stronger when configurations are grouped into structural classes. Let C be the space of typed configurations, and define a structural equivalence relation $\equiv$ over C such that

$$M_1 \equiv M_2 \tag{214}$$

if their typed motif structures are isomorphic under role-preserving and direction-preserving mappings. Motif isomorphism is defined over subgraphs preserving the attributes encoded in τ_V and τ_E. Structural classes thereby represent families of configurations sharing an identical architecture of admissibility.

If a transition t has been realized in some configuration M_* within a structural class, and another configuration M' belongs to the same class under $\equiv$, then

$$t \in P(M'). \tag{215}$$

Prediction therefore becomes transfer of admissibility across structurally equivalent graphs rather than extrapolation from empirical similarity alone.

Theorem of structural transferability. If two configurations belong to the same structural class, admissible transitions identified in one remain admissible in the other:

$$M_* \equiv M' \Rightarrow P(M_*) = P(M') \quad \text{up to class-preserving constraints.} \tag{216}$$

This theorem states that admissibility is transferable across structurally equivalent configurations. Prediction therefore generalizes by structure rather than by material identity.

The map also enables detection of structural lacunae, that is, subgraphs whose dependency coverage is incomplete but nearly sufficient. Let

$$\mathrm{Cover}_r(v) \tag{217}$$

measure local dependency coverage for node v over the required category set r. If for some r,

$$\mathrm{Cover}_r(v) < \mathrm{Threshold}_r \tag{218}$$

but approaches that threshold, then v marks a near-admissible state. Such nodes are latent tension points at which minimal structural augmentation, for example addition or re-typing of edges, may expand $P(M)$.

Motif detection further enhances prediction. Let M_k denote the set of k-node typed motifs known to support specific transition classes. For a given configuration M, compute

$$\mathrm{Occ}(M, M_k), \tag{219}$$

defined as the number of motif embeddings preserving typing constraints. If

$$\mathrm{Occ}(M, M_k) \geq \mathrm{Threshold}_k, \tag{220}$$

then M structurally supports the transitions associated with those motifs.

Another predictive metric is configurational readiness,

$$R(M) = \frac{N_T}{|V|}, \tag{221}$$

where N_T is the number of nodes satisfying $T(v) \geq T_{\mathrm{crit}}$. High $R(M)$ correlates with increased density of transition-supporting subgraphs. A predictive signature may therefore be expressed as

$$F_{\mathrm{pred}}(M) = \big(|P(M)|,\ R(M),\ \mathrm{Occ}(M, M_k)\big), \tag{222}$$

forming a structural descriptor of emergent potential.

Prediction in this framework does not rely on statistical trend projection. It rests on operator composition and structural invariance. Given two configurations with distinct material substrates but identical typed motif structure, their admissible transition sets coincide modulo environmental constraints. Structural permission, rather than empirical content, determines predictive transfer.

In practical terms, prediction reduces to structural computation. The causal map is queried for operator compositions supporting transitions, cross-level loop closure, dependency completeness, and amplification indicators. Where these structural conditions co-occur, emergence becomes admissible under minimal perturbation.

The causal map thereby becomes a generative instrument. It identifies not what will occur, but what is structurally permitted under current constraints. By computing $P(M)$, motif embeddings, and readiness indices, the system shifts from retrospective explanation to forward admissibility analysis. Prediction becomes evaluation of configurational permission landscapes rather than extrapolation of observed outcomes.

7.8 Connection to a computable model

The causal map of arising is a formally specified directed typed graph

$$M = (V, E, \tau_V, \tau_E) \tag{223}$$

designed for computational storage and algorithmic analysis. It is not merely a visual diagram but a machine-readable structure in which nodes and edges carry explicit ontological attributes. This representation enables systematic evaluation of admissibility, transition support, and configurational readiness.

Each node $v \in V$ is encoded as

$$\tau_V(v) = \big(\mathrm{Level}(v), \mathrm{Phase}(v), \mathrm{TypeRole}(v), \mathrm{Status}(v)\big). \tag{224}$$

Each edge $e \in E$ is encoded as

$$\tau_E(e) = \begin{pmatrix} \mathrm{InteractionModes}(e), & \mathrm{CausalDirections}(e), \\ \mathrm{DependencyType}(e), & \mathrm{Weight}(e), \\ \mathrm{OntologicalRole}(e) & \end{pmatrix}. \tag{225}$$

The graph is therefore a typed multi-relational structure. All structural operations are defined over these attributes rather than over unlabeled adjacency alone. OntologicalRole makes it possible to distinguish computationally between relations of first arising, persistence, and regime modulation.

Computationally, the graph may be represented as an adjacency list with typed metadata or as a property graph stored in graph databases such as Neo4j or TigerGraph. Let

$$\mathrm{Adj}(v) \tag{226}$$

denote the set of neighbors of v. Global structural density may be computed as

$$\rho = \frac{2|E|}{|V|}, \tag{227}$$

while directed density may be computed as

$$\rho_{\mathrm{dir}} = \frac{|E|}{|V|(|V|-1)}. \tag{228}$$

These quantities serve as global structural indicators.

Admissible-transition computation is defined as follows. Let T denote the set of possible transition types. For each transition $t \in T$, define a predicate function

$$A(M, t) \in \{0, 1\}. \tag{229}$$

The value $A(M, t) = 1$ if and only if the required DependencyTypes are covered, the required InteractionModes are present in supporting subgraphs, and the required CausalDirections satisfy level constraints. The admissible set is then

$$P(M) = \{\, t \in T \mid A(M, t) = 1 \,\}. \tag{230}$$

Motif detection is performed over typed subgraphs. Let M_k denote the set of k-node motif templates with role and direction constraints. Define

$$\mathrm{Occ}(M, M_k) \tag{231}$$

as the number of subgraph embeddings preserving the node and edge typings encoded in τ_V and τ_E. Subgraph matching may be implemented through backtracking or optimized graph-isomorphism algorithms. Presence of specific motifs correlates with support for particular transition classes.

Tension nodes are computable through a composite index $T(v)$. For node v, compute local metrics such as $\rho_F(v)$ for feedback density, $\rho_G(v)$ for cross-level connectivity, and closure(v) for participation in strongly connected components. Define

$$T(v) = a \cdot \rho_F(v) + b \cdot \rho_G(v) + c \cdot \mathrm{closure}(v). \tag{232}$$

Nodes satisfying

$$T(v) \geq T_{\text{crit}} \tag{233}$$

are marked as structurally sensitive.

A predictive evaluation function may therefore be defined as

$$F_{\text{pred}}(M) = \big(|P(M)|,\ R(M),\ \text{Occ}(M, M_k)\big), \tag{234}$$

where

$$R(M) = \frac{N_T}{|V|} \tag{235}$$

and N_T is the number of nodes satisfying $T(v) \geq T_{\text{crit}}$. This tuple forms a structural readiness signature independent of empirical content.

Algorithmically, admissibility evaluation proceeds in discrete stages:

1. compute dependency coverage for each node;
2. detect required motifs through typed subgraph search;
3. compute local tension indices;
4. evaluate $A(M, t)$ for each candidate transition t;
5. return $P(M)$ together with readiness metrics.

In pseudocode form, admissibility evaluation may be summarized as follows: for each $t \in T$, if

$$\text{CheckDependencies}(M, t)\ \wedge\ \text{CheckMotifs}(M, t)\ \wedge\ \text{CheckDirections}(M, t), \tag{236}$$

then include t in $P(M)$. This procedure is deterministic given the graph and rule definitions.

The model also supports integration with graph neural networks and structural learning methods. Feature vectors for nodes may include degree, local density, motif participation counts, and dependency-coverage scores. Such features enable classification of structural roles and detection of near-threshold configurations in large datasets.

At the architectural level, the computable model integrates four layers:

1. ontological typing of nodes and edges;
2. operator-based admissibility evaluation;
3. structural metrics for tension and readiness;
4. transition-rule libraries defining admissible transformation classes.

This layered structure allows simulation environments to operate not only on dynamic states but also on the architecture of permission itself.

Automated hypothesis generation follows directly from this framework. Given a configuration M, the system computes missing dependency categories, near-complete motifs, and minimal structural augmentations capable of expanding $P(M)$. Structural modifications may therefore be evaluated prior to empirical realization, transforming the causal map into a design instrument rather than a descriptive registry.

In this form, the causal map becomes a computational substrate for ontologically oriented reasoning systems. Such systems operate on structural admissibility landscapes rather than on empirical surface similarity. Emergence is evaluated as a function of graph structure rather than as an anomalous event. Prediction becomes systematic computation of permissible transitions within typed ontological graphs.

8 Causal completeness and the closure of arising conditions in ontological graphs

This section formalizes the concept of causal completeness as a measurable property of arising descriptions. The objective is not to enumerate all possible causes, but to determine whether the causal specification of an arising is structurally sufficient and epistemically grounded relative to its ontological role, level, phase, and type.

Causal completeness is treated as an internal property of an arising node within a structured graph. It measures whether the necessary categories of conditions are present, whether their influence is specified with sufficient precision, and whether the claimed causal relations are supported by evidence.

> An arising is causally complete not when nothing more can be said, but when nothing essential is missing. Completeness is therefore a property of closure rather than of exhaustiveness.

The problem of incomplete causality in expanding ontological systems

In an expanding ontological catalogue, new arisings are introduced continuously. Without a formal criterion of causal closure, the system accumulates nodes whose descriptions appear plausible yet omit essential causal conditions. Such incompleteness remains largely invisible when causality is represented merely as an unweighted list of parents.

Incomplete causality manifests in several characteristic structural pathologies: missing substrate conditions, absent contextual constraints, overgeneralized parent relations, and effects lacking specified mechanisms. These pathologies cannot be reliably detected by search or manual inspection once the system reaches significant scale.

A formal indicator of causal completeness is therefore required in order to distinguish arisings that are fully specified from those that remain structurally underdetermined.

Causal structure as weighted conditional coverage

Each arising is represented as a node with a finite set of parent relations. Each parent contributes a partial causal influence, expressed as a weight, and carries a degree of confidence reflecting the epistemic support for that relation. In addition, each parent relation is classified by dependency type and ontological role. Dependency type specifies what category of causal condition is contributed by the relation. Ontological role specifies whether this contribution belongs to first admissibility, persistence, or modulation of the child arising.

Let an arising be denoted by N, with parent set

$$P(N) = \{p_1, p_2, \ldots, p_k\}. \tag{237}$$

Each parent relation is assigned a normalized influence weight w_p, such that total causal influence is conserved:

$$\sum_{p \in P(N)} w_p = 1. \tag{238}$$

The weight expresses structural relevance within the causal configuration; it does not encode certainty. Epistemic support is represented independently by a confidence coefficient c_p associated with each relation.

Categories of causal conditions

Causal completeness is not evaluated at the level of individual parents, but at the level of condition categories that must be satisfied for an arising to be admissible.

The minimal set of causal condition categories is defined as follows:

Minimal categories of causal conditions

The minimal set of causal condition categories is represented through dependency types. Because each dependency type combines ontological definition, diagnostic criteria, phase-specific structural role, and practical weighting guidance, their presentation is divided into four complementary tables. Together, these tables provide the operational basis for evaluating causal completeness as weighted conditional coverage rather than as an undifferentiated parent list.

Conceptual identity and coverage mapping of dependency types

The following table presents the conceptual identity of each dependency type. It specifies the formal code, the ontological category name, the general definition, and the primary and secondary coverage dimensions through which the dependency contributes to causal completeness.

Code	**Name**	**Definition**	**Primary coverage**	**Secondary coverage**
DT_{SUB}	Substrate	Constitutive material or structural basis required for the phenomenon to be instantiable at all. The parent provides the physical or chemical carrier, building blocks, or persistent structural platform on which the child phenomenon can occur.	substrate	context

Code	Name	Definition	Primary coverage	Secondary coverage
DT_{BND}	Boundary	Delimitation and interface conditions separating, filtering, or gating interactions between an inside and an outside, or between compartments. The parent supplies boundary constraints shaping admissible flows and stabilizing regimes.	constraint	context
DT_{TMP}	Template	Guiding structure that biases assembly, copying, or pattern formation by providing a reference configuration reducing combinatorial ambiguity.	constraint	mechanism, context
DT_{CAT}	Catalyst	Catalytic agency that increases reaction feasibility by lowering activation barriers, increasing rates, or channeling pathways through specific transition states.	mechanism	constraint
DT_{REG}	Regulation	Control of rates, states, and regime selection through modulatory mechanisms such as switching, gating, homeostatic control, or parameter tuning.	feedback	constraint, mechanism
DT_{GRD}	Gradient	Dependence on sustained differences in potential across space, phases, or compartments, such as concentration, electrochemical, redox, or thermal gradients.	constraint	context, mechanism
DT_{CMP}	Compartment	Confinement and localization into a bounded domain that changes effective concentration, stochasticity, coupling, and protection from dilution.	context	constraint
DT_{INF}	Information	Reliance on encoded state or pattern, such as sequence, symbol, code, or memory, that constrains or selects system behavior.	constraint	feedback, context
DT_{SEL}	Selection	Population-level or ensemble-level filtering in which differential persistence, amplification, or stability reshapes the distribution of variants relevant to the child phenomenon.	feedback	constraint, context
DT_{RES}	Resource	Reliance on consumable throughput of matter or energy that fuels, feeds, or sustains the phenomenon.	substrate	context, mechanism
DT_{CON}	Constraint	Limiting conditions that restrict degrees of freedom and define an admissible region of state space, including stoichiometric limits, conservation laws, geometric constraints, and compatibility rules.	constraint	context
DT_{TRN}	Transport	Dependence on movement or exchange processes that deliver inputs, remove outputs, connect compartments, or maintain distributed fluxes.	mechanism	context, feedback

Code	Name	Definition	Primary coverage	Secondary coverage
DT_{ENV}	Environment	Reliance on exogenous conditions not internal to the represented system level, such as temperature, pressure, solvent composition, radiation, external fields, or ecological setting.	context	constraint, substrate

Table 19: Conceptual identity and coverage mapping of dependency types.

Formal criteria and detection tests

The next table presents the formal criteria and indicative detection tests associated with each dependency type. These criteria function as a diagnostic basis for determining whether a parent relation genuinely instantiates the dependency category attributed to it.

Dependency type	Formal criteria	Detection tests
DT_{SUB} — Substrate	Removal eliminates possibility rather than efficiency; contributes persistent material or structural prerequisites; allows functional substitution by equivalent substrate class; primarily concerns constitution rather than control.	Ablation removes the phenomenon under otherwise fixed conditions; reintroduction restores it without redesign; description emphasizes constitutive carriers more than control operators.
DT_{BND} — Boundary	Operates by restricting exchange with an external domain; requires inside/outside distinction or interface geometry; boundary properties enforce admissible fluxes; stability depends on boundary maintenance.	Changing permeability or selectivity alters the regime predictably; removing the boundary collapses gradients or localization; boundary perturbations act faster than internal reconfiguration.
DT_{TMP} — Template	Provides a reference pattern guiding formation; child outcome depends on parent configuration; increases fidelity or selectivity relative to non-templated formation; distinct from catalytic rate enhancement.	Changing the template changes the produced pattern predictably; removal collapses fidelity even if substrates remain; template-dependent copying appears as conditional mapping between parent pattern and child outcome.
DT_{CAT} — Catalyst	Primarily changes kinetics rather than only boundary conditions; catalytic effect may be sub-stoichiometric; parent is not consumed proportionally; specificity may appear as pathway channeling.	Rate enhancement occurs upon addition of the parent; inhibition or denaturation reduces throughput or changes pathway distribution; catalytic signatures such as turnover or activation-energy shift are observed.
DT_{REG} — Regulation	Acts on control parameters rather than substrate supply; child regime changes with parent control state; often involves feedback, gating, activation, inhibition, or set-point stabilization; may be context-dependent and nonlinear.	Perturbation changes operating regime; time series show control signatures such as overshoot, hysteresis, or stabilization; the child fails under parameter drift unless control is present.
DT_{GRD} — Gradient	Requires a maintained potential difference; acts as driving force or bias; gradient persistence depends on boundary or transport conditions; dependency is directional in potential space.	Collapsing the gradient removes or reverses the phenomenon; artificially establishing it induces or strengthens the effect; measured gradient magnitude correlates with effect size.

Dependency type	Formal criteria	Detection tests
DT_{CMP} — Compartment	Depends on confinement rather than only global concentration; compartmentalization changes rates, stability, or coupling through concentration and segregation effects; concerns domain structure rather than interface selectivity; persistence is enhanced by reduced mixing or dilution.	Removing confinement reduces efficiency or persistence; introducing compartments increases local throughput or stability; localization patterns correlate with activity.
DT_{INF} — Information	Dependency is informationally configuration-dependent; parent informational differences lead to systematically different outcomes; may be abstracted as code or state variable; supports reproducibility across different substrates realizing the same code.	Changing encoded state predictably changes outcome while bulk chemistry remains similar; information-preserving transformations preserve function; symbolic or state abstraction remains predictive.
DT_{SEL} — Selection	Operates on a variant ensemble with differential persistence or amplification; outcome is distributional rather than individual; variant properties map to persistence metrics; requires population semantics.	Repeated cycles produce frequency shifts; removing the filtering mechanism removes systematic trend; selection gradients are measurable as differential persistence or amplification rates.
DT_{RES} — Resource	Defined by consumable inputs and supply rates; child magnitude scales with availability; depletion dynamics matter; supply is measurable as input flow.	Limitation produces predictable decline or regime change; supplementation restores throughput without redesign; mass or energy balance shows explicit dependence on input flux.
DT_{CON} — Constraint	Best described as restriction of state space; remains meaningful without detailed dynamics; expressible as inequalities, invariants, or compatibility rules; phenomenon requires operation within constrained regime.	Relaxing the constraint increases degrees of freedom and destabilizes the regime; tightening reduces variability and may stabilize it; formal constraint representation predicts success and failure boundaries.
DT_{TRN} — Transport	Concerns flux-mediated coupling across space or domains; transport capacity and kinetics are relevant; depends on delivery or removal timescales; transport can create, sustain, or dissipate gradients.	Blocking transport alters the phenomenon even when substrates are present; transport-rate changes modulate stability or throughput; spatial heterogeneities track transport pathways.
DT_{ENV} — Environment	Originates outside the represented system boundary; defines feasibility regions or regime boundaries; stability depends on external ranges; influence is exogenous whether static or fluctuating.	External parameter sweeps produce predictable regime shifts; holding internal composition fixed while changing external conditions changes the phenomenon; description includes environmental ranges or setting-specific admissibility.

Table 20: Formal criteria and detection tests for dependency types.

Phase roles and typical causal profiles

The following table summarizes how each dependency type tends to function across level phases together with the interaction modes and causal directions with which it is most commonly associated. These values do not define necessity in every case, but they provide a structural profile useful for evaluating causal completeness and motif composition.

Dependency type	**Phase roles**	**Typical interaction modes**	**Typical causal directions**
DT_{SUB} — Substrate	LP_A: minimal material basis; LP_B: stabilizes assemblies; LP_C: maintains substrate pool; LP_D: limiting basis enforcing threshold ceilings.	$\mathrm{IM}_R, \mathrm{IM}_I$	$\mathrm{CD}_U, \mathrm{CD}_H$
DT_{BND} — Boundary	LP_A: early separation; LP_B: stabilizes interfaces; LP_C: sustains non-equilibrium organization; LP_D: critical control surface for transition.	$\mathrm{IM}_K, \mathrm{IM}_R$	$\mathrm{CD}_U, \mathrm{CD}_{DI}, \mathrm{CD}_H$
DT_{TMP} — Template	LP_A: selective pairing; LP_B: module alignment; LP_C: reproducible pattern stabilization; LP_D: persistence and propagation across threshold.	$\mathrm{IM}_K, \mathrm{IM}_I, \mathrm{IM}_R$	$\mathrm{CD}_H, \mathrm{CD}_{DI}, \mathrm{CD}_U$
DT_{CAT} — Catalyst	LP_A: enables feasible synthesis; LP_B: stabilizes throughput; LP_C: maintains catalytic closure; LP_D: supports threshold crossing under closure.	$\mathrm{IM}_R, \mathrm{IM}_I, \mathrm{IM}_C$	$\mathrm{CD}_U, \mathrm{CD}_H, \mathrm{CD}_{DI}$
DT_{REG} — Regulation	LP_A: primitive gating; LP_B: rate coordination; LP_C: homeostatic stabilization; LP_D: threshold triggering or suppression.	$\mathrm{IM}_F, \mathrm{IM}_K, \mathrm{IM}_C$	$\mathrm{CD}_D, \mathrm{CD}_{DI}, \mathrm{CD}_H$
DT_{GRD} — Gradient	LP_A: energetic biasing; LP_B: integrated module coupling; LP_C: non-equilibrium stabilization; LP_D: critical control factor for threshold crossing.	$\mathrm{IM}_K, \mathrm{IM}_R, \mathrm{IM}_F$	$\mathrm{CD}_H, \mathrm{CD}_U, \mathrm{CD}_{DI}$
DT_{CMP} — Compartment	LP_A: localized microdomains; LP_B: co-localization of modules; LP_C: retention and protected coupling; LP_D: protocellular closure and persistence.	$\mathrm{IM}_R, \mathrm{IM}_K, \mathrm{IM}_C$	$\mathrm{CD}_U, \mathrm{CD}_{DI}$
DT_{INF} — Information	LP_A: early state-dependent constraints; LP_B: coded module coordination; LP_C: memory and state-dependent regulation; LP_D: informational closure enabling threshold transition.	$\mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_R$	$\mathrm{CD}_{DI}, \mathrm{CD}_D, \mathrm{CD}_H$
DT_{SEL} — Selection	LP_A: weak differential stability; LP_B: filtering of integrated modules; LP_C: stabilizing ensemble feedback; LP_D: strong filtering in replication-like threshold regimes.	$\mathrm{IM}_F, \mathrm{IM}_K$	$\mathrm{CD}_{DI}, \mathrm{CD}_M, \mathrm{CD}_D$
DT_{RES} — Resource	LP_A: feasible formation under supply constraints; LP_B: stable throughput for coupling; LP_C: sustains non-equilibrium persistence; LP_D: critical sustainment threshold.	$\mathrm{IM}_I, \mathrm{IM}_R, \mathrm{IM}_C$	$\mathrm{CD}_U, \mathrm{CD}_H$
DT_{CON} — Constraint	LP_A: reduces combinatorial explosion; LP_B: enforces compatibility; LP_C: defines attractor basins; LP_D: determines critical boundaries.	IM_K	$\mathrm{CD}_H, \mathrm{CD}_{DI}, \mathrm{CD}_U$
DT_{TRN} — Transport	LP_A: delivery and concentration; LP_B: exchange among modules; LP_C: distributed coupling and flux balance; LP_D: bottleneck or gating condition for transition.	$\mathrm{IM}_R, \mathrm{IM}_I, \mathrm{IM}_F, \mathrm{IM}_C$	$\mathrm{CD}_H, \mathrm{CD}_U, \mathrm{CD}_{DI}$
DT_{ENV} — Environment	LP_A: admissibility window for initial formation; LP_B: stable operating envelope; LP_C: control of dissipation channels; LP_D: shifts threshold landscapes.	$\mathrm{IM}_K, \mathrm{IM}_I$	$\mathrm{CD}_{DC}, \mathrm{CD}_{DI}, \mathrm{CD}_U$

Dependency type	Phase roles	Typical interaction modes	Typical causal directions

Table 21: Phase roles and typical causal profiles of dependency types.

Activation conditions, failure modes, and weighting guidance

The final table summarizes practical guidance for using dependency types in causal specification. It includes activation conditions, common failure modes of misclassification, recommended necessity defaults, indicative weight ranges, and epistemic confidence guidance.

Dependency type	Activation conditions	Failure modes / confusions	Necessity	Weight	Confidence
DT_{SUB}	Carrier or building-block class is explicitly required; dependence is describable without control loops; substitution reasoning applies.	Confusing substrate with resource, boundary, or catalyst.	necessary	0.20–0.60	High when ablation/substitution evidence exists; moderate when inferred from constitutive chemistry or physics.
DT_{BND}	Selective exchange or containment is essential; interface geometry matters; perturbing permeability is a meaningful causal probe.	Confusing boundary with compartment, generic constraint, or environment.	necessary	0.15–0.50	High when interface manipulation changes regime; moderate when inferred from compartment necessity alone.
DT_{TMP}	Phenomenon implies copying, directed assembly, or pattern inheritance; dependence is configuration-sensitive; fidelity metrics are meaningful.	Confusing template with information, catalyst, or generic constraint.	necessary	0.10–0.45	High when conditional pattern mapping is measured; moderate when templating is inferred from complementarity.
DT_{CAT}	Phenomenon is rate-limited or barrier-limited without the parent; catalytic turnover reasoning applies; intervention is meaningful.	Confusing catalyst with scaffold, template, regulation, or substrate.	enabling	0.10–0.50	High with kinetic assays or inhibition data; moderate with plausible motifs but limited measurement.
DT_{REG}	Controlled balance of rates or concentrations is essential; control loops or gating logic are describable; evidence is modulation-based.	Confusing regulation with static constraint, selection, or transport.	contextual	0.05–0.35	High when modulation experiments show regime control; moderate when inferred from stability requirements.
DT_{GRD}	Phenomenon is directional and non-equilibrium; gradient magnitude is measurable; collapse or restoration interventions are meaningful.	Confusing gradient with resource, transport, or transient fluctuation.	enabling	0.05–0.35	High when gradient manipulation changes outcome; moderate when inferred from thermodynamic coupling.
DT_{CMP}	Localized networks, retention, or protection from dilution are essential; domain-based description is necessary; confinement affects rate or stability.	Confusing compartment with boundary, environment, or generic spatial heterogeneity.	enabling	0.05–0.40	High when compartment/no-compartment contrasts exist; moderate when inferred from known confinement benefits.
DT_{INF}	Outcome variability depends on encoded state rather than concentration; parent is modelable as pattern-carrying variable; informational interventions are meaningful.	Confusing information with template, pattern, or mere correlation.	contextual	0.05–0.35	High when state-dependent mapping is demonstrated; moderate when inferred from code-function relations.

Dependency type	Activation conditions	Failure modes / confusions	Necessity	Weight	Confidence
DT_{SEL}	A variant ensemble exists; persistence or amplification mechanism exists; outcome is measurable as frequency shift.	Confusing selection with regulation, constraint, or single-system control.	optional	0.05–0.25	High when frequency shifts and differential rates are measured; moderate when only conceptual selection arguments exist.
DT_{RES}	Phenomenon depends on input throughput and depletion sensitivity; supply interventions are meaningful; role concerns fueling rather than constitutive identity.	Confusing resource with substrate, gradient, or static constraint.	enabling	0.05–0.35	High with balance or limitation data; moderate when inferred from known energetic or material requirements.
DT_{CON}	Phenomenon is defined by admissibility boundaries; constraint statements are formulable; manipulating constraints predictably changes viability.	Confusing constraint with regulation, boundary, or generic correlation.	necessary	0.10–0.45	High when operationalized and validated; moderate when conceptual but not yet measured.
DT_{TRN}	Phenomenon is flux-limited or requires exchange coupling; blocking transport is meaningful; time-scale alignment matters.	Confusing transport with gradient, boundary, or mere compartmental localization.	enabling	0.05–0.35	High when flux blocking or modulation changes outcome; moderate when transport is inferred but unmeasured.
DT_{ENV}	A specific external setting or parameter range is required; environmental perturbations are interpretable as exogenous forcing; dependence is irreducible to internal factors.	Confusing environment with boundary, resource, or internally maintained set-points.	contextual	0.05–0.30	High when regime maps against external variables exist; moderate when dependence is inferred from plausibility.

Table 22: Activation conditions, failure modes, and weighting guidance for dependency types.

Each parent relation is assigned a dependency type that maps the relation to exactly one causal category.

Theorem of non-substitutability of dependency types. Distinct dependency types are not, in general, interchangeable in the constitution of an arising:

$$D_i \neq D_j \Rightarrow \neg\mathrm{Substitutable}(D_i, D_j). \tag{239}$$

This theorem states that different dependency types perform ontologically distinct causal functions. A missing substrate, boundary, template, catalyst, regulator, gradient, or other dependency category cannot in general be replaced by another category without altering the causal structure of the arising.

Parent relations may also be stratified by ontological role as *arising*, *maintenance*, or *modulation*. This distinction does not alter category coverage itself, but it prepares a layered extension in which completeness can be evaluated separately for first admissibility, persistence, and regime modulation of the child arising.

Normative requirements for an arising

Each arising declares a normative requirement profile specifying which causal categories must be covered. These requirements depend on level, phase, and type role, but are evaluated according to a uniform formal procedure.

Requirements are partitioned into mandatory, recommended, and optional sets. Mandatory requirements define structural admissibility. Recommended requirements refine robustness. Optional requirements capture higher-order stabilization.

Quantification of category coverage

For each causal category r, effective coverage is computed as the sum of weighted and confidence-modulated contributions of all parents assigned to that category:

$$\mathrm{Cover}_N(r) = \sum_{\substack{p \in P(N) \\ \mathrm{type}(p)=r}} \big(w_p \cdot c_p\big). \tag{240}$$

Coverage expresses how strongly a given category is causally represented in the description of the arising.

Law of minimal causal coverage. An arising is structurally admissible only if all mandatory causal-condition categories required for it are covered:

$$\forall r \in R_{\mathrm{must}}, \qquad \mathrm{Cover}_N(r) \geq T_{\mathrm{must}}. \tag{241}$$

This law states that admissibility depends on satisfaction of all mandatory causal categories. Missing mandatory coverage blocks structural completeness.

Thresholds and saturation

To prevent purely symbolic or negligible coverage from counting as satisfaction, minimal thresholds are introduced. A category is considered satisfied only if its coverage exceeds a predefined threshold.

Let T_{must} denote the threshold for mandatory categories and T_{should} the threshold for recommended categories. Define the saturation function by

$$\mathrm{Sat}(x;T) = \min\left(1, \frac{x}{T}\right). \tag{242}$$

Structural Completeness Index (SCI)

Structural completeness quantifies the extent to which the causal structure satisfies the normative requirements of the arising.

Define the mandatory score by

$$\mathrm{MustScore} = \frac{1}{|R_{\mathrm{must}}|} \sum_{r \in R_{\mathrm{must}}} \mathrm{Sat}\big(\mathrm{Cover}_N(r); T_{\mathrm{must}}\big), \tag{243}$$

and the recommended score by

$$\mathrm{ShouldScore} = \frac{1}{|R_{\mathrm{should}}|} \sum_{r \in R_{\mathrm{should}}} \mathrm{Sat}\big(\mathrm{Cover}_N(r); T_{\mathrm{should}}\big). \tag{244}$$

The Structural Completeness Index is then defined as a weighted combination prioritizing mandatory conditions:

$$\mathrm{SCI}_N = 0.8 \cdot \mathrm{MustScore} + 0.2 \cdot \mathrm{ShouldScore}. \tag{245}$$

Theorem of structural underdetermination. If at least one mandatory causal category fails to reach threshold, the arising remains structurally underdetermined:

$$\exists r \in R_{\mathrm{must}} : \mathrm{Cover}_N(r) < T_{\mathrm{must}} \Rightarrow \mathrm{Underdetermined}(N). \tag{246}$$

This theorem states that structural incompleteness is not a vague absence of detail, but an explicit failure of required causal coverage.

Epistemic Completeness Index (ECI)

Epistemic completeness evaluates the degree to which the causal claims are supported by evidence and scientific validation.

Three components are taken into account: the scientific status of the arising, the evidence supporting the node itself, and the evidence supporting the individual causal relations. Define

$$\mathrm{ECI}_N = 0.4 \cdot \mathrm{EdgeEvidence} + 0.4 \cdot \mathrm{NodeEvidence} + 0.2 \cdot \mathrm{StatusScore}. \tag{247}$$

Causal Completeness Index (CCI)

Overall causal completeness is defined as a convex combination of structural and epistemic completeness:

$$\mathrm{CCI}_N = 0.6 \cdot \mathrm{SCI}_N + 0.4 \cdot \mathrm{ECI}_N. \tag{248}$$

The index CCI provides a single scalar measure suitable for comparison, visualization, and filtering, while preserving access to its structural and epistemic components.

Classification of arisings by causal completeness

On the basis of SCI and ECI, arisings may be partitioned into distinct operational classes:

- causally complete and established;
- structurally complete but insufficiently evidenced;
- epistemically supported but structurally incomplete;
- exploratory or hypothesized;
- methodological placeholder.

Detection of missing causal conditions

An arising is diagnosed as causally incomplete when at least one mandatory category fails to reach its threshold. Define the set of missing categories by

$$\mathrm{Missing}_N = \{\, r \in R_{\mathrm{must}} \mid \mathrm{Cover}_N(r) < T_{\mathrm{must}} \,\}. \tag{249}$$

Law of explicit incompleteness. Structural incompleteness can be diagnosed as a non-empty set of mandatory causal categories whose coverage remains below threshold:

$$\mathrm{Missing}_N \neq \varnothing \Rightarrow \neg\mathrm{StructurallyComplete}(N). \tag{250}$$

This law states that incompleteness is explicitly representable within graph structure through the identified set of missing mandatory causal conditions.

Each missing category defines a concrete task: identification or introduction of a parent capable of contributing the required causal mass.

Ontological significance

Causal completeness does not assert truth or finality. It asserts structural closure relative to declared requirements. The index therefore transforms underdetermination from a subjective judgment into an explicit and navigable property of the graph.

9 Lagrangian admissibility and the calculation of transitions

A transition becomes calculable when the configuration can be expressed not only as a set of causal relations, but as a structured balance between transition-supporting and transition-limiting contributions. Within the present framework, a Lagrangian is treated as a compact operator of admissibility: it does not merely describe motion, but formalizes which transitions are structurally selectable within a given causal configuration.

9.1 Why a Lagrangian description is needed

The ontology of causal arisings already defines what a configuration is: a structured arrangement of nodes, parent relations, interaction modes, causal directions, ontological roles, and level–phase embedding. This makes it possible to classify transitions, compare heterogeneous phenomena, and reconstruct causal admissibility in graph form. Classification alone, however, does not yet provide a rule of calculation. A graph can show which relations are present, but it does not by itself quantify how strongly a configuration supports one transition rather than another, when a transition remains latent, or when it crosses into active realization.

A further layer is therefore required: a formal expression that compresses the dynamic balance internal to a configuration. Such a layer must register at least four things simultaneously. First, it must represent the contributions that actively drive transformation. Second, it must represent the contributions that constrain, stabilize, or resist transformation. Third, it must preserve the role of coupling terms through which multiple components act jointly rather than independently. Fourth, it must allow threshold-sensitive selection, so that not every structurally imaginable continuation is treated as equally admissible.

The Lagrangian is introduced here as the minimal formal schema satisfying these requirements. It is not restricted to classical mechanics or field theory. Within the present framework, the Lagrangian is generalized as an admissibility functional over a causal configuration. Its task is to express, in compact form, how a system balances realized transformation against configurational cost. The Lagrangian is therefore not read as a discipline-specific artifact of physics, but as a general operator for calculating transition-selection under structured constraints.

In mechanics, this balance is written as kinetic minus potential energy. That familiar form should be understood as one special case of a more general principle. Every organized system contains contributions that open transition pathways and contributions that price, channel, or restrict those pathways. The generalized Lagrangian formalism captures this balance by assigning positive, negative, and coupling terms to the active configuration. A transition then becomes calculable as a trajectory, regime shift, or threshold event selected under that balance rather than inferred from descriptive narrative alone.

The need for such a formal layer is already visible in the Level-0 reconstruction, where admissible oscillatory persistence is filtered through a variational admissibility functional rather than assumed as a ready-made physical law. There, the Lagrangian is treated explicitly as a structural selection device expressing the conditions under which oscillatory configurations remain admissible rather than collapsing as unstable deviations. This provides the correct logical point of departure for a broader theory of transitions: the variational form is not an optional mathematical ornament, but the formal mechanism through which admissibility is operationalized.

The same requirement follows from the internal ontology of SOMA. Interaction modes already distinguish cooperation, constraint, feedback, initiation, and realization as different operators of causal contribution. Constraint narrows state space and is already associated with admissibility conditions, inequality structure, and effective multiplier-like formulations. A generalized Lagrangian layer is therefore not external to the theory. It is the natural quantitative continuation of the existing ontology: once configurations are defined qualitatively, the next step is to express how their active operators are weighted, balanced, and selected in transition space.

A configuration therefore possesses two complementary descriptions. The first is topological–ontological: which nodes, roles, relations, and directions are present. The second is variational–operational: how these contributions are composed into a transition-selecting balance. The first description tells us what is structurally available. The second tells us which admissible continuation is dynamically favored, suppressed, stabilized, or threshold-enabled. Without the first, the Lagrangian floats without ontology. Without the second, the graph remains descriptive but not computational.

For this reason, the theory introduces a generalized transition Lagrangian as the formal bridge between configurational description and transition calculation. Its general role is to map a coded configuration into a transition functional whose stationary, threshold, or bifurcating structure determines whether a new arising remains latent, becomes active, or stabilizes as a persistent regime. In compact form, the task may be stated as follows:

$$\mathcal{L}_{tr} = \mathcal{D} - \mathcal{C} + \mathcal{K}. \tag{251}$$

Here $\mathcal{D}$ denotes the set of transition-driving contributions, $\mathcal{C}$ denotes the set of configurational costs and constraints, and $\mathcal{K}$

denotes coupling terms that cannot be reduced to either side alone. This expression is intentionally abstract at the present stage. Its purpose in the opening subsection is only to establish why such a form is necessary. Subsequent subsections will specify how $\mathcal{D}$, $\mathcal{C}$, and $\mathcal{K}$ are constructed from interaction modes, causal directions, ontological roles, carrier types, and threshold parameters.

The decisive point is methodological. A transition should not be calculated from labels alone, and it should not be reduced to a single universal physical energy metaphor. It should be calculated from the effective balance internal to the configuration itself. In some domains, the driving term will be kinetic- or throughput-like. In others, it will be catalytic throughput, informational gain, coordination yield, regulatory closure, or collective amplification. Likewise, the constraining term may correspond to potential energy, activation barriers, boundary cost, control burden, incoherence, instability, dissipation, conflict, or risk of structural disintegration. The Lagrangian formalism is introduced precisely because it can host all such balances without collapsing them into a single substance-like ontology.

This subsection therefore establishes the first principle of transition calculation: every arising that is to be computed rather than merely described must be represented by a balance functional over its active configuration. The remainder of the theory refines this principle into explicit construction rules for effective Lagrangians across different levels of organization.

9.2 General form of the transition Lagrangian

Once a configuration has been specified in ontological form, the next task is to construct the effective balance by which its admissible transitions can be evaluated. This requires a generalized transition Lagrangian. The purpose of this Lagrangian is not to reproduce the full mathematical machinery of classical or relativistic field theory at every level of organization. Its purpose is more general and more precise: it provides a compact form in which transition-supporting, transition-limiting, coupling, and threshold-sensitive contributions are brought into a single calculable expression.

The generalized transition Lagrangian is therefore defined as an effective admissibility functional over a configuration rather than as a universal law of nature. For a given phenomenon p, it operates over the active variables, carriers, and relations selected by the configurational code $\Gamma(p)$ and by its edge-level motifs $\kappa(r)$. In this sense, it is always local to a configuration class. It does not exist independently of the ontology. It is derived from the configuration as a compressed operational form of its transition logic.

The minimal generalized form is

$$\mathcal{L}_{tr}(p) = \mathcal{D}(p) - \mathcal{C}(p) + \mathcal{K}(p) + \mathcal{F}(p) + \mathcal{I}(p) + \mathcal{R}(p) - \mathcal{S}(p). \tag{252}$$

Each term expresses a distinct structural contribution. $\mathcal{D}(p)$ is the driving term: it collects those contributions that open or amplify transition pathways. $\mathcal{C}(p)$ is the configurational cost term: it collects those contributions that restrict, price, or resist transition. $\mathcal{K}(p)$ is the coupling term: it registers cooperative or cross-component effects that cannot be reduced to isolated additive contributions. $\mathcal{F}(p)$ is the feedback term: it captures closed-loop reinforcement or stabilization across the active configuration. $\mathcal{I}(p)$ is the initiation term: it represents threshold-crossing triggers that convert latent permissiveness into active unfolding. $\mathcal{R}(p)$ is the realization term: it maps structurally admissible and activated configurations into fixed or observable effect structure. $\mathcal{S}(p)$ is the suppression, or dissipation, term: it collects losses, incoherence, conflict, drift, uncontrolled dispersion, or destructive leakage that reduce persistence of the transition regime.

This decomposition is not arbitrary. It follows directly from the ontology of interaction modes already defined in the theory. Cooperation, constraint, feedback, initiation, and realization are treated there as the five fundamental operators of causal configuration, each with its own activation conditions, failure modes, and mathematical role. The generalized Lagrangian simply introduces a formal layer in which these operators become weighted contributions to transition selection. What is classified qualitatively in the interaction-mode ontology is written here quantitatively as part of an effective balance.

Within this framework, the classical form

$$L = T - V \tag{253}$$

appears as only one special case. There, T is a driving term associated with realized motion and V is a configurational cost landscape associated with positional dependence and restoring structure. The generalized form retains the same architecture while detaching it from a narrowly mechanical interpretation. At any level, a transition Lagrangian is constructed by identifying what in the configuration plays the role of transition drive and what plays the role of transition cost.

The general rule is as follows. A driving term is any contribution that increases the system's capacity to traverse a transition path, amplify an admissible mode, sustain throughput, or build closure. A cost term is any contribution that narrows the state space, imposes barrier structure, increases maintenance burden, raises incompatibility, or destabilizes persistence. A coupling term is any contribution whose effect appears only through coordinated participation of multiple carriers or substructures. A feedback term is any contribution whose magnitude depends on loop closure across the configuration. An initiation term is any

threshold-sensitive local contribution that activates a path already structurally permitted but not yet realized. A realization term is any contribution through which the configuration begins to express a stable effect, output, or newly fixed regime. A suppression term is any contribution that dissipates, fragments, or cancels transition-supporting organization.

In compact notation, the same structure can be written as a weighted sum over contributions extracted from the configuration:

$$\mathcal{L}_{tr}(p) = \sum_i \alpha_i D_i - \sum_j \beta_j C_j + \sum_k \gamma_k K_k + \sum_m \delta_m F_m + \sum_n \epsilon_n I_n + \sum_q \zeta_q R_q - \sum_u \eta_u S_u. \tag{254}$$

Here D_i, C_j, K_k, F_m, I_n, R_q, and S_u are effective terms derived from the active motif of the phenomenon, while the coefficients $\alpha_i, \beta_j, \gamma_k, \delta_m, \epsilon_n, \zeta_q, \eta_u$ encode relative strength, contextual relevance, carrier scaling, and regime sensitivity. These coefficients need not be fixed once and for all across domains. Their interpretation depends on the level and on the ontological nature of the counted carriers.

At this point, it becomes possible to specify more precisely what the analogues of kinetic and potential contributions are outside physics. The driving side of the Lagrangian is the domain-specific form of realized or realizable transition throughput. The cost side is the domain-specific form of configurational burden. In chemistry, the driving side may be reaction throughput, catalytic accessibility, or productive channel occupancy, whereas the cost side may be activation barriers, steric exclusion, off-path instability, and reaction-channel incompatibility. In cell biology, the driving side may be metabolic throughput, signaling propagation, differentiation advance, or repair progression, whereas the cost side may be membrane instability, genomic damage burden, stress load, viability risk, or boundary-maintenance demand. In neural systems, the driving side may be integration, propagation coherence, plastic update efficiency, or decision convergence, whereas the cost side may be dispersion, synchronization failure, metabolic burden, noise amplification, or unstable gain. In social systems, the driving side may be coordination throughput, cooperation yield, information aggregation, or institution-building capacity, whereas the cost side may be friction, conflict, mistrust, enforcement burden, and fragmentation of shared expectation structure.

The generalized Lagrangian therefore preserves the essential logic of subtraction without forcing all domains into literal energetic units. What is subtracted is not always potential energy. What is subtracted is the structured burden that makes transition more difficult, more costly, more unstable, or less admissible. This is the crucial point. A transition becomes calculable when the system-specific form of structural resistance is represented explicitly rather than hidden in descriptive language.

The form of the Lagrangian also depends on the phase of the level. In differentiation, initiation and minimal constraint terms often dominate because the problem is first admissible separation of components. In integration, coupling and compatibility terms increase in importance because the problem is stable assembly. In self-organization, feedback and regulation terms become central because the problem is persistence under internal circulation and perturbation. In threshold transition, realization and suppression terms become decisive because the system must cross from latent structural potency into a new stabilized regime. This phase dependence follows directly from the level-phase ontology already formalized in the theory.

The generalized form must therefore always be read as phase-sensitive. A phenomenon is not assigned one eternal Lagrangian independent of its developmental position. Instead, the active weight of each term depends on the phase in which the transition is being evaluated. This allows the same configurational family to exhibit different transition logic at different points in its organizational development.

A further refinement is required for quantitative causality. Many causal relations are not merely present or absent, but become active only when a sufficient number of carriers participates in the relevant motif. The theory already formalizes this through carrier groups, carrier types, and threshold parameters $N_{\min}$, N_{crit}, and N_{sat}, which determine when an edge is inactive, linear, threshold-transformative, or saturated. The generalized transition Lagrangian must therefore be written not only over state-like variables, but also over carrier-dependent activation structure. In general form,

$$\mathcal{L}_{tr}(p; N_x) = \mathcal{L}_{base}(p) + \mathcal{L}_{act}(N_x). \tag{255}$$

Here $\mathcal{L}_{base}(p)$ denotes the configurational balance independent of explicit carrier count, while $\mathcal{L}_{act}(N_x)$ denotes the carrier-dependent activation correction. This means that a configuration may be structurally well formed and yet remain subcritical, because the relevant carriers have not crossed the required activation or closure threshold. In this way, the generalized Lagrangian remains consistent with the distinction between structural potency and realized arising.

The result of this subsection may be stated as follows. A transition Lagrangian is an effective functional that maps a coded causal configuration into a weighted balance of transition-driving, transition-limiting, coupling, feedback, initiation, realization, and suppression terms, optionally modulated by carrier-dependent thresholds. In compact form,

$$\mathbb{L}\big(\Gamma(p)\big) \mapsto \mathcal{L}_{tr}(p). \tag{256}$$

The next subsection specifies how the terms of this functional are constructed from the ontology itself: from interaction modes,

causal directions, ontological roles, dependency types, and carrier structure. Only once that mapping is made explicit can transition calculation proceed in a systematic and reproducible way.

9.3 Mapping causal configurations into Lagrangian terms

Once the generalized transition Lagrangian has been introduced, the central methodological question becomes explicit: how exactly is a coded causal configuration translated into its Lagrangian terms? The answer cannot remain at the level of metaphor. If the theory is to support actual calculation, then each component of the configurational code must have a principled contribution rule. The transition Lagrangian must therefore be derived from the ontology itself rather than attached afterward as an interpretive overlay.

The required starting point is already present in the formal schema of an arising. Each node is defined by level, phase, and primary type role. Each parent relation is defined by causal directions, interaction modes, weight, necessity, dependency type, ontological role, and optional quantization parameters including carrier group, carrier type, and thresholds such as $N_{\min}$, N_{crit}, and N_{sat}. This means that the causal graph is not a bare topology. It already contains the information needed to construct an effective balance functional, provided that translation rules are specified for each ontological field.

The mapping proceeds in layers. First, interaction modes determine the operator class of a term: whether it contributes as cooperation, constraint, feedback, initiation, or realization. Second, causal directions determine the orientation of the term: whether it acts as upward construction, downward constraint, horizontal coupling, diagonal loop closure, decoupled effective control, or metacausal admissibility framing. Third, ontological role determines whether a relation contributes primarily to the birth of the child arising, to its maintenance, or to its modulation. Fourth, dependency type determines the specific semantic class of the contribution: substrate, boundary, template, catalyst, regulation, gradient, information, selection, resource, transport, environment, and so forth. Fifth, carrier group and carrier type determine how the contribution scales quantitatively and whether threshold logic must be attached. Finally, level and phase determine the global weighting context within which all such terms are interpreted.

In compact form, for a phenomenon p with parent relations $r \in \text{Parents}(p)$, the effective transition Lagrangian is constructed as an aggregate over edge-level terms:

$$\mathcal{L}_{tr}(p) = \sum_{r \in \text{Parents}(p)} \mathcal{T}(r \to p). \tag{257}$$

Each edge-term $\mathcal{T}(r \to p)$ is itself a composed contribution:

$$\mathcal{T}(r \to p) = w_r \cdot \Omega_{\text{phase}}(p) \cdot \Omega_{\text{level}}(p) \cdot \Omega_{\text{role}}(r) \cdot \Omega_{\text{dep}}(r) \cdot \Omega_{\text{dir}}(r) \cdot \Omega_{\text{mode}}(r) \cdot \Omega_{\text{quant}}(r). \tag{258}$$

Here w_r is the explicit relation weight stored in the parent record. $\Omega_{\text{phase}}(p)$ is the phase-sensitive weighting envelope for the child phenomenon. $\Omega_{\text{level}}(p)$ is the level-sensitive scaling envelope. $\Omega_{\text{role}}(r)$ maps ontological role into arising, maintenance, or modulation coefficients. $\Omega_{\text{dep}}(r)$ maps dependency type into a semantic contribution class. $\Omega_{\text{dir}}(r)$ maps causal direction into orientation-sensitive term structure. $\Omega_{\text{mode}}(r)$ maps interaction mode into operator class. $\Omega_{\text{quant}}(r)$ maps carrier-dependent threshold activation into multiplicative or switching structure. The exact analytic form may vary by domain, but the decomposition itself remains fixed.

The first and most basic mapping is from interaction modes to Lagrangian operator classes. This follows directly from the formal ontology of interaction modes, where cooperation is defined as local mutual enhancement, constraint as state-space narrowing, feedback as closed-loop influence, initiation as threshold-triggered activation, and realization as fixation of an observable effect. These definitions are already mathematical in orientation and can therefore be translated into effective Lagrangian terms without introducing an alien formalism.

The mapping is given by the following rule set:

$$\text{IM}_C \mapsto +\mathcal{K}_{\text{coop}}, \tag{259}$$

$$\text{IM}_K \mapsto -\mathcal{C}_{\text{constr}}, \tag{260}$$

$$\text{IM}_F \mapsto +\mathcal{F}_{\text{loop}} \quad \text{or} \quad -\mathcal{F}_{\text{stab}}, \tag{261}$$

$$\text{IM}_I \mapsto +\mathcal{I}_{\text{thr}}, \tag{262}$$

$$\text{IM}_R \mapsto +\mathcal{R}_{\text{fix}}. \tag{263}$$

Cooperation contributes positively because it increases compatibility, mutual reinforcement, clustering, and local assembly potential. Constraint contributes negatively because it narrows admissible state space and prices incompatible trajectories.

Feedback contributes with variable sign because positive feedback may amplify and stabilize regime entry, whereas negative feedback may damp, retain, or regulate a regime. Initiation contributes as a threshold-sensitive trigger term. Realization contributes as a fixation term mapping permissive structure into observable or stabilized effect.

This yields the first direct translation from qualitative ontology into effective calculus. For a relation r, the mode contribution can be written as

$$\Omega_{\text{mode}}(r) = \sum_{m \in IM(r)} \sigma_m \, T_m, \tag{264}$$

where σ_m is the sign and T_m is the analytic form associated with mode m. A simple default assignment is

$$\sigma_{\text{IM}_C} = +1, \qquad \sigma_{\text{IM}_K} = -1, \qquad \sigma_{\text{IM}_I} = +1, \qquad \sigma_{\text{IM}_R} = +1, \tag{265}$$

while σ_{IM_F} remains context-dependent and is determined by loop polarity and phase.

The second mapping is from causal directions to orientation structure. Causal directions do not merely label where influence comes from; they determine how the contribution enters the effective balance. Upward causation contributes constructive terms from lower-level carriers into higher-level organization. Downward causation contributes constraint or control terms from macrostate to microdynamics. Horizontal causation contributes peer-coupling terms. Diagonal causation contributes loop-coupled multiscale terms. Decoupled causation contributes effective coarse-grained terms that are insensitive to explicit micro-resolution. Metacausation contributes admissibility filters at the rule level rather than ordinary dynamical terms.

The directional mapping may therefore be written as

$$\text{CD}_U \mapsto \mathcal{D}_{\text{up}}, \tag{266}$$

$$\text{CD}_D \mapsto \mathcal{C}_{\text{down}}, \tag{267}$$

$$\text{CD}_H \mapsto \mathcal{K}_{\text{peer}}, \tag{268}$$

$$\text{CD}_{DI} \mapsto \mathcal{F}_{\text{cross}}, \tag{269}$$

$$\text{CD}_{DC} \mapsto \mathcal{E}_{\text{eff}}, \tag{270}$$

$$\text{CD}_M \mapsto \mathcal{M}_{\text{meta}}. \tag{271}$$

These should not be read as rigid one-to-one identities, but as default placement rules. Upward terms usually contribute to constructive and driving structure. Downward terms usually contribute to control and admissibility restriction. Horizontal terms usually contribute to coupling and local coordination. Diagonal terms usually contribute to closed-loop, multiscale, or regime-sensitive structure. Decoupled terms usually contribute as effective parameters. Metacausal terms do not directly change state variables but alter which terms are admissible in the first place.

The third mapping is from ontological role to lifecycle weighting. The schema explicitly distinguishes whether a parent relation participates in the child arising as *arising*, *maintenance*, or *modulation*. This distinction is critical because not every parent contributes in the same way to transition calculation. A relation may be indispensable for the birth of a phenomenon, yet only weakly involved in its later persistence. Another may be required for maintenance but not for first emergence. A third may modulate the regime without constituting it.

The default mapping is

$$OR = \textit{arising} \mapsto \lambda_A, \tag{272}$$

$$OR = \textit{maintenance} \mapsto \lambda_M, \tag{273}$$

$$OR = \textit{modulation} \mapsto \lambda_{\text{Mod}}, \tag{274}$$

with

$$\lambda_A > \lambda_M > \lambda_{\text{Mod}} \tag{275}$$

during threshold-entry analysis, while in persistence analysis the ordering may shift toward λ_M. This allows the same configuration to be evaluated differently depending on whether the target question concerns origination, stabilization, or adaptive reconfiguration.

The fourth mapping is from dependency type to semantic term family. Dependency types define whether a parent contributes as substrate, boundary, template, catalyst, regulation, gradient, information, selection, resource, transport, or environment. It is here that the generalized Lagrangian acquires domain-specific content without surrendering formal uniformity. Substrate dependencies usually contribute constitutive or carrier-availability terms. Boundary dependencies usually contribute admissibility constraints and interface costs. Template dependencies contribute pattern-guiding or fidelity terms. Catalyst dependencies contribute throughput-enhancing or barrier-lowering terms. Regulation dependencies contribute feedback and control terms. Gradient dependencies contribute directional potential terms. Information dependencies contribute state-selection and encoded-guidance terms. Selection dependencies contribute filtering terms. Resource dependencies contribute availability and consumption terms.

Transport dependencies contribute flow-support terms. Environmental dependencies contribute contextual load and boundary conditions.

A compact semantic mapping may therefore be written as

$$\mathrm{DT}_{SUB} \mapsto \mathcal{B}_{\mathrm{car}}, \tag{276}$$

$$\mathrm{DT}_{BND} \mapsto -\mathcal{C}_{\mathrm{bnd}}, \tag{277}$$

$$\mathrm{DT}_{TMP} \mapsto \mathcal{G}_{\mathrm{tmpl}}, \tag{278}$$

$$\mathrm{DT}_{CAT} \mapsto +\mathcal{D}_{\mathrm{cat}}, \tag{279}$$

$$\mathrm{DT}_{REG} \mapsto \mathcal{F}_{\mathrm{reg}}, \tag{280}$$

$$\mathrm{DT}_{GRD} \mapsto \mathcal{P}_{\mathrm{grad}}, \tag{281}$$

$$\mathrm{DT}_{INF} \mapsto \mathcal{I}_{\mathrm{info}}, \tag{282}$$

$$\mathrm{DT}_{SEL} \mapsto -\mathcal{C}_{\mathrm{sel}}, \tag{283}$$

$$\mathrm{DT}_{RES} \mapsto \mathcal{A}_{\mathrm{res}} - \mathcal{S}_{\mathrm{dep}}, \tag{284}$$

$$\mathrm{DT}_{TRN} \mapsto \mathcal{J}_{\mathrm{flow}}, \tag{285}$$

$$\mathrm{DT}_{ENV} \mapsto \mathcal{E}_{\mathrm{ctx}}. \tag{286}$$

These mappings specify what kind of Lagrangian term a dependency contributes, but not yet its magnitude. Magnitude is supplied by explicit weight, necessity, carrier scaling, and phase-sensitive envelopes.

The fifth mapping is from necessity and explicit weight to relation amplitude. The schema already contains both Weight and Necessity for each parent relation. Weight gives the direct relative contribution. Necessity indicates whether the contribution is *necessary*, *enabling*, *contextual*, or *optional*. These can be combined multiplicatively:

$$a_r = w_r \cdot \nu_r, \tag{287}$$

with a default necessity scale such as

$$\nu_{\mathrm{necessary}} = 1.0, \qquad \nu_{\mathrm{enabling}} = 0.75, \qquad \nu_{\mathrm{contextual}} = 0.4, \qquad \nu_{\mathrm{optional}} = 0.2. \tag{288}$$

This rule preserves the explicit graph weighting while distinguishing constitutive from merely supportive factors.

The sixth mapping is from carrier structure and thresholds to activation form. Carrier groups and carrier types determine what kind of entity is being counted, while N_{min}, N_{crit}, and N_{sat} determine whether the relation is inactive, linear, threshold-transformative, or saturated. The effective activation factor may therefore be written as a piecewise function:

$$\Omega_{\mathrm{quant}}(r; N_x) = 0, \qquad N_x < N_{\mathrm{min}}, \tag{289}$$

$$\Omega_{\mathrm{quant}}(r; N_x) = \phi_{\mathrm{lin}}(N_x), \qquad N_{\mathrm{min}} \le N_x < N_{\mathrm{crit}}, \tag{290}$$

$$\Omega_{\mathrm{quant}}(r; N_x) = \phi_{\mathrm{thr}}(N_x), \qquad N_{\mathrm{crit}} \le N_x < N_{\mathrm{sat}}, \tag{291}$$

$$\Omega_{\mathrm{quant}}(r; N_x) = \phi_{\mathrm{sat}}(N_x), \qquad N_x \ge N_{\mathrm{sat}}. \tag{292}$$

The analytic form of ϕ depends on carrier ontology. For mode carriers it may correspond to closure logic. For body carriers it may correspond to low-N topological thresholds. For functional material carriers it may correspond to stoichiometric or collective transitions. For topological carriers it may correspond to cycle or connectivity closure. For informational carriers it may correspond to expressivity thresholds. For agent or population carriers it may correspond to coalition, norm, quorum, or market-formation thresholds. This preserves the ontological specificity of quantity rather than collapsing every threshold into raw counting.

The seventh mapping is phase-sensitive reweighting. Level phases are already defined as differentiation, integration, self-organization, and threshold transition, each with characteristic dominant mechanisms. This means that the same edge-term should not be interpreted identically in every phase. Initiation and early constraint matter more in differentiation. Cooperation and compatibility matter more in integration. Feedback and regulation matter more in self-organization. Realization and suppression balance matter more in threshold transition.

A simple phase envelope may be introduced as

$$\Omega_{\mathrm{phase}}(\mathrm{LP}_A) = (\chi_I, \chi_K, \chi_C, \chi_F, \chi_R), \tag{293}$$

$$\Omega_{\mathrm{phase}}(\mathrm{LP}_B) = (\chi_C, \chi_K, \chi_I, \chi_F, \chi_R), \tag{294}$$

$$\Omega_{\mathrm{phase}}(\mathrm{LP}_C) = (\chi_F, \chi_K, \chi_C, \chi_R, \chi_I), \tag{295}$$

$$\Omega_{\text{phase}}(\text{LP}_D) = (\chi_R, \chi_F, \chi_K, \chi_I, \chi_S), \tag{296}$$

where the ordering indicates relative dominance rather than fixed numerical values. In implementation, these envelopes can be transformed into actual coefficients per mode family. The transition Lagrangian thereby becomes phase-aware by construction.

Putting all mappings together, the default edge-term for a parent relation can be written explicitly as

$$\mathcal{T}(r \to p) = w_r \cdot \nu_r \cdot \Omega_{\text{phase}}(p) \cdot \Omega_{\text{quant}}(r; N_x) \cdot \Big[\sum_{m \in IM(r)} \sigma_m T_m \Big] \cdot \Big[\sum_{d \in CD(r)} \rho_d U_d \Big] \cdot \Big[\sum_{s \in DT(r)} \tau_s V_s \Big] \cdot \lambda_{OR(r)}. \tag{297}$$

Here T_m are mode operators, U_d are directional operators, V_s are dependency-class operators, ρ_d are directional coefficients, τ_s are dependency coefficients, and $\lambda_{OR(r)}$ is the ontological-role coefficient. The total transition Lagrangian is then obtained by summing all such edge-terms, optionally together with node-level regularization or phase-level global terms.

This construction gives the theory an explicit operational bridge from graph structure to calculable transition balance. It also explains why the existing schema was already close to a variational formalism even before this subsection was added. The schema does not merely list parents. It specifies how parents participate, in what direction, by what mode, under what dependency class, with what weight, in what lifecycle role, and under what threshold logic. That is already the information a generalized Lagrangian requires. What the present subsection adds is the formal rule by which that information becomes a transition functional.

One final methodological point is essential. The mapping defined here does not imply that every phenomenon will immediately admit a closed-form analytic Lagrangian. In many cases the result will be an effective symbolic functional, a simulation-ready weighted operator sum, or a numerically estimated admissibility surface rather than a neat textbook equation. This does not weaken the framework. On the contrary, it is precisely because the ontology is richer than any single physical domain that the resulting Lagrangian layer must remain flexible enough to appear as analytic, discrete, graph-dynamical, or hybrid depending on the configuration.

The next subsection can therefore move to the concrete semantic core of the matter: if the Lagrangian is built from transition-supporting and transition-limiting terms, then what exactly occupies the subtractive side in different domains of organization? That is the point at which the generalized form becomes fully interpretable across physics, chemistry, biology, neural systems, and social structures.

9.4 Domain-specific interpretation of subtractive structure

The generalized transition Lagrangian becomes genuinely useful only when the subtractive side of the balance is interpreted with domain precision. It is not sufficient to claim that every system contains a driving term and a constraining term. One must specify what exactly plays the role of configurational burden in different organizational regimes. In mechanics this burden is conventionally expressed through potential structure. Outside mechanics, however, the subtractive term must be reinterpreted in a manner faithful to the ontology of the relevant domain rather than forcing all systems into literal energy language.

The key principle is the following: the subtractive term of a transition Lagrangian is the structured cost of admissibility. It represents whatever, in a given configuration, narrows the accessible transition space, increases the burden of maintaining the transition path, penalizes instability, or blocks realization of a higher-order regime. In some domains this appears as explicit potential. In others it appears as activation barriers, viability burden, incoherence, coordination friction, informational fragmentation, or other forms of structured resistance. What remains invariant is not the substance of the term, but its role: it prices or limits transition.

This is fully consistent with the existing ontology of interaction modes and dependency types. Constraint already functions as the narrowing of admissible state space, and its mathematical role is associated with inequalities, admissible subspaces, and multiplier-like structure. The task here is to interpret that general logic of constraint in domain-specific form. The subtractive term is therefore not an optional metaphorical extension. It is the quantitative continuation of constraint, boundary, regulation, selection, and other limiting dependency classes as they appear in concrete domains.

Physics: potential landscape and configurational constraint

In physics, the subtractive structure has the clearest and historically earliest form. In classical mechanics it is expressed as potential energy, which defines the configurational landscape through which motion occurs. Potential is not merely a stored quantity. It encodes how the current position or configuration of a system restricts and shapes admissible trajectories.

Gravitational elevation, elastic deformation, electrostatic arrangement, field gradients, and boundary conditions all contribute to the structure subtracted from the driving side of motion.

At a more general physical level, what is subtracted is not only potential in the narrow textbook sense, but configurational constraint more broadly. The subtractive term includes restoring structure, geometric exclusion, field-induced admissibility, and localization cost. In the Level-0 reconstruction, this logic is already explicit: the variational admissibility functional acts as a structural filter, and the mass-like scalar term enters as an internal closure parameter distinguishing bounded oscillatory organization from scale-free propagation rather than functioning as a primitive physical substance. Even there, the subtractive side already serves as a closure and boundedness burden imposed on otherwise propagative variation.

The physical subtractive term may therefore be written in generalized form as

$$\mathcal{C}_{\text{phys}} = V_{\text{pot}} + C_{\text{geom}} + C_{\text{bnd}} + C_{\text{disp}}. \tag{298}$$

Here V_{pot} denotes the conventional potential landscape, C_{geom} denotes configurational or metric constraint, C_{bnd} denotes boundary-related admissibility cost, and C_{disp} denotes dispersion or loss of localization. The physical meaning of subtraction is thus the cost of occupying, traversing, or maintaining a configuration under the laws and constraints of the physical regime. A transition becomes admissible only if the driving contribution exceeds, balances, or is variationally selected against this structured burden.

Chemistry: activation barriers and reaction-channel exclusion

In chemistry, the subtractive structure is not primarily spatial position but reactional difficulty. A chemical system does not usually fail to transition because it lacks coordinates. It fails because the relevant reaction path is blocked, energetically unfavorable, geometrically excluded, sterically hindered, electronically mismatched, or kinetically inaccessible. The subtractive term in chemistry therefore corresponds to the burden imposed by the reaction landscape.

This includes activation barriers, transition-state difficulty, off-pathway instability, competing-channel exclusion, charge-distribution incompatibility, solvent mismatch, and steric obstruction. In enzymatic and catalytic contexts this becomes especially clear. Catalysis operates not by adding a substance-like motion to the system, but by lowering precisely those subtractive burdens that would otherwise prevent the desired transformation. That is why catalytic dependency belongs on the driving side, whereas activation barrier and pathway exclusion belong on the subtractive side.

The existing level structure of chemistry already reflects this logic. Molecular formation, ionic organization, resonance, hydrogen bonding, polarization, and crystallization all depend on interaction architectures whose admissibility is shaped by stable electromagnetic regime, electronic organization, and local compatibility constraints. At the biochemical level, protein folding, enzymatic catalysis, active sites, metabolic cycles, and template-guided reactions all rely on selective opening and closing of reaction channels rather than on bulk energetic availability alone.

The chemical subtractive term may therefore be expressed as

$$\mathcal{C}_{\text{chem}} = B_{\text{act}} + E_{\text{excl}} + I_{\text{steric}} + M_{\text{misfit}} + U_{\text{offpath}}. \tag{299}$$

Here B_{act} denotes activation-barrier structure, E_{excl} denotes reaction-channel exclusion, I_{steric} denotes steric incompatibility, M_{misfit} denotes electronic or geometric mismatch, and U_{offpath} denotes instability associated with undesired or off-pathway configurations. What is subtracted in chemistry is therefore the structured burden that prevents molecular transformation from taking the desired route. A chemically effective Lagrangian does not subtract generic potential alone; it subtracts the full landscape of reactional difficulty.

Biology: viability costs, maintenance burden, and boundary instability

In biological systems, the subtractive structure becomes broader and more deeply systemic. A living configuration is not defined only by transformation throughput. It must also preserve viability, boundary integrity, genomic stability, metabolic sufficiency, internal regulation, and controlled exchange with the environment. A biological transition therefore always occurs under a burden that is simultaneously functional and organizational. The system must not only change; it must remain viable, bounded, and coherent enough for that change to remain admissible.

This means that what is subtracted in biology is the total viability cost of the transition path. Every developmental shift, differentiation event, signaling regime, repair process, or tissue transformation consumes organizational capacity and risks

destabilizing the system. At the cellular level, this includes membrane instability, redox stress, osmotic drift, transport burden, replication-error load, and regulatory fragility. At multicellular levels, it includes morphogenetic mispatterning, repair burden, inflammatory overshoot, extracellular-matrix failure, and systemic-control destabilization. The theory already formalizes many of these through boundary, regulation, transport, repair, maintenance, and environmental dependencies in cellular and tissue-level structures.

This is precisely why a purely kinetic reading fails in biology. The driving side may be metabolic throughput, differentiation advance, regenerative rebuilding, vascular formation, or organ integration. But the subtractive side consists of the burdens that make such changes biologically costly or dangerous. A system may possess enough local activity to change and yet still fail to realize the transition because the viability burden exceeds what the configuration can sustain.

The biological subtractive term may therefore be written as

$$\mathcal{C}_{\text{bio}} = V_{\text{cost}} + M_{\text{burden}} + B_{\text{instab}} + R_{\text{frag}} + D_{\text{damage}}. \tag{300}$$

Here V_{cost} denotes viability cost, M_{burden} denotes maintenance burden, B_{instab} denotes boundary instability, R_{frag} denotes regulatory fragility, and D_{damage} denotes accumulated damage burden. In specific contexts additional terms may be needed, such as inflammatory cost, immune conflict, morphogenetic mismatch, or transport insufficiency. But the general biological logic remains stable: what is subtracted is the burden of remaining viable while transforming.

Neural systems: incoherence, dispersion, metabolic cost, and synchronization burden

In neural systems, the subtractive structure becomes a burden on integration, timing, and system-wide coherence. Neural organization is not merely a matter of firing or signal throughput. A neural transition must preserve or construct enough coordinated structure for information to propagate, synchronize, stabilize, and bias later states in a usable way. What blocks neural transitions is therefore not only a lack of signal, but also dispersion, timing mismatch, incoherence, unstable gain, noise amplification, metabolic overcost, or destructive desynchronization.

The level structure of neural phenomena already makes this visible. Formation of neural networks establishes a persistent connectivity substrate. Plasticity modifies it. Oscillatory organization provides temporal coordination. Decision processes depend on evidence accumulation and competitive stabilization. Perception depends on integrative coherence under noisy multimodal input. Affective and neuromodulatory states globally bias processing regimes. Synchronization and criticality govern large-scale dynamical readiness. In all these cases, the transition-supporting side cannot be understood without its subtractive counterpart: the costs of losing coherence, failing synchronization, dissipating signal, or exceeding sustainable metabolic and modulatory bounds.

This means that the neural subtractive term should be understood as the burden of maintaining organized coordination in a distributed excitable system. It includes incoherence across populations, dispersion of activity, unstable phase relations, metabolic overcost, and synchronization failure. It may also include plastic over-instability when the network loses retention structure through either excessive updating or excessive rigidity.

The neural subtractive term may therefore be written as

$$\mathcal{C}_{\text{neural}} = I_{\text{inc}} + D_{\text{disp}} + M_{\text{met}} + S_{\text{sync}} + N_{\text{noise}}. \tag{301}$$

Here I_{inc} denotes incoherence cost, D_{disp} denotes dispersion cost, M_{met} denotes metabolic burden, S_{sync} denotes synchronization burden or phase-instability cost, and N_{noise} denotes noise-driven degradation of signal integration. In some contexts one may also include threshold misalignment, gain instability, or maladaptive loop amplification. What is subtracted in neural systems is therefore the full burden of sustaining coherent distributed processing under biological constraints.

Social systems: coordination friction, conflict, enforcement cost, and informational fragmentation

In social systems, the subtractive structure becomes explicitly relational and collective. A social transition does not fail because a physical object cannot move or because a molecule cannot overcome a barrier. It fails because coordination is costly, trust is incomplete, information is fragmented, incentives diverge, conflict grows, enforcement burdens accumulate, and stable common expectations are not achieved. The subtractive side of a social Lagrangian is therefore the burden of maintaining collective admissibility.

This is already visible in the higher social levels of the ontology. Social organization of animals, stratification, recurrent social practices, resource coordination, collective decision-making, social norms, collective intelligence, market dynamics, and institutional patterns all arise through configurations whose persistence depends on distributed alignment under frictional conditions. None of these phenomena can be treated as pure cooperation. Every one of them is shaped by what must be subtracted: conflict, mistrust, coordination delay, signaling ambiguity, free-rider burden, enforcement cost, and fragmentation of the common informational field.

What is subtracted in a social domain is therefore the total frictional burden preventing a distributed population from behaving as a more coherent organized whole. Coordination friction includes delays, mismatch of local expectations, and action-alignment cost. Conflict cost includes active opposition and incompatible incentives. Enforcement cost includes the resources required to stabilize norms, roles, and compliance. Informational fragmentation includes loss of shared state visibility, communicative distortion, and inability to aggregate distributed signals into coherent collective judgment.

The social subtractive term may therefore be written as

$$\mathcal{C}_{\text{soc}} = F_{\text{coord}} + C_{\text{conf}} + E_{\text{enf}} + I_{\text{frag}} + T_{\text{trust}}. \tag{302}$$

Here F_{coord} denotes coordination friction, C_{conf} denotes conflict cost, E_{enf} denotes enforcement burden, I_{frag} denotes informational fragmentation, and T_{trust} denotes the cost imposed by insufficient trust or expectation instability. Depending on the domain, one may add resource asymmetry, institutional brittleness, or signaling overload. The central social principle nevertheless remains stable: what is subtracted is the frictional and fragmenting burden that blocks collective stabilization.

Unifying rule of interpretation

These domain differences do not dissolve the unity of the formalism. They clarify it. The generalized Lagrangian remains one and the same at the structural level because in every case the subtractive term plays an identical role: it expresses the burden of admissibility. What changes is the ontology of that burden. Physics subtracts potential landscape and configurational constraint. Chemistry subtracts reactional difficulty and exclusion. Biology subtracts viability burden and instability of maintenance. Neural systems subtract incoherence, dispersion, and metabolic–synchronization cost. Social systems subtract friction, conflict, enforcement load, and informational fragmentation.

This is precisely why the generalized Lagrangian is more useful than a naive extension of $T - V$. It preserves the formal architecture of transition selection while allowing each domain to specify its own structured cost field. The theory therefore avoids two opposite errors at once. It does not collapse all systems into literal physical energy. But neither does it abandon the variational insight that transitions are selected under a balance of enabling and limiting contributions.

The practical implication is straightforward. For any phenomenon p, construction of an effective transition Lagrangian requires identification not only of the driving terms, but also of the domain-correct structure of subtraction. The first question is never simply what moves the system forward. The equally important question is what exact form of structural burden the system must overcome, sustain, or variationally accommodate in order for the transition to become admissible. Only when both sides are specified does the configuration become calculable rather than merely classifiable.

The next step of the theory is therefore to show how such terms become transition criteria: how the effective balance yields latent permissiveness, subcritical activation, threshold crossing, regime stabilization, or saturation. In other words, once the subtractive structure has been domain-specified, the theory can proceed from term interpretation to actual transition conditions.

9.5 Transition criteria, thresholds, and regime change

A transition Lagrangian is not introduced merely to generate a trajectory. Its deeper task is to determine whether a qualitatively distinct arising remains structurally latent, becomes locally active, stabilizes into a persistent regime, or saturates without further ontological novelty. For that reason, the theory requires explicit transition criteria. These criteria must distinguish between admissible variation within an already given regime and actual crossing into a new regime of organization. The generalized Lagrangian must therefore be read together with threshold logic, closure conditions, and saturation structure rather than only with stationarity in the narrow variational sense.

The basic principle is the following. A calculable transition is not defined by movement alone, but by the relation between a configuration and its critical conditions of admissibility. A system may have a well-defined transition Lagrangian and still remain below the threshold at which a new arising becomes realizable. It may also cross a threshold and yet fail to stabilize if

closure is absent or if the resulting regime is too fragile to persist. Conversely, a system may enter a saturated state in which additional growth changes intensity or scale but no longer changes the ontological regime. A full theory of transitions must therefore formalize at least five distinct criteria: stationary admissibility, threshold crossing, bifurcation, closure, and saturation.

Stationary admissibility

The first criterion is stationary admissibility. This is the minimal variational condition under which a configuration selects a dynamically consistent path or state rather than dispersing into unstable alternatives. In the physical tradition, this appears as stationarity of the action. In the present framework, the same logic is generalized: a transition becomes admissible when the effective Lagrangian admits a locally stable or structurally self-consistent extremal under the active configuration.

Formally, for a phenomenon p with effective action $S_{tr}[p]$, stationary admissibility is expressed as

$$\delta S_{tr}[p] = 0. \tag{303}$$

This condition alone is not sufficient for regime change, but it is necessary. It states that the configuration supports a coherent continuation rather than immediate collapse. In the Level-0 reconstruction this role is explicit: the admissibility variational functional acts as a structural filter separating stable oscillatory configurations from unstable deviation, and stationarity functions there as a criterion of structural coherence rather than merely as a ready-made physical law. The same general rule applies across domains: if the effective action does not admit local stationarity, then the proposed transition is not even minimally admissible.

At this stage the system may still be subcritical. A stationary path may exist without producing a new arising. It may describe only a linear, reversible, or weakly perturbed continuation within the old regime. That is why stationary admissibility is the first filter rather than the final criterion.

Threshold crossing

The second criterion is threshold crossing. A new regime does not become active merely because a stationary path exists. It becomes active when the configuration crosses a critical boundary beyond which the old regime is no longer sufficient to contain the dynamics. In the ontology of the theory, this corresponds to the threshold-transition logic already embedded in the phase architecture, where LP_D marks the point at which a new organizational logic becomes admissible.

Threshold crossing can be expressed in two complementary ways. The first is variational: a control parameter enters a critical region at which the sign, curvature, or minimizer structure of the effective Lagrangian changes. The second is carrier-based: the active carriers cross one of the quantized thresholds associated with the relevant edge or motif. The schema already formalizes this through $N_{\min}$, N_{crit}, and N_{sat}, which determine whether a causal mechanism is undefined, linear, threshold-transformative, or saturated.

The most general threshold condition may therefore be written as

$$\Theta(p) = \Theta_{\mathrm{var}}(p) \cup \Theta_{\mathrm{quant}}(p), \tag{304}$$

where $\Theta_{\mathrm{var}}(p)$ is the set of variational critical conditions and $\Theta_{\mathrm{quant}}(p)$ is the set of carrier-count critical conditions. A transition becomes actively accessible when at least one necessary threshold set is crossed while the others remain compatible with stabilization.

The minimal carrier condition is

$$N_x \geq N_{\min}. \tag{305}$$

This does not yet imply regime change. It means only that the causal edge or motif is structurally defined. The first genuine regime-shift condition is

$$N_x \geq N_{\mathrm{crit}}. \tag{306}$$

This is the point at which numerical growth no longer merely scales an existing mechanism but changes its admissibility class. Depending on carrier ontology, this may mean closure, collective synchronization, catalytic-network viability, multicellular coordination, quorum behavior, or market-like aggregation. Threshold crossing is therefore the formal point at which a system ceases to behave as a weak continuation of the previous regime and begins to open a new regime of organization.

Bifurcation and regime branching

The third criterion is bifurcation. Threshold crossing does not always lead to one uniquely determined continuation. In many systems, crossing a critical parameter reorganizes admissible state space in such a way that multiple continuations become available, only some of which stabilize into the new regime. This is the formal meaning of bifurcation in the present framework: a configuration reaches a point at which one regime branches into several admissible continuations, each with different closure, stability, or persistence properties.

A bifurcation condition can be read off from the effective Lagrangian when the local curvature or Hessian structure changes in a way that alters the number or type of admissible extrema. In the simplest symbolic form,

$$\det H(\mathcal{L}_{tr}) = 0, \tag{307}$$

where $H(\mathcal{L}_{tr})$ is the Hessian, or effective second-variation operator, of the transition Lagrangian. At such a point, the old regime loses uniqueness or stability, and multiple branches become possible. In discrete or graph-dynamical settings the same logic may be expressed as a topology change in the attractor or admissibility landscape rather than as a smooth bifurcation in the classical differential sense.

Bifurcation is especially important because it distinguishes regime change from mere monotonic escalation. A system may cross a threshold and yet not produce one inevitable outcome. Instead, it enters a branch-sensitive region in which small contextual differences, initiation conditions, or feedback signatures determine which continuation is realized. This is why initiation and modulation terms remain important even after threshold crossing: they bias branch selection rather than merely activating the regime.

Closure as the criterion of ontological persistence

The fourth criterion is closure. This is the decisive criterion for distinguishing an active but transient transition from a genuinely stabilized arising. A system may cross a threshold and even enter a new branch, but if the resulting configuration does not achieve sufficient closure, the new regime will not persist as an ontological unit. Closure means that the resulting organization is supported by a sufficiently dense contour of internal relations such that its persistence is no longer reducible to a single external push or to a temporary coincidence.

This logic is already central to the theory. At Level 0, the argument that dyadic balance is insufficient and that triadic resonant closure constitutes the first irreducible contour of stable localized organization makes explicit that ontological novelty requires more than simple pairwise balance: it requires closed relational support. The same principle generalizes across levels. A new regime stabilizes only when its active relations form a closure structure capable of maintaining that regime under small perturbation.

Closure can be expressed abstractly as the condition that the realized regime generates sufficient internal support to reproduce or retain its own admissibility conditions. In compact symbolic form,

$$\mathcal{Q}_{cl}(p) \geq Q_{\mathrm{crit}}, \tag{308}$$

where $\mathcal{Q}_{cl}(p)$ is an effective closure measure and Q_{crit} is the minimum closure required for regime persistence. The exact form of $\mathcal{Q}_{cl}$ depends on domain. In physics it may correspond to resonance closure or bound-state retention. In chemistry it may correspond to cycle closure or catalytic self-support. In biology it may correspond to viability-maintaining feedback closure. In neural systems it may correspond to sustained integrative coherence. In social systems it may correspond to recurrent coordination supported by norms, roles, and feedback-stabilized expectations.

Closure is therefore the criterion by which a threshold-crossed branch ceases to be merely possible and becomes ontologically persistent. Without closure there may be activation, but not durable arising.

Saturation and the end of ontological novelty within a regime

The fifth criterion is saturation. Not every increase in carriers, intensity, scale, or throughput generates a new regime. Many systems enter a state in which further growth increases magnitude but no longer changes ontological type. The theory already encodes this by means of N_{sat}, the saturation threshold beyond which further increases do not introduce regime novelty but merely strengthen or scale the existing one.

This distinction is essential for separating emergence from accumulation. A regime may require a critical number of carriers in order to become admissible, but beyond a higher threshold it may simply deepen or enlarge without becoming qualitatively

different. Saturation therefore marks the point at which ontological novelty has already been achieved and additional growth produces only parametric extension.

The simplest formal statement is

$$N_x \geq N_{\text{sat}} \Rightarrow \frac{\partial R_e}{\partial N_x} \approx 0, \tag{309}$$

where R_e is the regime function of the edge or motif. Once saturation is reached, qualitative regime identity remains approximately invariant under further increase of N_x. In variational terms, this corresponds to a flattening of novelty yield in the effective admissibility landscape. The system may still become stronger, larger, faster, or denser, but it does not thereby cross into a further distinct organizational regime.

Saturation is especially important for preventing false positives in emergence analysis. Without a saturation criterion, every incremental increase after threshold crossing could be misread as a new ontological event. The theory avoids that error by explicitly distinguishing threshold novelty from post-threshold scaling.

Putting the criteria together

These five criteria can now be combined into a general transition logic. Let p be a candidate arising described by a transition Lagrangian $\mathcal{L}_{tr}(p)$, carrier state N_x, and closure measure $\mathcal{Q}_{cl}(p)$. Then the transition status of p can be classified in stages:

- **Undefined or inactive:** $N_x < N_{\min}$. The mechanism is not yet structurally formed.
- **Stationarily admissible but subcritical:** $\delta S_{tr} = 0$ and $N_{\min} \leq N_x < N_{\text{crit}}$. A coherent path exists, but no new regime is yet activated.
- **Threshold-active:** $N_x \geq N_{\text{crit}}$, or an equivalent variational critical condition has been crossed. A new regime becomes actively accessible.
- **Branch-sensitive:** the system enters a bifurcation region in which multiple continuations are admissible.
- **Stabilized arising:** one branch attains sufficient closure, $\mathcal{Q}_{cl}(p) \geq Q_{\text{crit}}$, and persists as a new regime.
- **Saturated regime:** $N_x \geq N_{\text{sat}}$, so further growth no longer changes ontological type.

This sequence corresponds directly to the distinction already present in the theory between structural potency, active unfolding, and realized organizational transition. The transition Lagrangian does not replace the ontology of phases and configurations. It provides the formal means to calculate where in that sequence a given configuration currently stands.

Transition criterion as a regime function

For implementation purposes, all five criteria can be combined into a regime-classification function:

$$\mathcal{R}_{tr}(p) = f\big(\delta S_{tr},\ N_x,\ H(\mathcal{L}_{tr}),\ \mathcal{Q}_{cl},\ N_{\min},\ N_{\text{crit}},\ N_{\text{sat}}\big). \tag{310}$$

The output of $\mathcal{R}_{tr}(p)$ is not simply a trajectory. It is a regime label: *inactive*, *admissible*, *subcritical*, *threshold-active*, *bifurcating*, *stabilized*, or *saturated*. This is the key methodological gain. The Lagrangian layer allows the theory to compute not only how a system moves, but whether a new arising is actually coming into being, whether it remains merely potential, and whether it has crossed into persistent ontological reality.

Why trajectory is not enough

This distinction matters because complex systems often exhibit motion without transition, activation without persistence, and amplification without emergence. A molecular system can fluctuate without reacting. A cell can activate stress programs without differentiating. A neural system can oscillate without producing a new integrative regime. A social group can intensify interaction without crossing into norm formation or institution-like stability. In all such cases, trajectory alone is an insufficient diagnostic. What matters is whether the trajectory satisfies the full set of criteria for regime change.

The transition Lagrangian therefore becomes the formal core of a broader criterion system. Stationarity identifies coherent admissibility. Threshold crossing identifies activation of a new regime. Bifurcation identifies branch-sensitive reorganization.

Closure identifies persistence as an ontological unit. Saturation identifies the end of novelty within the current regime. Taken together, these criteria transform the Lagrangian from a trajectory-generating formalism into a genuine calculus of arising.

The next subsection can now address the remaining conceptual link: how these criteria relate to structural potency. That is the point at which the theory explicitly distinguishes a configuration that already contains the admissibility conditions for a transition from one in which the transition has actually been realized.

9.6 Relation between Lagrangian selection and structural potency

The generalized transition Lagrangian becomes theoretically decisive only when it is connected to the ontology of structural potency. Without that connection, the Lagrangian would describe only already activated dynamics. The broader theory, however, requires something more precise: it must distinguish between a configuration that already contains the admissibility conditions for a transition and a configuration in which that transition has actually crossed into realized organizational form. This distinction is precisely the distinction between potency and realization.

Structural potency denotes neither an unrealized fiction nor a merely verbal possibility, but an objectively structured allowance embedded in a configuration. A potent configuration is one in which the required roles, relations, directions, interaction operators, and threshold-supporting conditions are already assembled in such a way that a specific transition has become admissible, even if it has not yet been activated. Potency therefore belongs to ontology rather than to psychology or epistemic uncertainty. It is a real property of the configuration itself: the system is arranged in a way that permits a transition, but the transition has not yet unfolded or been fixed as a new regime. This is already implicit in the theory's account of structural potency as a configuration-level allowance that precedes manifestation while still constraining what can arise from that structure.

The role of the Lagrangian is to make this distinction calculable. A generalized transition Lagrangian does not only classify active trajectories. It defines an admissibility landscape over the configuration. Within that landscape, some trajectories or regime branches are forbidden, some are permitted but subcritical, some are threshold-accessible, and some are realized and stabilized. The Lagrangian must therefore be read in two distinct modes. In the first mode, it functions as a selector of latent admissibility. In the second mode, it functions as the active balance governing realized unfolding.

The first mode corresponds to latent admissibility. A configuration is latently admissible when its effective transition Lagrangian already supports a coherent candidate regime, but the activation conditions for realization have not yet been crossed. In formal terms, the system possesses a structured variational minimum, branch, or admissible continuation in its transition landscape, but the corresponding regime has not yet become active in dynamics. The system is therefore not indifferent with respect to future transition. It is selectively prepared.

This can be stated in compact form as

$$\exists \pi^* \in \Pi(p) \ : \ \delta S_{tr}[\pi^*] = 0 \ \wedge \ \Theta_{\text{real}}(p) \text{ not yet crossed.} \tag{311}$$

Here $\Pi(p)$ denotes the set of admissible transition paths or regime branches for phenomenon p, and $\Theta_{\text{real}}(p)$ denotes the activation conditions required for actual realization. This formula expresses the precise meaning of potency in Lagrangian terms: the system already contains a stationary admissible continuation, but realization is not yet active. Potency is therefore not outside the Lagrangian. It is a specific region of the Lagrangian landscape: a region of structured permission without completed activation.

This point is crucial because it prevents confusion between possibility in the weak sense and potency in the strong ontological sense. Weak possibility means only that nothing in logic immediately forbids an event. Structural potency means that the configuration itself has already assembled the support required for a specific transition class. In other words, potency is a constrained and typed possibility. It belongs to the internal architecture of the system. The generalized Lagrangian reveals this because it assigns lower admissibility cost, viable branch structure, and threshold-relevant organization to some continuations rather than to others.

Realization begins when the latent admissibility encoded in the Lagrangian ceases to remain merely permissible and becomes dynamically active. This does not require that the resulting regime is already fully stabilized. It requires only that the configuration has crossed from structural permission into active unfolding. This is the role of initiation and threshold crossing. In the ontology of interaction modes, initiation converts latent permissiveness into active configuration, while realization fixes the resulting effect into an observable or stable regime. The generalized Lagrangian captures this through the appearance of threshold-active and realization terms that were previously only latent contributors.

The simplest expression of this passage is

$$\text{Pot}(p) \Rightarrow \text{Act}(p) \quad \text{iff} \quad \Theta_{\text{act}}(p) \text{ is crossed,} \tag{312}$$

where $\mathrm{Pot}(p)$ denotes structural potency and $\mathrm{Act}(p)$ denotes activated unfolding. This means that potency is the precondition of realization, but is not identical with it. A configuration can be potent without being active. But no realized regime can arise without a prior potent configuration, whether that potency was accumulated gradually, assembled abruptly, or became visible only at the moment of crossing.

The distinction can be sharpened further by separating three stages rather than two. First, there is latent admissibility. Second, there is activated regime entry. Third, there is realized arising in the full ontological sense, which requires persistence, closure, and stabilization. This tripartite distinction is necessary because many systems pass from potency into activation without successfully stabilizing the new regime. A reaction can be initiated without yielding a durable product. A cell can begin differentiative reconfiguration without locking into a stable new identity. A neural network can transiently enter a highly integrated state without establishing a reproducible higher-order mode. A social group can activate coordination without forming a persistent institution or norm structure.

These three stages may therefore be written as

$$\text{Latent Admissibility} \;\rightarrow\; \text{Activated Transition} \;\rightarrow\; \text{Realized Arising}, \tag{313}$$

or, more formally,

$$LA(p) \rightarrow AT(p) \rightarrow RA(p). \tag{314}$$

The first stage, $LA(p)$, holds when the Lagrangian already admits a structurally coherent branch. The second stage, $AT(p)$, holds when threshold-sensitive activation has begun. The third stage, $RA(p)$, holds only when the activated branch attains sufficient closure and persistence to count as a new arising rather than a transient excursion.

This distinction allows the relation between potency and realization to be stated with greater precision. Potency corresponds to latent admissibility. Realization corresponds not to any movement whatsoever, but to an activated and sufficiently stabilized regime. The difference is therefore not merely temporal. It is structural. In potency, the configuration contains the conditions of transition. In realization, those conditions have become operational, crossed threshold, and produced a regime that now participates causally as a new node in the system's organization.

The generalized Lagrangian makes this distinction measurable because each stage corresponds to a different regime of the same functional. In latent admissibility, the relevant branch exists but remains below critical activation. In activated transition, the branch is entered but may still remain fragile or branch-sensitive. In realized arising, the branch has acquired sufficient closure and retention structure to persist as a new ontological unit. This can be summarized by combining the previous criteria:

$$LA(p): \quad \delta S_{tr} = 0, \qquad N_x \geq N_{\min}, \qquad N_x < N_{\mathrm{crit}}, \tag{315}$$

$$AT(p): \quad \delta S_{tr} = 0, \qquad N_x \geq N_{\mathrm{crit}}, \qquad \mathcal{Q}_{cl} < Q_{\mathrm{crit}} \text{ or the branch remains unstable}, \tag{316}$$

$$RA(p): \quad \delta S_{tr} = 0, \qquad N_x \geq N_{\mathrm{crit}}, \qquad \mathcal{Q}_{cl} \geq Q_{\mathrm{crit}}. \tag{317}$$

These conditions are schematic, but they display the core logic. Latent admissibility begins when the mechanism is structurally available. Activated transition begins when the critical threshold is crossed. Realized arising begins when the activated branch acquires sufficient closure to persist as a new regime. Potency is therefore neither reducible to mere structural presence nor identical with finished emergence. It is the intermediate ontological condition in which the configuration has become transition-ready.

This also explains why structural potency should not be confused with hidden actuality. Potency is not a completed arising waiting behind a veil. The new regime does not already exist in full. What exists is a configuration with a Lagrangian landscape that already privileges or permits a specific class of transition. The content of realization remains open within the admissible branch space. That openness is especially important in bifurcating systems, where potency may support several possible continuations rather than one predetermined outcome. In such cases, the Lagrangian does not encode one fixed future event. It encodes a field of admissible futures together with their distinct costs, thresholds, and closure conditions.

For this reason, structural potency should be interpreted as admissible structured possibility rather than deterministic predestination. A potent configuration narrows what can happen next, but it does not erase branch selection, context sensitivity, or feedback-dependent path divergence. The generalized Lagrangian captures exactly this mixture of openness and structure. It defines the possible regime space, weights its branches, and specifies what must be crossed for one of them to become active and then real.

The distinction becomes especially important in multi-level systems. A lower-level configuration may become potent for a higher-level arising long before the higher-level node is realized. Catalytic networks may become potent for proto-cellular closure before actual bounded cellular organization appears. Tissue coordination may become potent for organogenesis before stable organ objects arise. Neural synchronization and integrative branching may become potent for conscious-access regimes before reportable conscious experience is stabilized. Social exchange and recurrent practice may become potent for institutionalization

before institutions become persistent objects or patterns. In all such cases, the Lagrangian formalism makes it possible to speak rigorously about partial readiness without falsely claiming that the higher-level arising is already present.

The conceptual payoff is substantial. The theory can now distinguish four different statuses of a candidate transition. A transition may be impossible because the configuration lacks potency. It may be potent but latent because activation has not begun. It may be active but unrealized because closure is insufficient. Or it may be realized because the activated regime has stabilized as a new causal unit. This gives the ontology a principled vocabulary for discussing incomplete, partial, emergent, precritical, and realized transitions without conflating them.

The relation between Lagrangian selection and structural potency can therefore be summarized in one sentence: the generalized Lagrangian defines the admissibility landscape of a configuration, potency is the existence of a transition-ready branch within that landscape, and realization is the activation and stabilization of that branch into a persistent new regime. This relation is fully consistent with the broader theory of causal arisings, in which configurations do not merely contain finished results, but define structured allowances through which new levels become admissible before they become real.

The next subsection can therefore move from pure theory to compact worked examples. This will make the distinction between latent admissibility, activated transition, and realized arising visible across several levels through the same formal structure but with different domain-specific terms.

9.7 Worked cross-level examples

The formal apparatus developed above becomes clearer when applied to compact examples drawn from different levels of organization. The purpose of these examples is not to provide a fully empirical model for each phenomenon, but to show how the same transition logic can be instantiated across heterogeneous domains. In each case, the procedure is the same: identify the configurational code, translate its active relations into effective Lagrangian terms, distinguish driving and subtractive structure, and then specify the transition criteria separating latent admissibility from realized arising.

Example 1. Level-0 admissibility filter

The Level-0 framework provides the cleanest minimal case because the transition logic is already explicitly variational. The relevant arisings are Temporal Distinction, Spatial Distinction, Compensation Energy, Admissible Wave Coherence, and the Admissibility Variational Functional, followed by Triadic Resonant Closure, Configurational Density, and the CRT-node. The configuration can be compressed as follows:

$$\Gamma_{L0} = \big(L_0, \{\mathrm{LP}_A, \mathrm{LP}_B, \mathrm{LP}_C\}, \{\mathrm{TR}_{PROP}, \mathrm{TR}_{PR}, \mathrm{TR}_{PAT}, \mathrm{TR}_{OBJ}\}, \{\mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_I, \mathrm{IM}_R, \mathrm{IM}_C\}, \{\mathrm{CD}_U, \mathrm{CD}_D, \mathrm{CD}_H, \mathrm{CD}_{DI}\}\big). \tag{318}$$

At the first variational stage, the effective transition Lagrangian is not yet a physical field Lagrangian in the later empirical sense. It is an admissibility filter over ordered distinction, localization, and compensation. A compact effective form is

$$\mathcal{L}_{L0} = D_{\mathrm{osc}}(\psi) - C_{\mathrm{comp}}(\psi) - C_{\mathrm{disp}}(\psi) + I_{\mathrm{dist}}(\partial_t, \partial_x). \tag{319}$$

Here $D_{\mathrm{osc}}(\psi)$ denotes the persistence of coherent oscillatory support, $C_{\mathrm{comp}}(\psi)$ denotes compensatory burden, $C_{\mathrm{disp}}(\psi)$ denotes dispersive loss, and I_{dist} denotes the initiation structure opened by temporal and spatial distinction.

The first threshold is not yet objecthood, but stable oscillatory admissibility:

$$\delta S_{L0} = 0. \tag{320}$$

The next threshold is triadic closure. At this point the theory explicitly rejects dyadic sufficiency and requires a three-mode closed contour for irreducible localization:

$$N_{\mathrm{mode}} \geq N_{\mathrm{crit}} = 3. \tag{321}$$

Closure is then measured not by mere oscillation but by concentrated resonant support:

$$\mathcal{Q}_{cl}^{CRT} = f(\Gamma, \Psi, \rho_{\mathrm{conf}}) \geq Q_{\mathrm{crit}}. \tag{322}$$

Latent admissibility at this level means that the variational filter already supports bounded coherent modes, but the triadic contour has not yet formed. Realized arising begins only when triadic closure and sufficient configurational density stabilize the CRT-node as the first persistent localized unit.

Example 2. Protein folding and enzymatic catalysis

At Level 3, a compact biochemical chain can be built from Protein Folding and Functional Conformation, Enzyme Active Sites, and Enzymatic Catalysis. This example is especially useful because it displays all three stages of the theory: latent admissibility, activation, and realized functional arising.

A minimal configurational code for the folding-to-catalysis chain is

$$\kappa_1 = \big(\{\mathrm{IM}_C, \mathrm{IM}_K\}; \mathrm{TR}_{OBJ} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_U\}; \textit{arising}; L_3, \mathrm{LP}_B\big), \tag{323}$$

$$\kappa_2 = \big(\{\mathrm{IM}_K, \mathrm{IM}_R\}; \mathrm{TR}_{PR} \to \mathrm{TR}_{PROP}; \{\mathrm{CD}_D\}; \textit{arising}; L_3, \mathrm{LP}_C\big), \tag{324}$$

$$\kappa_3 = \big(\{\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_R\}; \mathrm{TR}_{PROP} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_H, \mathrm{CD}_{DI}\}; \textit{arising}; L_3, \mathrm{LP}_C\big). \tag{325}$$

The effective transition Lagrangian may be written as

$$\mathcal{L}_{prot} = D_{\mathrm{fold}} + D_{\mathrm{bind}} + K_{\mathrm{align}} - C_{\mathrm{misfold}} - C_{\mathrm{barrier}} - C_{\mathrm{offpath}}. \tag{326}$$

Here D_{fold} denotes the drive toward entry into a functional conformational basin, D_{bind} denotes substrate-compatibility throughput, K_{align} denotes cooperative active-site alignment, C_{misfold} denotes conformational instability or misfolding burden, C_{barrier} denotes activation-barrier cost, and C_{offpath} denotes off-pathway reaction or structural mismatch burden.

Latent admissibility exists when the polypeptide sequence and local interaction architecture already support entry into a functional basin, but the conformation has not yet stabilized:

$$LA_{\mathrm{prot}}: \quad \delta S_{\mathrm{prot}} = 0, \qquad N_{\mathrm{func}} \geq N_{\mathrm{min}}, \qquad \mathcal{Q}_{cl}^{conf} < Q_{\mathrm{crit}}. \tag{327}$$

Activated transition begins when the folded conformation is reached and active-site structure becomes chemically available. Realized arising occurs when the catalytic regime becomes persistent and reproducible:

$$RA_{\mathrm{cat}}: \quad \mathcal{Q}_{cl}^{cat} \geq Q_{\mathrm{crit}}, \qquad N_{\mathrm{turnover}} \geq N_{\mathrm{crit}}. \tag{328}$$

This example shows why catalysis is not reducible to the mere existence of a protein object. The arising is realized only when folding, local property concentration, and pathway-lowering structure jointly overcome the subtractive burden of barrier and mismatch.

Example 3. Cellular regulation and gene-expression control

At Level 4, a compact example can be built from Prokaryotic Cell Organization, Cellular Signaling and Communication, and Genetic Regulation and Gene Expression. This example is important because it shows how transition calculation works in a domain where the subtractive side is dominated by viability and regulatory burden rather than by physical potential or chemical barrier alone.

A compact configurational code is

$$\kappa_1 = \big(\{\mathrm{IM}_C, \mathrm{IM}_K, \mathrm{IM}_R\}; \mathrm{TR}_{OBJ} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_U\}; \textit{maintenance}; L_4, \mathrm{LP}_B\big), \tag{329}$$

$$\kappa_2 = \big(\{\mathrm{IM}_F, \mathrm{IM}_R\}; \mathrm{TR}_{PR} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_H, \mathrm{CD}_{DI}\}; \textit{arising}; L_4, \mathrm{LP}_C\big), \tag{330}$$

$$\kappa_3 = \big(\{\mathrm{IM}_K, \mathrm{IM}_F, \mathrm{IM}_R\}; \mathrm{TR}_{PR} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_D, \mathrm{CD}_{DI}\}; \textit{maintenance}; L_4, \mathrm{LP}_C\big). \tag{331}$$

The effective transition Lagrangian is

$$\mathcal{L}_{cell} = D_{\mathrm{signal}} + D_{\mathrm{expr}} + F_{\mathrm{reg}} - C_{\mathrm{noise}} - C_{\mathrm{viab}} - C_{\mathrm{bnd}} - C_{\mathrm{damage}}. \tag{332}$$

Here D_{signal} denotes signaling-propagation drive, D_{expr} denotes regulated-expression throughput, F_{reg} denotes feedback-based control closure, C_{noise} denotes signaling ambiguity and stochastic interference, C_{viab} denotes viability burden, C_{bnd} denotes boundary-maintenance burden, and C_{damage} denotes damage load or genome-instability cost.

Latent admissibility here means that the cell already possesses the boundary, template, and regulatory substrate for a new control regime, but the relevant feedback architecture has not yet crossed into stable whole-cell coordination. Activated transition occurs when signaling and regulatory loops exceed their critical coordination threshold. Realized arising occurs when a persistent regulated regime emerges, such as stable gene-expression control or a durable specialized cellular state:

$$N_{\text{reg}} \geq N_{\text{crit}}, \qquad \mathcal{Q}_{cl}^{reg} \geq Q_{\text{crit}}. \tag{333}$$

This makes clear why cellular regulation is not simply a process of local chemistry. It is a transition into a regime in which signaling, feedback, boundary integrity, and viability burden become jointly balanced at the scale of the whole cell.

Example 4. Neural synchronization and critical coordination

At Level 6, a compact chain can be built from Formation of Neural Networks, Brain Oscillations and Rhythms, Global Neurotransmitter Modulation, and Network Synchronization and Criticality. This example shows particularly clearly how latent admissibility differs from realized large-scale integration.

A compact configurational code is

$$\kappa_1 = \big(\{\text{IM}_C, \text{IM}_R\}; \text{TR}_{PAT} \to \text{TR}_{PAT}; \{\text{CD}_H\}; \textit{arising}; L_6, \text{LP}_B\big), \tag{334}$$

$$\kappa_2 = \big(\{\text{IM}_F, \text{IM}_R\}; \text{TR}_{PAT} \to \text{TR}_{PAT}; \{\text{CD}_{DI}\}; \textit{modulation}; L_6, \text{LP}_B\big), \tag{335}$$

$$\kappa_3 = \big(\{\text{IM}_F, \text{IM}_K\}; \text{TR}_{PROP} \to \text{TR}_{PAT}; \{\text{CD}_D, \text{CD}_{DI}\}; \textit{modulation}; L_6, \text{LP}_C\big). \tag{336}$$

The effective transition Lagrangian is

$$\mathcal{L}_{neural} = D_{\text{prop}} + D_{\text{sync}} + F_{\text{loop}} + M_{\text{mod}} - C_{\text{inc}} - C_{\text{disp}} - C_{\text{met}} - C_{\text{phase}}. \tag{337}$$

Here D_{prop} denotes distributed propagation capacity, D_{sync} denotes synchronization drive, F_{loop} denotes recurrent loop support, M_{mod} denotes neuromodulatory gain control, C_{inc} denotes incoherence cost, C_{disp} denotes dispersion burden, C_{met} denotes metabolic cost, and C_{phase} denotes phase-misalignment or synchronization burden.

Latent admissibility exists when network architecture and oscillatory infrastructure already support the possibility of large-scale synchrony, but the system has not yet entered a coherent near-critical regime. Activated transition begins when recurrent coupling and modulation push the network across the critical synchronization threshold. Realized arising occurs only when large-scale coordination acquires sufficient closure to persist as a distinct dynamical regime rather than as a transient oscillatory episode:

$$N_{\text{sync}} \geq N_{\text{crit}}, \qquad \rho_{\text{loop}} \geq \rho_{\text{crit}}, \qquad \mathcal{Q}_{cl}^{net} \geq Q_{\text{crit}}. \tag{338}$$

This example shows that neural synchronization is not equivalent to mere rhythmic activity. Rhythms may already be present while the regime remains subcritical. The realized arising is the stabilized network-level coordination pattern that emerges only once distributed coherence overcomes incoherence, dispersion, and metabolic–synchronization burden.

Example 5. Collective decision-making in social systems

At Level 7, a compact social example can be constructed from Social Organization of Animals or human-group structure, Recurrent Social Practices, Social Exchange and Resource Coordination, and Collective Decision-Making. This example is important because it shows how the same formal structure extends into social domains, where the subtractive side is explicitly frictional rather than energetic.

A compact configurational code is

$$\kappa_1 = \big(\{\text{IM}_C, \text{IM}_R\}; \text{TR}_{PAT} \to \text{TR}_{PR}; \{\text{CD}_H\}; \textit{maintenance}; L_7, \text{LP}_B\big), \tag{339}$$

$$\kappa_2 = \big(\{\mathrm{IM}_K, \mathrm{IM}_F\}; \mathrm{TR}_{PAT} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_D, \mathrm{CD}_{DI}\}; \textit{maintenance}; L_7, \mathrm{LP}_B\big), \tag{340}$$

$$\kappa_3 = \big(\{\mathrm{IM}_I, \mathrm{IM}_R\}; \mathrm{TR}_{PR} \to \mathrm{TR}_{PR}; \{\mathrm{CD}_{DI}\}; \textit{arising}; L_7, \mathrm{LP}_C\big). \tag{341}$$

The effective transition Lagrangian is

$$\mathcal{L}_{soc} = D_{\mathrm{coord}} + D_{\mathrm{agg}} + F_{\mathrm{expect}} - C_{\mathrm{fric}} - C_{\mathrm{conf}} - C_{\mathrm{enf}} - C_{\mathrm{frag}}. \tag{342}$$

Here D_{coord} denotes coordination throughput, D_{agg} denotes aggregation of distributed signals, F_{expect} denotes feedback through shared expectations and recurrent practice, C_{fric} denotes coordination friction, C_{conf} denotes conflict burden, C_{enf} denotes enforcement cost, and C_{frag} denotes informational fragmentation. These terms are not introduced ad hoc. They follow directly from the social nodes in which practices, exchange, role structure, and collective decision are treated as process- and pattern-level arisings requiring distributed coordination under organizational burden.

Latent admissibility exists when the group already possesses sufficient stable interaction structure and continuity of practice for a collective decision regime to be possible, but signal aggregation remains below effective threshold. Activated transition begins when quorum, influence alignment, or convergence dynamics cross the critical point. Realized arising occurs when the selected outcome is not merely momentarily aggregated but becomes group-effective and reorganizes subsequent coordination:

$$N_{\mathrm{group}} \geq N_{\mathrm{crit}}, \qquad \mathcal{A}_{\mathrm{sig}} \geq A_{\mathrm{crit}}, \qquad \mathcal{Q}_{cl}^{soc} \geq Q_{\mathrm{crit}}. \tag{343}$$

This example makes visible the generality of the framework. A social decision regime is structurally analogous to lower-level transitions: there is a coded configuration, a balance of driving and subtractive terms, a threshold, a branch-sensitive convergence region, and a closure criterion determining whether the outcome remains episodic or becomes causally effective at group level.

Cross-level comparison

Taken together, these examples show that the generalized transition formalism is neither domain-specific rhetoric nor an empty abstraction. At Level 0 the Lagrangian filters admissible coherent variation and triadic closure. In biochemistry it balances folding and catalytic drive against barrier and misfold burden. In cellular regulation it balances signaling and control against viability and boundary cost. In neural systems it balances distributed synchronization against incoherence and metabolic burden. In social systems it balances coordination and aggregation against friction, conflict, enforcement cost, and fragmentation.

The same formal structure recurs in every case:

$$\mathcal{L}_{tr} = \mathcal{D} - \mathcal{C} + \mathcal{K} + \mathcal{F} + \mathcal{I} + \mathcal{R} - \mathcal{S}. \tag{344}$$

What changes is not the logic of transition selection, but the ontology of the terms. This is exactly what the theory requires: one formal architecture, many domain-specific realizations. The examples therefore justify the use of the generalized Lagrangian as a true cross-level operator of transition admissibility rather than as a physical metaphor loosely borrowed for other sciences.

The next subsection must address the methodological limit of this framework: where the generalized Lagrangian can be written directly, where it must remain effective or simulation-based, and where the configuration itself must first be refined before any honest variational compression becomes possible.

9.8 Methodological limits of the generalized Lagrangian

The generalized transition Lagrangian is introduced as a formal bridge between configurational ontology and transition calculation. Its strength lies in compression: it allows a heterogeneous causal configuration to be rewritten as a structured balance of driving, constraining, coupling, feedback, initiation, realization, and suppression terms. Precisely because the framework is generalized across many levels of organization, however, it cannot be applied in the same epistemic manner everywhere. In some domains the Lagrangian can be written directly from already formalized variables. In other domains it can only be written as an effective or phenomenological approximation. In still other cases, no honest Lagrangian should yet be written at all, because the underlying configuration has not been specified with sufficient ontological clarity.

This limitation is not a defect of the framework. It is a methodological requirement imposed by the ontology itself. The theory of causal arisings already distinguishes between structural description and realized transition, between qualitative typing

and quantitative activation, and between structural potency and realized regime. The same discipline must apply here: one must not assign a compact variational form to a system whose operative roles, directions, dependency classes, carrier ontology, and threshold structure remain indeterminate. A generalized Lagrangian is legitimate only when the relevant configuration is sufficiently resolved to support meaningful compression.

Direct applicability: domains with already formalized state variables and admissibility structure

The generalized Lagrangian is most directly applicable in domains where the system already possesses a stable and explicit state space, well-defined variables, interpretable couplings, and mathematically legible admissibility structure. This includes classical mechanics, field theory, statistical-field descriptions, many reaction-kinetic systems, some population-dynamical models, and selected control-theoretic biological or neural subdomains. In such cases the variables are already explicit enough that driving and subtractive terms can be assigned without major ontological ambiguity.

The Level-0 framework provides the clearest example of this kind of direct applicability. There the admissibility variational functional is introduced not as metaphor but as the explicit structural filter through which coherent oscillatory configurations are selected, extended, and stabilized, with the logic of compensation, hyperbolic admissibility, scalar closure, and triadic resonance already written in a variational language. At that level, the generalized Lagrangian is not merely heuristic. It is native to the construction.

A weaker but still direct applicability exists in domains such as protein folding, catalytic channeling, cellular transport, electrophysiology, and some network-dynamical settings, where the relevant variables may not yield a single canonical Lagrangian but still support a disciplined effective balance between transition drive and configurational burden. In such cases the Lagrangian is direct in the sense that the variable set and causal operator classes are sufficiently explicit to support formal construction, even if the resulting functional is not unique.

The methodological criterion for direct applicability may be stated as follows. A generalized Lagrangian is directly admissible when four conditions are met simultaneously: the relevant variables are explicit; the dominant interaction modes are identifiable; the principal constraints and closure terms can be specified; and the threshold logic is attached to a known carrier ontology rather than to undifferentiated counting. If these conditions hold, then the transition functional may be written as a legitimate first-order representation of the configuration rather than as a speculative summary.

Effective applicability: domains requiring phenomenological or simulation-based approximation

A second class of cases includes systems in which a generalized Lagrangian is meaningful, but only as an effective approximation. This is the appropriate mode for many biological, neural, and social systems. In such domains the operative variables are real, but they are typically coarse-grained, composite, path-dependent, and context-sensitive. The system may still be treated through a balance functional, but that functional must not be mistaken for an exact microscopic law. It is an effective compression of an already typed causal configuration.

This mode of applicability is already anticipated by the theory's treatment of decoupled causation, in which macro-dynamics remain valid across micro-variation and are therefore represented through effective parameters rather than through full micro-reconstruction. The generalized Lagrangian frequently operates in precisely this decoupled sense outside foundational physics. It does not reconstruct every lower-level event. It captures the admissibility balance of the higher-order regime that becomes visible at the selected descriptive level.

Examples include tissue repair, organ-level homeostasis, neural synchronization, collective decision-making, norm stabilization, market dynamics, and institutional persistence. In all such cases the transition logic can be modeled through driving and subtractive terms, but the terms themselves are already effective aggregates: viability burden rather than every molecular perturbation, incoherence rather than every spike event, coordination friction rather than every local interaction history, informational fragmentation rather than every symbolic token. The causal configuration supports variational compression, but only after ontological aggregation has already occurred.

In this effective regime, the correct methodological attitude is one of restrained realism. The Lagrangian is real as a model of transition selection at the chosen level, but it is not ontologically fundamental. It is valid within a regime, under a given granularity, and relative to a defined carrier structure. The same phenomenon may admit multiple effective Lagrangians depending on whether one models it at the scale of local repair, organ homeostasis, whole-organism regulation, network synchrony,

or collective decision dynamics. This plurality is not a failure. It is the expected consequence of multi-level causality and scale-dependent admissibility.

The methodological criterion here is that the Lagrangian may be written when the coarse-grained variables are sufficiently stable, the operator classes remain interpretable, and the resulting functional can be validated against threshold behavior, closure, or regime persistence. In such cases the generalized Lagrangian should be treated as a regime functional rather than as an exact law of the full substrate.

Pre-Lagrangian cases: when the configuration must be clarified first

The most important limit of the framework concerns cases in which the configuration is not yet sufficiently specified. In such cases a generalized Lagrangian should not be written directly. The first task is not calculation but ontological clarification. One must determine what the actual nodes are, what type roles they occupy, which parent relations are constitutive, which interaction modes are dominant, how causal directions are oriented, which dependency types are active, and what carrier ontology governs threshold structure. Until this has been done, any compact balance functional risks being rhetorically elegant but structurally false.

This is especially important in domains where the literature is rich in description but poor in ontological separation. The theory explicitly warns against conflating objects, processes, patterns, properties, and effects within a single node, and against collapsing heterogeneous causal roles into undifferentiated relations. If that discipline is not observed, a Lagrangian written over such a configuration would not be a compression of the actual structure. It would instead be a compression of conceptual confusion.

Typical pre-Lagrangian situations include speculative cross-domain analogies, underdefined social constructs, poorly individuated cognitive states, vaguely described technological ecosystems, and emerging biological phenomena whose carrier structure and dependency classes remain unresolved. In such cases the correct procedure is to return to configurational analysis. One must first refine $\Gamma(p)$, expand the edge-level motifs $\kappa(r)$, assign interaction modes and causal directions more carefully, identify carrier groups and carrier types, and only then ask whether a meaningful effective balance can be written.

This point is methodologically non-negotiable. A generalized Lagrangian is never a substitute for ontological analysis. It is a compression of ontological analysis only after that analysis has become structurally mature enough. If the configuration remains ambiguous, the correct output is not a premature equation but a refined graph, a clarified dependency structure, or a simulation-ready motif library that still precedes variational compression.

How to decide which regime applies

The framework therefore requires an explicit decision rule before constructing a generalized Lagrangian. The first question is whether the phenomenon already has an interpretable state space or regime space. The second is whether the dominant causal operators are identifiable in terms of interaction modes, causal directions, ontological roles, and dependency classes. The third is whether carrier-sensitive threshold logic is specified or at least inferable. The fourth is whether the intended functional will describe the phenomenon directly, effectively, or only after further configurational work.

This yields a three-way methodological classification:

- **Direct Lagrangian regime:** variables, constraints, and thresholds are explicit enough to support a formal transition functional immediately.
- **Effective Lagrangian regime:** the phenomenon is best treated through coarse-grained or phenomenological variables, with the functional interpreted as a regime approximation.
- **Pre-Lagrangian regime:** the configuration is still too ambiguous, and the task is first to refine the ontology of the arising before compressing it into a variational form.

This classification should itself be treated as part of the theory of admissibility. Just as a transition may be latent, active, stabilized, or saturated, the Lagrangian description itself may be direct, effective, or premature. A responsible theory must classify not only phenomena, but also the legitimacy of its own formal tools.

The role of simulation and hybrid formalism

Between direct and effective regimes there exists an important intermediate practical zone: simulation-based Lagrangian approximation. In many complex systems the balance functional is easier to construct operationally than analytically. One may assign weighted driving, constraint, coupling, and threshold terms directly to graph motifs, simulate the resulting dynamics, and only afterward infer an approximate effective functional from the observed regime structure. This is especially relevant in domains where the theory already includes graph architecture, carrier thresholds, and multi-level causal motifs as formal objects of representation.

In such cases the generalized Lagrangian functions as a hybrid formalism. Part of it may be analytic, part discrete, part graph-dynamical, and part learned from simulation. This does not violate the theory. The essential requirement is not that the functional be smooth or closed-form, but that it remain a disciplined representation of transition selection grounded in the causal configuration. A symbolic weighted operator sum, a graph-based admissibility score, or a simulation-derived effective potential may all count as legitimate Lagrangian forms provided they preserve the ontology of the configuration together with its threshold logic.

Final methodological principle

The generalized Lagrangian should therefore be treated neither as a universal master equation nor as a loose metaphor for any kind of change. It is a formal compression tool whose legitimacy depends on the maturity of the underlying configuration. Where variables and admissibility structure are explicit, the Lagrangian can be applied directly. Where only higher-level regime structure is stable, it should be applied effectively. Where the configuration itself remains underdefined, the correct task is not compression but ontological clarification.

This principle keeps the theory aligned with its own foundations. SOMA does not begin from formulas and then force reality into them. It begins from structured causal arisings and asks which formal compressions become admissible once the configuration is sufficiently specified. The generalized Lagrangian is one of the strongest such compressions, but only when used under the correct methodological conditions. In this sense, its limit is also its strength: it cannot replace ontology, but once ontology is sufficiently mature, it becomes one of the most powerful instruments for calculating how transitions become admissible, active, and real.

With this, the present block is complete. It has established why a generalized Lagrangian is needed, how it is constructed from causal configurations, what the subtractive side means across domains, how transitions are identified through thresholds and closure, how latent potency differs from realized arising, where the formalism can be applied directly, where only effectively, and where ontology must still precede calculation.

10 Conclusion and further development of the theory

The present manuscript has established the current formal core of the Theory of Causal Arisings. It has introduced the ontological grammar of arisings, the architecture of causal configurations, the principle of structural potency, the notion of causal completeness, and the generalized Lagrangian framework for transition selection across heterogeneous domains. Taken together, these components define a unified basis for describing how new organizational regimes become admissible, how they are stabilized, and how they can be analyzed in a typed multi-level ontology.

At the same time, the theory remains under active development. The framework presented here should therefore be understood not as a finished closed system, but as a structurally coherent and expandable foundation. Several major components are still being developed in order to extend the theory from its current descriptive and formal core toward broader predictive, quantitative, and cybernetic scope.

The next stages of development will include at least the following sections.

Emergent quantitative characteristics in models of self-organization. This section will address the numerical descriptors that arise in self-organizing systems and cannot be reduced to isolated local variables alone. The goal is to formalize how higher-order measurable characteristics emerge from configurational closure, distributed interaction, and threshold-sensitive coordination.

Predictive architecture: construction of hypotheses about non-existing emergent phenomena. This section will extend the theory from analysis of realized arisings to the structured generation of hypotheses about arisings that have not yet been observed. The aim is to define how admissibility rules, configurational motifs, and structural analogies can be used to infer plausible but as yet unrealized emergent forms.

Formula-based formalization of arising. This section will develop a more explicit symbolic and mathematical language for expressing arisings, transitions, and configurational dependencies in formula form. Its purpose is to move from descriptive formalization toward a more systematic calculus of arising.

Ontocybernetic model of complexity. This section will formulate complexity in explicitly ontocybernetic terms, integrating causal ontology with regulation, control, feedback, and multi-level adaptive organization. The goal is to show how the theory of arisings can be extended into a broader model of structured complexity and recursive system governance.

These forthcoming developments are not external additions to the theory. They are direct continuations of the ontology established in the present work. The theory therefore remains open in the strong methodological sense: open not because it is unfinished in a vague way, but because its own structure already points beyond its current formulation toward further formal, predictive, and computational elaboration.

The work on the theory is ongoing.